



U.S. Department
of Transportation

**Federal Aviation
Administration**

Advisory Circular

TITLE 14 OF THE CODE OF FEDERAL REGULATIONS (14 CFR) GUIDANCE MATERIAL

Subject: ACCEPTABLE METHODS,
TECHNIQUES, AND PRACTICES—AIRCRAFT
INSPECTION AND REPAIR

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1. **PURPOSE.** This advisory circular (AC) contains methods, techniques, and practices acceptable to the Administrator for the inspection and repair of nonpressurized areas of civil aircraft, only when there are no manufacturer repair or maintenance instructions. This data generally pertains to minor repairs. The repairs identified in this AC may only be used as a basis for FAA approval for major repairs. The repair data may also be used as approved data, and the AC chapter, page, and paragraph listed in block 8 of FAA form 337 when:
 - a. the user has determined that it is appropriate to the product being repaired;
 - b. it is directly applicable to the repair being made; and
 - c. it is not contrary to manufacturer's data.
2. **CANCELLATION.** The AC 43.13-1A dated 1988 is canceled.
3. **REFERENCE:** Title 14 of the Code of Federal Regulations part 43, section 43.13(a) states that each person performing maintenance, alteration, or preventive maintenance on an aircraft, engine, propeller, or appliance shall use the methods, techniques, and practices prescribed in the current manufacturer's maintenance manual or Instructions for Continued Airworthiness prepared by its manufacturer, or other methods, techniques, or practices acceptable to the Administrator, except as noted in section 43.16. FAA inspectors are prepared to answer questions that may arise in this regard. Persons engaged in the inspection and repair of civil aircraft should be familiar with 14 CFR part 43, Maintenance, Preventive Maintenance, Rebuilding, and Alteration, and part 65, Subparts A, D, and E of Certification: Airmen Other Than Flight Crewmembers, and the applicable airworthiness requirements under which the aircraft was type certificated.
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5. **COMMENTS INVITED.** Comments regarding this AC should be directed to DOT/FAA; ATTN: Airworthiness Programs Branch, AFS-610; PO Box 25082; Oklahoma City, OK 73125

Acting Deputy Director, Flight Standards Service

CONTENTS

Paragraph

Page

CHAPTER 1. WOOD STRUCTURE

SECTION 1. MATERIALS AND PRACTICES

1-1.	General.....	1-1
1-2.	Woods.....	1-1
	Figure 1-1. Relative Shrinkage of Wood Member Due to Drying.....	1-1
	Table 1-1. Selection and Properties of Aircraft Wood.....	1-3
	Figure 1-2. Tapering of Faceplate	1-3
1-3.	Modified Wood Products	1-3
1-4.	Adhesives.....	1-3
1-5.	Bonding Precautions.....	1-5
1-6.	Preparation of Wood Surfaces for Bonding.....	1-5
1-7.	Applying the Adhesive	1-6
1-8.	Assembly Time in Bonding.....	1-6
1-9.	Bonding Temperature.....	1-6
1-10.	Clamping Pressure.....	1-7
1-11.	Method of Applying Pressure.....	1-7
1-12.—1-17.	[RESERVED.].....	1-8

SECTION 2. HEALTH AND SAFETY

1-18.	General.....	1-9
1-19.	Sanding in Areas of Existing Bond Joints and Finishes.....	1-9
1-20.	Handling of Adhesives and Finishes	1-9
1-21.—1-26.	[RESERVED.].....	1-9

SECTION 3. INSPECTION

1-27.	General.....	1-11
1-28.	Types of Deterioration and Damage	1-11
1-29.	Inspection Methods	1-11
	Figure 1-3. Likely Areas to Incur Structural Damage	1-13
1-30.—1-35.	[RESERVED.].....	1-14

SECTION 4. REPAIRS

1-36.	General.....	1-15
1-37.	Replacement of Drain Holes and Skin Stiffeners.....	1-15
1-38.	Control Surface Flutter Precautions	1-15
1-39.	Scarf Joints.....	1-15
	Figure 1-4. Consideration of Grain Direction When Making Scarf Joints	1-15
1-40.	Splicing of Spars.....	1-15
	Figure 1-5. Method of Splicing Solid or Laminated Rectangular Spars.....	1-17
	Figure 1-6. Method of Splicing Solid “I” Spars	1-18
	Figure 1-7. Repairs to Built-Up “I” Spar.....	1-19
	Figure 1-8. Method of Splicing Box Spar Flanges (Plate Method)	1-20
1-41.	Spar Replacement.....	1-16
1-42.	Splicing of Box Spar Webs	1-16
	Figure 1-9. Method of Splicing Box Spar Webs.....	1-21
1-43.	Replacing Solid-Type Spars With Laminated-Type Spars	1-16

CONTENTS (CONTINUED)

Paragraph	Page
1-44. Spar Longitudinal Cracks and Local Damage.....	1-16
Figure 1-10. Method of Reinforcing a Longitudinal Crack and/or Local Damage in a Solid Spar.....	1-22
1-45. Elongated Holes in Spars	1-18
1-46. Rib Repairs	1-18
Figure 1-11. Repair of Wood Ribs	1-23
Figure 1-12. Typical Wing Compression Rib Repairs	1-24
1-47. Plywood Skin Repair.....	1-20
1-48. Determination of Single or Double Curvature	1-20
1-49. Repairs to Single Curvature Plywood Skin.....	1-20
Table 1-2. Minimum Recommended Bend Radii for Aircraft Plywood.....	1-25
1-50. Repairs to Double Curvature Plywood Skin	1-22
1-51. Types of Patches.....	1-25
Figure 1-13. Splayed Patch.....	1-27
Figure 1-14. Surface Patches	1-28
Figure 1-15. Scarf Patches (Back of Skin Accessible).....	1-29
Figure 1-16. Scarf Patches (Back of Skin Not Accessible).....	1-30
Figure 1-17. Oval Plug Patch Assembly.....	1-32
Figure 1-18. Round Plug Patch Assembly.....	1-33
1-52. Fabric Patch.....	1-31
1-53.—1-63. [RESERVED.].....	1-33

SECTION 5. FINISHING WOOD STRUCTURES

1-64. General.....	1-35
1-65. Acceptable Finishes	1-35
1-66. Precautions.....	1-35
1-67. Finishing of Interior Surfaces	1-36
1-68. Finishing of Exterior Surfaces	1-36
1-69. Finishing of End Grain Surfaces	1-36
1-70. Finishing with Fabric or Tape	1-37
1-71. Sealing of Bolt Holes.....	1-37
1-72.—1-79. [RESERVED.].....	1-37

CHAPTER 2. FABRIC COVERING

SECTION 1. PRACTICES AND PRECAUTIONS

2-1. General.....	2-1
2-2. Problem Areas	2-1
2-3. Aircraft Fabric-Synthetic	2-1
2-4. Aircraft Fabric-Natural	2-2
Table 2-1. Cotton and Linen Fabrics.....	2-3
2-5. Recovering Aircraft	2-2
Table 2-2. Cotton and Linen, Tapes and Threads.....	2-4
2-6. Preparation of the Structure for Covering.....	2-2
2-7. Fabric Seams.....	2-5
Figure 2-1. Fabric Seams.....	2-7
Figure 2-2. Typical Methods of Attaching Fabric.....	2-8
2-8. Covering Methods	2-9

CONTENTS (CONTINUED)

Paragraph	Page
2-9. Reinforcing Tape	2-10
Figure 2-3. Exploded Side View of Rib	2-10
2-10. Lacing	2-11
Figure 2-4. Standard External Modified Seine Knot Used for Single and Double Rib Lacing.....	2-11
Figure 2-5. Starting Stitch for Rib Lacing.....	2-12
Figure 2-6. Standard Single-Loop Lacing.....	2-13
Figure 2-7. Standard Knot for Double-Loop Lacing.....	2-14
Figure 2-8. Standard Double-Loop Lacing (Optional).....	2-15
Figure 2-9a. Alternate Sequence to Tie a Modified Seine Knot for Rib Lacing	2-16
Figure 2-9b. Alternate Sequence to Tie a Modified Seine Knot for Rib Lacing	2-17
Figure 2-9c. Alternate Sequence to Tie a Modified Seine Knot for Rib Lacing	2-18
Figure 2-10. Splice Knot.....	2-19
Figure 2-11. Blindstitch Lacing - Square Knot Secured with Half Hitches.....	2-20
2-11. Stitch Spacing	2-19
Figure 2-12. Fabric Attachment Spacing.....	2-21
2-12. Fasteners	2-21
2-13. Finishing Tape	2-22
2-14. Inspection Rings and Drain Grommets	2-22
2-15.—2-19. [RESERVED.].....	2-23

SECTION 2. APPLICATION OF DOPE

2-20. General.....	2-25
Table 2-3. Safety Tips for Dope/Paint.....	2-26
Table 2-4. Tips for Doping.....	2-26
2-21. Dope Application Procedure (Natural Fabrics)	2-26
2-22. Covering Over Plywood.....	2-28
2-23. Coating Application Defects	2-29
2-24.—2-29. [RESERVED.].....	2-31

SECTION 3. INSPECTION AND TESTING

2-30. General.....	2-33
2-31. Fabric Identification	2-34
2-32. Coating Identification.....	2-34
2-33. Strength Criteria for Aircraft Fabric.....	2-35
2-34. Fabric Testing.....	2-35
2-35. Rejuvenation of Dope Film	2-36
2-36.—2-41. [RESERVED.].....	2-36

SECTION 4. REPAIRS TO FABRIC COVERING

2-42. General.....	2-37
2-43. Repair of Tears and Access Openings.....	2-37
Figure 2-13 Repair of Tears in Fabric	2-38
2-44. Sewn-Patch Repair	2-38
Figure 2-14 Hand-Stitch Detail.....	2-39
2-45. Doped-On Patch Repair	2-39
2-46.—2-51. [RESERVED.].....	2-40

CONTENTS (CONTINUED)

Paragraph

Page

CHAPTER 3. FIBERGLASS AND PLASTICS

SECTION 1. REPAIR OF LIGHT LOAD LAMINATE STRUCTURES

3-1.	General.....	3-1
3-2.	Fiberglass Laminate Repairs	3-1
	Figure 3-1. Typical Laminate (Facing) Repair	3-2
	Figure 3-2. Typical Core and Facing Repair	3-2
	Figure 3-3. Typical Stepped Joint Repair	3-3
3-3.	Repairing Holes	3-3
	Figure 3-4. Preparing the Fiberglass Sandwich	3-3
	Figure 3-5. Scarfed Repair to a Nonstructural Laminated Fiberglass Component	3-4
	Figure 3-6. Balanced Layup Same as Original Number of Plies, Plus One Extra Ply	3-4
	Figure 3-7. Typical Scarf Joint Repair	3-5
	Figure 3-8. Carefully Cut Through Each Layer of Fiberglass Cloth and Remove it From the Damaged Area	3-5
3-4.	Sample Bagging and Curing Process	3-5
	Figure 3-9. Sample Bagging Layup Cross Section.....	3-6
3-5.—3-9.	[RESERVED.].....	3-8

SECTION 2. METALLIC SANDWICH SECONDARY STRUCTURE REPAIRS

3-10.	Repairs to Metallic Sandwich Secondary Structure.....	3-9
	Figure 3-10. Typical Types of Core Replacement	3-10
	Figure 3-11. Typical Undercut Core Material Cavity	3-10
	Figure 3-12. Cross Section of Bonded and Bolted Overlap Repairs	3-11
	Figure 3-13. Honeycomb Core Removal.....	3-11
	Figure 3-14. Removing Honeycomb Core from a Tapered Control Surface.....	3-11
	Figure 3-15. Removal of Core with Core Knife	3-11
	Figure 3-16. Details of Core Repair	3-12
3-11.	Finishing	3-12
3-12.—3-17.	[RESERVED.].....	3-13

SECTION 3. TRANSPARENT PLASTICS

3-18.	General.....	3-15
3-19.	Storage and Handling	3-15
3-20.	Forming Procedures and Techniques.....	3-15
3-21.	Heating.....	3-16
	Figure 3-17. Hanging of Acrylic Sheets	3-16
3-22.	Forms	3-16
3-23.	Forming Methods	3-17
	Table 3-1. Typical Temperatures for Forming Acrylic Sheets	3-17
	Figure 3-18. Drill Having an Included Angle of Approximately 150°, Used to Drill Acrylic Plastics.....	3-18
	Figure 3-19. Unibit Drill for Drilling Acrylic Plastics	3-18
	Figure 3-20. Applying Pressure to Acrylic Plastics.....	3-19

CONTENTS (CONTINUED)

Paragraph	Page
3-24. Repair of Plastics.....	3-19
Figure 3-21. Stop-Drilling Cracks	3-20
Figure 3-22. Surface Patches	3-20
Figure 3-23. Plug Patch Repair.....	3-21
3-25. Cleaning and Polishing Transparent Plastic.....	3-21
3-26. Replacement Panels.....	3-21
3-27. Installation Procedures	3-22
3-28.—3-39. [RESERVED.].....	3-22

SECTION 4. WINDSHIELDS, ENCLOSURES, AND WINDOWS

3-40. General.....	3-23
Figure 3-24. Temporary Repairs to Cracked Windshields or Windows.....	3-24
3-41. Protection.....	3-24
3-42. Windshield Installation.....	3-24
3-43.—3-47. [RESERVED.].....	3-25

CHAPTER 4. METAL STRUCTURE, WELDING, AND BRAZING

SECTION 1. IDENTIFICATION OF METALS

4-1. General.....	4-1
Table 4-1. Ferrous (Iron) Alloy Materials.....	4-1
4-2. Identification of Steel Stock	4-1
Table 4-2. Numerical System for Steel Identification.....	4-2
Table 4-3. Examples of Stainless and Heat-Resistant Steels Nominal Composition (Percent).....	4-3
4-3. Interchangeability of Steel Tubing.....	4-2
4-4. Identification of Aluminum	4-2
Table 4-4. Basic Temper Designations and Subdivisions from Aluminum Alloys.....	4-4
4-5.—4-15. [RESERVED.].....	4-4

SECTION 2. TESTING OF METALS

4-16. Hardness Testing	4-5
4-17. Rockwell Hardness Test.....	4-5
4-18. Brinell Hardness Test	4-5
4-19. Vickers Hardness Test.....	4-6
4-20. Microhardness Testing	4-6
4-21. Indentations	4-7
Figure 4-1. Comparison of Indentation Made by Knoop and Vickers Indenters in the Same Work Metal and at the Same Loads.....	4-7
4-22. Magnetic Testing	4-8
4-23. Aluminum Testing.....	4-8
Table 4-5. Hardness Values for Aluminum Alloys. (Reference MIL-H-6088G.).....	4-8
4-24.—4-35. [RESERVED.].....	4-8

CONTENTS (CONTINUED)

Paragraph

Page

SECTION 3. PRECAUTIONARY MEASURES

4-36.	Flutter and Vibration Precautions.....	4-9
4-37.	Load Factors for Repairs	4-10
4-38.	Transfer of Stresses Within a Structure.....	4-10
4-39.—4-49.	[RESERVED.].....	4-10

SECTION 4. METAL REPAIR PROCEDURES

4-50.	General.....	4-11
4-51.	Riveted (or Bolted) Steel Truss-Type Structures.....	4-11
4-52.	Aluminum Alloy Structures.....	4-11
4-53.	Selection of Aluminum for Replacement Parts.....	4-12
4-54.	Heat Treatment of Aluminum Alloy Parts	4-12
4-55.	Bending Metal	4-13
	Table 4-6. Recommended Radii for 90-Degree Bends in Aluminum Alloys.....	4-14
4-56.	Setback.....	4-13
	Figure 4-2. Setback for a 90-Degree Bend.....	4-13
	Table 4-7. K-Chart for Determining Setback for Bends Other Than 90 Degrees	4-15
	Figure 4-3. Methods of Determining Setbacks for Bends Other Than 90 Degrees	4-16
4-57.	Riveting.....	4-14
	Figure 4-4. Rivet Identification and Part Number Breakdown	4-16
	Table 4-8. Aircraft Rivet Identification.....	4-17
	Figure 4-5. Rivet Hole Spacing and Edge Distance for Single-Lap Sheet Splices	4-20
	Figure 4-6. Riveting Practice and Rivet Imperfections	4-21
	Figure 4-7. Self Plugging Friction-Lock Cherry Rivets	4-22
	Figure 4-8. Mechanical-Lock (Bulbed Cherry) Cherry Rivet.....	4-23
	Figure 4-9. Cherry Max Rivet.....	4-24
	Figure 4-10. Olymic-Lok Rivet	4-25
	Figure 4-11. Huck Rivet	4-26
4-58.	Repair Methods and Precautions for Aluminum Structure	4-25
	Figure 4-12. Typical Repair Method for Tubular Members of Aluminum Alloy.....	4-28
	Figure 4-13. Typical Repair for Buckled or Cracked Metal Wing Rib Capstrips.....	4-29
	Figure 4-14. Typical Metal Rib Repairs (Usually Found on Small and Medium-Size Aircraft).....	4-30
	Figure 4-15. Typical Repairs of Trailing Edges	4-31
	Figure 4-16. Typical Repairs of Stressed Sheet Metal Coverings	4-32
	Figure 4-17. Typical Stringer and Flange Splices	4-33
	Figure 4-18. Example of Stringer Splice (Material-2017 Alloy)	4-34
	Figure 4-19. Application of Typical Flange Splices and Reinforcement.....	4-35
	Table 4-9. Number of Rivets Required for Splices (Single-Lap Joint) in Bare 2014-T6, 2024-T3, 2024-T3, and 7075-T6 Sheet, Clad 2014-T6, 2024-T3, 2024-T36, and 7075-T6 Sheet, 2024-T4, and 7075-T6 Plate, Bar, Rod, Tube, and Extrusions, 2014-T6 Extrusions	4-37
	Table 4-10. Number of Rivets Required for Splices (Single-Lap Joint) in 2017, 1017 Alclad, 2024-T3 Alclad Sheet, Plate, Bar, Rod, Tube, and Extrusions	4-38
	Table 4-11. Number of Rivets Required for Splices (Single-Lap Joint) in 5032 (All Hardnesses) Sheet	4-39

CONTENTS (CONTINUED)

Paragraph	Page
4-59. Repairing Cracked Members.....	4-40
Figure 4-20. Example of Intermediate Frame Stringer Splice (Material 2017-T3 AL Alloy).....	4-41
Figure 4-21. Typical Methods of Repairing Cracked Leading and Trailing Edges and Rib Intersections	4-42
Figure 4-22. Typical Methods of Replacing Cracked Member at Fittings	4-43
Figure 4-23. Typical Methods of Repairing Cracked Frame and Stiffener Combination	4-44
Figure 4-24. Typical Repairs to Rudder and to Fuselage at Tail Post	4-45
4-60. Steel and Aluminum Fittings	4-40
Figure 4-25. Typical Methods of Repairing Elongated or Worn Bolt Holes.....	4-46
4-61. Castings.....	4-40
4-62. Selective Plating in Aircraft Maintenance.....	4-40
4-63.—4-73. [RESERVED.].....	4-51

SECTION 5. WELDING AND BRAZING

4-74. General.....	4-53
Table 4-12. Current and Polarity Selection for Inert Gas Welding	4-54
Figure 4-26. Common Defects to Avoid when Fitting and Welding Aircraft Certification Cluster.....	4-55
Table 4-13. Torch Tip Sizes.....	4-55
Figure 4-27. Butt-Welding Strength Calculations	4-56
4-75. Equipment Selection.....	4-53
4-76. Accurately Identify the Type of Material to be Repaired	4-54
4-77. Preparation for Welding	4-54
4-78. Inspection of a Completed Weld.....	4-54
4-79. Microfissures	4-55
4-80. Nondestructive Testing.....	4-55
4-81. Practices to Guard Against	4-55
4-82. Torch Size (Oxyacetylene welding)	4-55
4-83. Welding Rods and Electrodes	4-56
4-84. Rosette Welds.....	4-56
4-85. Heat-Treated Members.....	4-56
4-86. Types of Welding	4-56
Figure 4-28. Basic Gas-Welding Flames: Each has Distinctive Shape, Color and Sound. Neutral Flame is the Most Used.....	4-57
Figure 4-29. Set TIG Welder to DC Current, Straight Polarity for Welding Mild Steel, Stainless Steel and Titanium.....	4-58
Figure 4-30. Set TIG to AC Current for Welding Aluminum and Magnesium	4-58
4-87. Electric-Resistance Welding	4-58
Figure 4-31. In Spot Welding, Heat is Produced by Electrical Resistance Between Copper Electrodes. Pressure is Simultaneously Applied to Electrode Tips to Force Metal Together to Complete Fusing Process. Spot-Weld-Nugget Size is Directly Related to Tip Size	4-58
4-88. Brazing.....	4-59
Table 4-14. Calculated Allowable Strength of Base Metal.....	4-60
Figure 4-32. Silver Brazing Joints	4-60
Figure 4-33. Electric Soldering Iron.....	4-61
4-89. Aircraft Parts Not to be Welded.....	4-62
4-90. Welding Rod Selection.....	4-62
Table 4-15. Chart Showing Welding Filler Rod Selection	4-63
4-91. Repair of Tubular Members	4-62
Figure 4-34. Finger Patch Repairs for Members Dented at a Cluster	4-64
4-92. Repair by Welded Sleeve	4-65
Figure 4-35. Members Dented in a Bay (Repairs by Welded Sleeve)	4-66

CONTENTS (CONTINUED)

Paragraph	Page
4-93. Repair by Bolted Sleeve	4-65
4-94. Welded-Patch Repair.....	4-65
Figure 4-36. Welded Patch Repair	4-67
4-95. Splicing Tubing by Inner-Sleeve Method	4-65
Figure 4-37. Splicing by Inner-Sleeve Method	4-67
4-96. Splicing Tubing by Outer-Sleeve Method	4-65
Figure 4-38. Splicing by Outer-Sleeve Method (Replacement by Welded Outside Sleeve).....	4-69
Figure 4-39. Tube Replacement at a Station by Welded Outer Sleeves	4-69

CONTENTS (CONTINUED)

Paragraph	Page
4-97. Splicing Using Larger Diameter Replacement Tubes.....	4-68
Figure 4-40. Splicing Using Larger Diameter Replacement Tube.....	4-70
4-98. Repairs at Built-In Fuselage Fittings.....	4-71
Figure 4-41. Repairs at Built-In Fuselage Fittings	4-72
4-99. Engine-Mount Repairs.....	4-71
4-100. Built-Up Tubular Wing or Tail-Spars.....	4-73
Figure 4-42. Streamline Tube Splice Using Round Tube (Applicable to Landing Gear)	4-74
4-101. Wing-Brace Struts and Tail-Brace Struts.....	4-73
Figure 4-43. Streamline Tube Splice Using Split Sleeve (Applicable to Wing and Tail Surface Brace Struts and Other Members)	4-75
4-102. Landing Gear Repair	4-73
Figure 4-44. Streamline Tube Splice Using Split Insert (Applicable to Landing Gear).....	4-76
Figure 4-45. Streamline Tube Splice Using Plates (Applicable to Landing Gear).....	4-77
Figure 4-46. Representative Types of Repairable Axle Assemblies.....	4-78
Figure 4-47. Landing Gear Assemblies that Cannot be Repaired by Welding.....	4-79
4-103. Repairs to Welded Assemblies.....	4-74
4-104. Stainless Steel Structure	4-75
4-105.—4-110. [RESERVED.].....	4-79

SECTION 6. WELDING AND BRAZING SAFETY

4-111. General.....	4-81
4-112. Fire and Explosion Safety	4-81
4-113. Welding Work Area	4-81
4-114. Fire Protection	4-81
4-115. Protective Apparel.....	4-82
4-116. First-Aid Kits.....	4-82
4-117.—4-128. [RESERVED.].....	4-82

CHAPTER 5. NONDESTRUCTIVE INSPECTION (NDI)

SECTION 1. GENERAL

5-1. General.....	5-1
5-2. Approved Procedures	5-1
5-3. NDT Levels	5-1
5-4. Training, Qualification, and Certification	5-2
5-5. Flaws.....	5-2
5-6. Selecting the NDI Method.....	5-5
Table 5-1. Advantages and Disadvantages of NDI Methods.....	5-7
5-7. Types of Inspections.....	5-6
5-8.—5-14. [RESERVED.].....	5-7

SECTION 2. VISUAL INSPECTION

5-15. General.....	5-9
5-16. Simple Visual Inspection Aids	5-9
5-17. Borescopes.....	5-10
Figure 5-1. Typical Borescope Designs	5-11

CONTENTS (CONTINUED)

Paragraph	Page
5-18. Visual Inspection Procedures	5-10
Figure 5-2. Using a flashlight to Inspect for Cracks.....	5-13
5-19.—5-24. [RESERVED.].....	5-13

SECTION 3. EDDY CURRENT INSPECTION

5-25. Eddy Current Inspection.....	5-15
Figure 5-3. Generating an Eddy Current	5-15
Figure 5-4. Detecting an Eddy Current.....	5-15
Figure 5-5. Typical Instrument Displays	5-16
5-26. Eddy Current Coils and Probes	5-15
5-27. Field Application of Eddy Current Inspection	5-15
5-28. Surface Inspection	5-16
Figure 5-6. Typical Surface Cracks	5-16
5-29. Subsurface Inspection.....	5-17
Figure 5-7. Typical Subsurface Cracks	5-18
5-30. Corrosion Inspection	5-17
Figure 5-8. Typical Structural Corrosion.....	5-18
5-31. Establishing Eddy Current Inspection Procedures.....	5-17
5-32.—5-39. [RESERVED.].....	5-18

SECTION 4. MAGNETIC PARTICLE INSPECTION

5-40. General.....	5-19
5-41. Principles of Operation.....	5-19
Figure 5-9. Magnetic field Disrupted	5-19
Figure 5-10. Crack Detection by Magnetic Particle Inspection	5-19
5-42. Applications.....	5-20
5-43. Electrical Magnetizing Equipment.....	5-20
5-44. Materials Used in Magnetic Particle Inspection	5-20
Table 5-2. Listing of Commonly Accepted Standards and Specifications for Magnetic Particle Inspection	5-21
5-45. Preparation of Surface	5-22
5-46. Methods of Examination	5-22
5-47. Application of Magnetic Particles.....	5-22
5-48. Magnetization	5-23
Figure 5-11. Circular Magnetization	5-23
Figure 5-12. Longitudinal Magnetization.....	5-24
5-49. Determination of Field Strength.....	5-25
5-50. Special Examination Techniques	5-25
5-51. Demagnetization and Post-Examination Cleaning.....	5-27
5-52.—5-59. [RESERVED.].....	5-28

SECTION 5. PENETRANT INSPECTION

5-60. General.....	5-29
Figure 5-13. Penetrant and Developer Action.....	5-30
5-61. Equipment Used in the Penetrant Inspection Process.....	5-29
Table 5-3. Classification of Penetrant Inspection Materials Covered by MIL-I-25135E	5-30

CONTENTS (CONTINUED)

Paragraph	Page
5-62. Basic Steps to Perform Penetration Inspection	5-29
Table 5-4. Fluorescent and Visible Penetrant Inspection General Processing Procedures Flowsheet	5-31
Table 5-5. Pre-Cleaning Methods for Penetrant Inspection	5-32
5-63. Cleaners and Applicators	5-33
5-64. Technical Standards	5-34
Table 5-6. Listing of Commonly Accepted Standards and Specifications for Penetrant Inspection	5-35
5-65.—5-72. [RESERVED.]	5-35

SECTION 6. RADIOGRAPHY (X-RAY) INSPECTION

5-73. General	5-37
Figure 5-14. Radiography	5-38
5-74. Limitations	5-37
5-75. Film or Paper Radiography	5-37
5-76. Real-Time Radiography	5-37
5-77. Advantage of Real-Time Radiography Over Film Radiography	5-37
5-78. Computed Tomography (CT)	5-37
5-79. Uses of Radiography	5-38
5-80. Comparison with Other NDI Methods	5-38
5-81. Flaws	5-39
5-82. Field Inspection	5-39
5-83. Safety	5-39
5-84.—5-88. [RESERVED.]	5-39

SECTION 7. ULTRASONIC INSPECTION

5-89. General	5-41
Figure 5-15. Ultrasound	5-42
5-90. Sound Reflection	5-41
5-91. Ultrasonic Inspection Techniques	5-41
Figure 5-16. Pulse-Echo and through-Transmission Ultrasonic Inspection Techniques	5-42
5-92. Flaw Detection	5-41
5-93. Basic Equipment	5-41
Figure 5-17. Typical Portable Ultrasonic Inspection Instrument	5-43
Figure 5-18. Example of Position fixture and Shoe	5-44
Figure 5-19. Example of the Use if a Transducer Positioning Fixture	5-44
Figure 5-20. Example of a Typical Reference Standard	5-44
5-94. Inspection of Bonded Structures	5-43
Figure 5-21. Examples of Bonded Structure Configurations and Suggested Inspection Coverage	5-47
Table 5-7. Acceptable Ultrasonic Inspection Methods Associated with the Example Bonded Structure Configurations Shown in Figure 5-21.	5-48
Table 5-8. Ultrasonic Inspection Methods for Bonded Structures	5-49
5-95. Bond Testing Instruments	5-46
5-96. Thickness Measurements	5-50
5-97. Leak Testing	5-50
5-98.—5-104. [RESERVED.]	5-51

CONTENTS (CONTINUED)

Paragraph Page

SECTION 8. TAP TESTING

5-105.	General.....	5-53
	Figure 5-22. Sample of Special Tap Hammer	5-53
5-106.—5-111.	[RESERVED.].....	5-53

SECTION 9. ACOUSTIC-EMISSION

5-112.	General.....	5-53
5-113.	Applications.....	5-54
5-114.—5-119.	[RESERVED.].....	5-54

SECTION 10. THERMOGRAPHY

5-120.	General.....	5-54
5-121.—5-126.	[RESERVED.].....	5-54

SECTION 11. HOLOGRAPHY

5-127.	General.....	5-54
5-128.—5-133.	[RESERVED.].....	5-54

SECTION 12. SHEAROGRAPHY

5-134.	General.....	5-55
5-135.—5-140.	[RESERVED.].....	5-55

CHAPTER 6. CORROSION, INSPECTION & PROTECTION

SECTION 1. GENERAL

6-1.	General.....	6-1
	Figure 6-1. Simplified Corrosion Cell Showing Conditions which Must Exist for Electrochemical Corrosion	6-2
	Figure 6-2. Elimination of Corrosion by Application of an Organic Film to Metal Surface	6-2
6-2.	Factors Influencing Corrosion.....	6-1
6-3.	Common Corrosive Agents	6-3
6-4.	Micro-Organisms.....	6-3
6-5.—6-10.	[RESERVED.].....	6-4

SECTION 2. TYPES OF CORROSION

6-11.	General.....	6-5
	Figure 6-3. Corrosion Attack	6-5
6-12.	General Surface Corrosion	6-5
	Figure 6-4. General Surface Corrosion.....	6-5
6-13.	Pitting Corrosion.....	6-5
	Figure 6-5(a). Pitting Corrosion (External View)	6-5
	Figure 6-5(b). Pitting Corrosion (Magnified Cross Section)	6-5

CONTENTS (CONTINUED)

Paragraph	Page
6-14. Concentration Cell Corrosion.....	6-6
Figure 6-6. Concentration Cell Corrosion	6-6
6-15. Active-Passive Cells	6-6
Figure 6-7. Active-Passive Cell	6-7
6-16. Filiform Corrosion.....	6-7
Figure 6-8. Filiform Corrosion	6-7
6-17. Intergranular Corrosion	6-7
Figure 6-9. Intergranular Corrosion of 7050-T6 Aluminum Adjacent to Steel Fastener.....	6-8
6-18. Exfoliation Corrosion	6-7
Figure 6-10. Exfoliation Corrosion.....	6-8
6-19. Galvanic Corrosion.....	6-8
Figure 6-11. Galvanic Corrosion of Magnesium Adjacent to Steel Fastener	6-9
6-20. Stress Corrosion Cracking.....	6-8
Figure 6-12. Stress Corrosion Cracking	6-9
6-21. Fatigue Corrosion	6-9
6-22. Fretting Corrosion.....	6-10
Figure 6-13. Fretting Corrosion	6-10
6-23.—6-28. [RESERVED.].....	6-10

SECTION 3. CORROSION PROTECTION MEASURES FOR BASIC MATERIALS

6-29. General.....	6-11
6-30. Anodizing and Related Processes	6-11
6-31. Plating	6-11
6-32. Phosphate Rust-Proofing.....	6-11
6-33. Chrome-Pickle Treatment	6-11
6-34. Dichromate Treatment.....	6-11
6-35. Stannate Immersion Treatment.....	6-11
6-36. Galvanic Anodizing Treatment	6-12
6-37. Cladding.....	6-12
6-38. Metal Spraying	6-12
6-39. Shot-Peening.....	6-12
6-40. Organic Coatings	6-12
6-41. Dope Proofing	6-12
6-42. Tube Interiors	6-12
6-43.—6-49. [RESERVED.].....	6-12

SECTION 4. CORROSION PREVENTIVE MAINTENANCE

6-50. Guidelines: All Aircraft.....	6-13
6-51. Guidelines: Aircraft Operating Over Salt Water.....	6-13
6-52.—6-62. [RESERVED.].....	6-14

SECTION 5. VISUAL CORROSION INSPECTION GUIDE FOR AIRCRAFT

6-63. General.....	6-15
6-64. Exhaust Trail Areas	6-15
6-65. Battery Compartments and Battery Vent Openings	6-15
6-66. Lavatories and Galleys	6-15
6-67. Bilge Areas	6-15
6-68. Wheel Wells and Landing Gear	6-15

CONTENTS (CONTINUED)

Paragraph	Page
6-69. External Skin Areas	6-15
6-70. Water Entrapment Areas	6-16
6-71. Engine Frontal Areas	6-16
6-72. Electronic Package Compartments	6-16
6-73. Flexible Hose Assemblies	6-16
6-74. Sandwich Panels	6-16
6-75. Control Cables	6-16
6-76. Integral Fuel Cells	6-16
6-77. Electrical Connectors	6-16
6-78.—6-88. [RESERVED.]	6-16

SECTION 6. CORROSION REMOVAL PROCEDURES

6-89. General	6-17
6-90. Safety Precautions	6-17
6-91. Corrosion Control Work Procedures	6-18
6-92.—6-112. [RESERVED.]	6-19

SECTION 7. BASIC CORROSION REMOVAL TECHNIQUES

6-113. General	6-21
6-114. Preparations for Rework	6-21
6-115. Fairing or Blending Reworked Areas	6-21
Table 6-1. Abrasives for Corrosion Removal	6-22
Figure 6-14. Typical Example of Acceptable Cleanup of Corrosion Pits	6-23
Figure 6-15. Blendout of Corrosion as a Single Depression	6-24
Figure 6-16. Blendout of Multiple Pits in a Corroded Area	6-24
6-116. Corrosion Removal by Blasting	6-22
6-117. Cleaners, Polishes, and Brighteners	6-22
6-118. Standard Methods	6-23
6-119.—6-131. [RESERVED.]	6-24

SECTION 8. ALUMINUM AND ALUMINUM ALLOYS

6-132. General	6-25
6-133. Special Treatment of Anodized Surfaces	6-25
6-134. Repair of Aluminum Alloy Sheet Metal	6-25
6-135. Corrosion Removal Around Countersunk Fasteners in Aluminum Alloy	6-26
6-136. Examples of Removing Corrosion From Aluminum and Aluminum Alloys	6-26
6-137.—6-147. [RESERVED.]	6-28

SECTION 9. MAGNESIUM AND MAGNESIUM ALLOYS

6-148. General	6-29
6-149. Treatment of Wrought Magnesium Sheets and Forgings	6-29
6-150. Repair of Magnesium Sheet Metal After Extensive Corrosion Removal	6-29
6-151. In-Place Treatment of Magnesium Castings	6-30
6-152. Example of Removing Corrosion from Magnesium	6-30
6-153.—6-163. [RESERVED.]	6-33

CONTENTS (CONTINUED)

Paragraph

Page

SECTION 10. FERROUS METALS

6-164. General.....	6-35
6-165. Special Treatment of High Strength Steel	6-35
6-166. Special Treatment of Stainless Steel	6-35
6-167. Example of Removing Corrosion from Ferrous Metals.....	6-36
6-168.—6-178. [RESERVED.].....	6-36

SECTION 11. OTHER METALS AND ALLOYS

6-179. Noble Metal Coatings - Cleanup and Restoration.....	6-37
6-180. Copper and Copper Alloys.....	6-37
6-181. Titanium and Titanium Alloys	6-37
6-182.—6-192. [RESERVED.].....	6-38

SECTION 12. PLATED PARTS

6-193. Chromium and Nickel Plated Parts	6-39
6-194. Cadmium and Zinc Plated Parts	6-39
6-195.—6-205. [RESERVED.].....	6-39

SECTION 13. CORROSION PROOFING OF LAND PLANES CONVERTED TO SEA PLANES

6-206. General.....	6-41
6-207. Necessary Minimum Precautions.....	6-41
6-208. Recommended Precautions	6-41
6-209.—6-219. [RESERVED.].....	6-42

SECTION 14. HANDLING AND CARE OF AIRCRAFT RECOVERED FROM WATER IMMERSION

6-220. General.....	6-43
6-221. Initial Fresh Water or Detergent Wash	6-43
6-222. Reciprocating Engines and Propellers.....	6-43
6-223. Airframe.....	6-43
6-224.—6-234. [RESERVED.].....	6-44

CHAPTER 7. AIRCRAFT HARDWARE, CONTROL CABLES, AND TURNBUCKLES

SECTION 1. RIVETS

7-1. General.....	7-1
7-2. Material Applications.....	7-1
7-3.—7-13. [RESERVED.].....	7-2

SECTION 2. SCREWS

7-14. General.....	7-3
7-15. Structural Screws.....	7-3
7-16. Machine Screws.....	7-3

CONTENTS (CONTINUED)

Paragraph	Page
7-17. Panhead Screws (NAS600 through NAS606, NAS610 through NAS616, NAS623, and NAS1402 through NAS1406).....	7-3
7-18. Self-Tapping Screws.....	7-3
7-19. Wood Screws.....	7-4
7-20.—7-33. [RESERVED.].....	7-4

SECTION 3. BOLTS

7-34. General.....	7-5
7-35. Bolts.....	7-5
7-36. Identification.....	7-5
Figure 7-1. Typical Aircraft Bolt Markings	7-6
7-37. Grip Length.....	7-5
7-38. Locking or Safelying of Bolts	7-6
7-39. Bolt Fit	7-6
7-40. Torques	7-6
Figure 7-2. Torque Wrench with Various Adapters.....	7-8
Table 7-1. Recommended Torque Values (Inch-Pounds).....	7-9
7-41. Standard Aircraft Hex Head Bolts (AN3 through AN20)	7-7
7-42. Drilled Head Bolts (AN73 through AN81).....	7-7
7-43. Engine Bolts	7-7
7-44. Close-Tolerance Bolts	7-9
7-45. Internal Wrenching Bolts (NAS144 through NAS158 and NAS172 through NAS176).....	7-10
7-46. Internal Wrenching Bolts (MS20004 through MS20024) and Six Hole, Drilled Socket Head Bolts (AN148551 through AN149350)	7-10
7-47. Twelve Point, External Wrenching Bolts, (NAS624 through NAS644)	7-10
7-48. Close-Tolerance Shear Bolts (NAS434).....	7-10
7-49. NAS6200 Series Bolts.....	7-10
7-50. Clevis Bolts (AN21 through AN36).....	7-10
7-51. Eyebolts (AN42 through AN49)	7-10
7-52.—7-62. [RESERVED.].....	7-10

SECTION 4. NUTS

7-63. General.....	7-11
7-64. Self-Locking Nuts.....	7-11
Table 7-2. Minimum Prevailing Torque Values for Reused Self-Locking Nuts	7-12
7-65. Nut Identification - Finishes	7-11
7-66. Castle Nut (AN310).....	7-12
7-67. Castellated Shear Nut (AN320).....	7-13
7-68. Plain Nut (AN315 and AN335).....	7-13
7-69. Light Hex Nuts (AN340 and AN345).....	7-13
7-70. Checknut (AN316)	7-13
7-71. Wingnuts (AN350).....	7-13
7-72. Sheet Spring Nuts (AN365)	7-13
7-73.—7-84. [RESERVED.].....	7-13

CONTENTS (CONTINUED)

Paragraph	Page
SECTION 5. WASHERS	
7-85. General.....	7-15
7-86. Plain Washers (AN960 and AN970).....	7-15
7-87. Lockwashers (AN935 and AN936).....	7-15
7-88. Ball Socket and Seat Washers (AN950 and AN955)	7-15
7-89. Taper Pin Washers (AN975).....	7-15
7-90.—7-100. [RESERVED.].....	7-15
SECTION 6. PINS	
7-101 Taper Pins.....	7-17
7-102 Flathead Pins (AN392 through AN406)	7-17
7-103 Cotter Pins (AN380).....	7-17
7-104 Spring Pins.....	7-17
7-105 Quick-Release Pins.....	7-17
7-106.—7-121. [RESERVED.].....	7-17
SECTION 7. SAFETYING	
7-122. General.....	7-19
7-123. Safety Wire	7-19
Figure 7-3. Securing Screws, Nuts, Bolts, and Snaprings.....	7-20
Figure 7-3a. Wire Twisting by Hand.....	7-20
7-124. Safety-Wiring Procedures	7-20
7-125. Twisting With Special Tools.....	7-21
Figure 7-4. Use of a Typical Wire Twister.....	7-22
7-126. Securing Oil Caps, Drain Cocks, and Valves.....	7-21
Figure 7-4a. Securing Oil Caps, Drain Cocks, and Valves	7-22
Figure 7-5. Safety-Wiring Procedures.....	7-23
Figure 7-5a. Safety-Wiring Procedures.....	7-24
Figure 7-5b. Safety-Wiring Procedures.....	7-25
7-127. Securing With Cotter Pins.....	7-26
Figure 7-6. Securing with Cotter Pins.....	7-26
Figure 7-7. Alternate Method for Securing with Cotter Pins	7-26
7-128.—7-139. [RESERVED.].....	7-26
SECTION 8. INSPECTION AND REPAIR OF CONTROL CABLES AND TURNBUCKLES	
7-140. General.....	7-27
7-141. Cable Definitions.....	7-27
7-142. Flexible Cables	7-27
Table 7-3. Flexible Cable Construction and Physical Properties.....	7-28
Figure 7-8. Flexible Cable Cross Section.....	7-29
7-143. Nylon-Coated Cables.....	7-27
7-144. Nonflexible Cables	7-28
Table 7-4. Nonflexible Cable Construction and Physical Properties.....	7-29
Figure 7-9. Nonflexible Cable Cross Section.....	7-29
7-145. Cable Specifications	7-29
7-146. Cable Proof Loads.....	7-29
7-147. Replacement of Cables	7-30

CONTENTS (CONTINUED)

Paragraph	Page
7-148. Mechanically-Fabricated Cable Assemblies	7-30
Table 7-5. Straight-Shank Terminal Dimensions. (Cross Reference AN to MS: AN-666 to MS2159, AN-667 to MS 20667, AN-668 to MS 20668, AN-669 to MS 21260.)	7-31
Figure 7-10. Insertion of Cable Into Terminal.....	7-31
Figure 7-11. Gauging Terminal Shank After Swaging.....	7-31
Figure 7-12. Typical Cable Splices	7-32
Figure 7-13. Typical Terminal Gauge	7-32
Figure 7-14. Typical Thimble-Eye Splice	7-33
Table 7-6. Copper Oval Sleeve Data.....	7-34
Table 7-7. Copper Stop Sleeve Data	7-34
Figure 7-15. Typical Terminal Gauge	7-34
7-149. Cable System Inspection	7-35
Figure 7-16. Cable Inspection Technique.....	7-35
Figure 7-17. Cable Wear Patterns.....	7-36
Figure 7-18. Worn Cable (Replacement Necessary).....	7-36
Figure 7-19. Internal Cable Wear	7-37
Figure 7-20. Pulley Wear Patterns.....	7-38
7-150. Corrosion and Rust Prevention.....	7-38
Figure 7-21. Corrosion.....	7-39
7-151. Wire Splices.....	7-39
Figure 7-22. Manufacturer's Wire Splice	7-39
7-152. Cable Maintenance	7-39
7-153. Cable Tension Adjustment	7-39
7-154.—7-164. [RESERVED.].....	7-39

SECTION 9. TURNBUCKLES

7-165. General.....	7-41
7-166. Turnbuckle Installation.....	7-41
Figure 7-23. Turnbuckle Thread Tolerance	7-41
7-167. Witness Hole.....	7-41
7-168.—7-178. [RESERVED.].....	7-41

SECTION 10. SAFETY METHODS FOR TURNBUCKLES

7-179. General.....	7-43
7-180. Double-Wrap Method.	7-43
Figure 7-24. Safetying Turnbuckles	7-44
7-181. Single-Wrap Method.	7-43
7-182. Safety-Wire Secured Turnbuckles.....	7-44
Table 7-8. Turnbuckle Safetying Guide	7-45
Figure 7-25. Securing Turnbuckles	7-46
7-183. Special Locking Devices.	7-45
Figure 7-26. Clip-Type Locking Device	7-46
7-184. Assembling and Securing Clip-Locking Turnbuckles	7-45
Table 7-9. Locking-Clip Application.....	7-47
Figure 7-27. Assembling and Securing Clip-Locking Turnbuckles.....	7-47
7-185.—7-195. [RESERVED.].....	7-47

CONTENTS (CONTINUED)

Paragraph Page

SECTION 11. HARDWARE IDENTIFICATION TABLES

Table 7-10.	Table of Rivets	7-49
Table 7-11.	Table of Screws.....	7-50
Table 7-12.	Table of Bolts.....	7-57
Table 7-13.	Table of Nuts.....	7-63
Table 7-14.	Table of Washers	7-69
Table 7-15.	Table of Pins	7-71
7-196.—7-206.	[RESERVED.].....	7-74

CHAPTER 8. ENGINES, FUEL, EXHAUST, AND PROPELLERS

SECTION 1. ENGINES

8-1.	General.....	8-1
8-2.	Special Inspection.....	8-1
8-3.	Crankshaft Inspection and Repair Requirements	8-3
8-4.	Replacement Parts in Certificated Engines	8-3
8-5.	Oil System Lines Inspection.....	8-4
8-6.	Oil Filter Inspection.....	8-4
8-7.	Cylinder Hold-Down Nuts and Cap Screws	8-4
8-8.	Reuse of Safetying Devices.....	8-4
8-9.	Self-Locking Nuts for Aircraft Engines and Accessories	8-4
8-10.	Metallizing.....	8-5
8-11.	Plating.....	8-5
	Table 8-1. Current Engine and Maximum Permissible Cylinder Barrel Oversize	8-6
8-12.	Corrosion	8-6
8-13.	Engine Run-In.....	8-6
8-14.	Compression Testing of Aircraft Engine Cylinders	8-6
	Figure 8-1. Schematic of Differential Pressure Compression Tester	8-7
8-15.	Spark Plugs.....	8-8
	Figure 8-2. Chart of Spark Plug Temperature Ranges	8-9
	Figure 8-3. Hot and Cold Spark Plugs.....	8-10
	Figure 8-4. Spark Plug Reach.....	8-10
	Figure 8-5. Method of Checking Spark Plug Gap.....	8-10
8-16.	Operational Problems.....	8-11
	Figure 8-6. Typical Carbon-Fouled Spark Plug	8-12
	Figure 8-6a. Typical Lead-Fouled Spark Plug.....	8-12
	Figure 8-6b. Typical Oil-Fouled Spark Plug	8-13
	Figure 8-6c. Typical Spark Plug with Cracked Core Nose	8-13
	Figure 8-6d. Typical Worn Out Spark Plug	8-14
	Figure 8-6e. Typical Spark Plug with Bridged Electrodes	8-14
	Figure 8-7. Spark Plug Well Flashover	8-14
8-17.	Spark Plug Pre-Reconditioning Inspection	8-14
8-18.	Ignition Harnesses Inspection.....	8-14
	Figure 8-8. Typical Method of Clamping Leads.....	8-15

CONTENTS (CONTINUED)

Paragraph	Page
8-19. Magneto Inspection	8-15
Figure 8-9. Normal Contact Point	8-16
Figure 8-10. Point with Minor Irregularities	8-16
Figure 8-11. Point with Well-Defined Mound	8-16
8-20. Magneto-to-Engine Timing	8-17
8-21.—8-29. [RESERVED.].....	8-17

SECTION 2. FUEL SYSTEMS

8-30. General.....	8-19
8-31. Fuel Lines and Fittings	8-19
Figure 8-12. Location of Clamps at Tube Bends	8-20
8-32. Fuel Tanks and Cells	8-19
8-33. Fuel Tank Caps, Vents, and Overflow Lines	8-20
8-34. Fuel Cross-Feed, Firewall Shutoff, and Tank Selector Valves.....	8-21
8-35. Fuel Pumps	8-21
8-36. Fuel Filters, Strainers, and Drains	8-21
8-37. Indicator Systems.....	8-22
8-38. Fuel System Precautions.....	8-22
8-39.—8-44. [RESERVED.].....	8-22

SECTION 3. EXHAUST SYSTEMS

8-45. General.....	8-23
8-46. Muffler/Heat Exchanger Failures.....	8-23
Figure 8-13. Typical Muffler Wall Fatigue Failure at Exhaust Outlet. (A. Complete Muffler Assembly with Heat Shroud Removed; B. Detail View of Failure.)	8-23
Figure 8-14. Typical Muffler Wall Failure. (A. Complete Muffler Assembly with Heat Shroud Removed; B. Detail View of Failure; C. Cross Section of Failed Muffler.)	8-24
Figure 8-15. Typical Muffler Wall Fatigue Failure. (A. Complete Muffler Assembly with Heat Shroud Partially Removed; B. Detailed View of Failure.)	8-25
Figure 8-16. Typical Fatigue Failure of Muffler End Plate at Stack Inlet	8-25
8-47. Manifold/Stack Failures	8-23
8-48. Internal Muffler Failures	8-23
Figure 8-17. Section of a Muffler Showing Typical Internal Baffle Failure.....	8-25
Figure 8-18. Loose Pieces of a Failed Internal Baffle.....	8-25
Figure 8-19. Failed Internal Baffle Partially Obstructing the Muffler Outlet	8-27
Figure 8-20. Failed Internal Baffle Completely Obstructing the Muffler Outlet	8-27
Figure 8-21a. Example of Exhaust Outlet Guard	8-27
Figure 8-21b. Example of Exhaust Outlet Guard Installed	8-27
8-49. Inspection.....	8-26
Figure 8-22. Effect of Improperly Positioned Exhaust Pipe/Muffler Clamp	8-28
Figure 8-23. Primary Inspection Areas.....	8-28

CONTENTS (CONTINUED)

Paragraph	Page
8-50. Repairs	8-26
8-51. Turbo-Supercharger.....	8-27
8-52. Augmentor Systems.....	8-28
8-53.—8-70. [RESERVED.].....	8-28

SECTION 4. REPAIR OF METAL PROPELLERS

8-71. General.....	8-29
8-72. Steel Blades	8-29
8-73. Aluminum Propeller Repairs	8-29
Figure 8-24. Method of Repairing Surface Cracks, Nicks, etc., on Aluminum-Alloy Propellers	8-30
Figure 8-25. Correct and Incorrect Method of Reworking Leading Edge of Aluminum-Alloy Propellers.....	8-30
Figure 8-26. Method of Repairing Damaged tip of Aluminum-Alloy Propellers.....	8-31
8-74. Repair Limits	8-31
Figure 8-27. Example 1. Determine the Repair Width Limits.	8-32
Figure 8-28. Example 2. Determine the Repair Thickness Limits	8-34
8-75. Steel Hubs and Hub Parts	8-33
8-76. Propeller Hub and Flange Repair	8-33
Figure 8-29. Repair of Fixed-Pitch Hub and Propeller with Elongated or Damaged Bolt Holes	8-35
8-77. Control Systems.....	8-34
8-78. Deicing Systems	8-34
8-79.—8-90. [RESERVED.].....	8-35

SECTION 5. INSPECTION OF PROPELLERS

8-91. General.....	8-37
Table 8-2. Sample Manufacturer's Propeller Inspection Checklist	8-38
8-92. Wood or Composition Propellers and Blades.....	8-37
8-93. Metal Propellers and Blades.....	8-38
8-94. Propeller Hub.....	8-39
8-95. Tachometer Inspection	8-40
8-96.—8-106. [RESERVED.].....	8-40

SECTION 6. PROPELLER TRACKING AND VIBRATION

8-107 General.....	8-41
8-108 Propeller Tracking Check.....	8-41
Figure 8-30. Propeller Tracking (Wood Block or Cowling Fixture Shown)	8-42
8-109. Vibration.....	8-41
8-110.—8-129. [RESERVED.].....	8-42

CONTENTS (CONTINUED)

Paragraph

Page

CHAPTER 9. AIRCRAFT SYSTEMS AND COMPONENTS**SECTION 1. INSPECTION AND MAINTENANCE OF LANDING GEAR**

9-1.	General.....	9-1
9-2.	General Inspection.....	9-1
9-3.	Cleaning and Lubricating	9-1
9-4.	Fixed-Gear Inspection	9-1
	Table 9-1. Bungee Cord Color Codes	9-2
9-5.	Inspection of Retractable Landing Gear.....	9-3
9-6.	Emergency Systems.....	9-3
9-7.	Landing Gear Components.....	9-4
9-8.	Floats and Skis.....	9-5
	Figure 9-1. A Typical Ski Installation	9-6
9-9.	Inspection and Repair of Floats and Skis	9-7
9-10.	Types of Landing Gear Problems.....	9-8
	Figure 9-2. Typical Bolt Cracks	9-8
	Figure 9-3. Typical Cracks Near Bolt Holes	9-8
	Figure 9-4. Typical Bolt Hole Cracks	9-9
	Figure 9-5. Typical Rod-End Cracks.....	9-9
	Figure 9-6. Typical Torque Tube Bolt Hole Elongation.....	9-9
9-11.	Special Inspections	9-9
9-12.	Retraction Tests	9-9
9-13.	Tire and Tube Maintenance.....	9-9
9-14.	Tire Inspection and Repair	9-10
	Figure 9-7. Examples of Tread Wear Indicating Over-Inflation and Under-Inflation.....	9-11
9-15.	Inflation of Tires	9-11
9-16.	Personal Safety	9-11
9-17.	Disassemble the Wheel.....	9-11
9-18.	Reassembling the Wheel	9-12
9-19.	Slippage	9-12
9-20.	Wheel Inspection	9-12
9-21.	Wheel Installation.....	9-12a
9-22.—9-24.	[RESERVED.].....	9-12

SECTION 2. HYDRAULIC SYSTEMS

9-25.	General.....	9-13
9-26.	Purposes of Hydraulic Systems	9-13
9-27.	Types of Hydraulic Fluid.....	9-13
9-28.	Handling Hydraulic Fluids	9-14
9-29.	Hydraulic System Maintenance Practices	9-15
9-30.	Hydraulic Lines and Fittings	9-18
	Table 9-2. Tube Data.....	9-19
	Figure 9-8. Hose Assembly Instructions (Can Be Used for Low Pressure Hydraulic Fluid, and Oil Line Applications).....	9-21

CONTENTS (CONTINUED)

Paragraph	Page
Table 9-3. Aircraft Hose Specifications	9-22
Figure 9-9. Proper Hose Installations	9-23
Figure 9-10. Minimum Bend Radii.....	9-24
Table 9-4. Ball Diameters for Testing Hose Restrictions or Kinking.....	9-25
Figure 9-11. Suggested Handling of Preformed Hose.....	9-25
9-31.—9-36. [RESERVED.].....	9-26

SECTION 3. EMERGENCY EQUIPMENT

9-37. Life Rafts	9-27
9-38. Life Raft Inspections.....	9-27
Figure 9-12. Inflation Valve	9-29
9-39. Survival Kit Inspection.....	9-28
9-40. Special Inspections	9-31
9-41. Inspection Record.....	9-32
9-42. Raft Repairs	9-32
Figure 9-13. Repair Dimensions.....	9-33
9-43. Life Preservers	9-34
9-44. Life Preserver Inspection.....	9-35
9-45. Repair of Life Preservers.....	9-36
9-46. Miscellaneous Equipment	9-36
9-47. Oxygen Systems	9-37
9-48. Inspection.....	9-37
9-49. Maintenance.....	9-38
Figure 9-14. Oxygen Cylinder Damage.....	9-39
Figure 9-15. Cylinder Brackets and Clamps.....	9-39
9-50. Functional Testing After Repair.....	9-41
9-51. Service Requirements --Oxygen Cylinders.....	9-41
Table 9-5. Table of Filling Pressures	9-43
9-52.—9-59. [RESERVED.].....	9-43

SECTION 4. CABIN INTERIOR

9-60. General.....	9-45
9-61. Car-3 Aircraft Interior	9-45
9-62. Part 23 Aircraft Interior.....	9-45
9-63. Source of Information.....	9-46
9-64. Upholstery and/or Belts.....	9-46
9-65.—9-70. [RESERVED.].....	9-46

CHAPTER 10. WEIGHT AND BALANCE

SECTION 1. TERMINOLOGY

10-1. General.....	10-1
10-2. Terminology	10-1
Figure 10-1. Typical Datum Locations.....	10-2
Figure 10-2. Illustration of Arm (or Moment Arm)	10-3
Figure 10-3. Example of Moment Computation	10-4
Figure 10-4. Empty Weight Center of Gravity Formulas.....	10-5
Figure 10-5. Empty Weight and Empty Center of Gravity - Tail-Wheel Type Aircraft.....	10-6

CONTENTS (CONTINUED)

Paragraph	Page
Figure 10-6. Empty Weight and Empty Weight Center of Gravity - Nosewheel Type Aircraft	10-7
Figure 10-7. Operating Center of Gravity Range	10-8
Figure 10-8. Weighing Point Centerline	10-9
10-3.—10-13. [RESERVED.]	10-9

SECTION 2. WEIGHING PROCEDURES

10-14. General	10-11
10-15. Procedures	10-11
Figure 10-9. Empty Weight and Empty Weight Center of Gravity When Aircraft is Weighed With Oil	10-12a
10-15a. Repairs and Alterations	10-11
10-15b. Annual or 100-Hour Inspection	10-11
10-16. Weight and Balance Computations	10-11
10-17. Weight and Balance Extreme Conditions	10-13
Figure 10-10. Example of Check of Most Forward Weight and Balance Extreme	10-14
Figure 10-11. Example of Check of Most Rearward Weight and Balance Extreme	10-16
10-18. Loading Conditions and/or Placards	10-15
Figure 10-12. Loading Conditions: Determination of the Number of Passengers and Baggage Permissible with Full Fuel	10-17
Figure 10-13. Loading Conditions: Determination of the Fuel and Baggage Permissible with Maximum Passengers	10-18
Figure 10-14. Loading Conditions: Determination of the Fuel and the Number and Location of Passengers Permissible with Maximum Baggage	10-19
10-19. Equipment List	10-15
10-20. Equipment Change	10-15
Figure 10-15. Effects of the Addition of Equipment Items on Balance	10-20
Figure 10-16. Example of Moment and Weight Changes Resulting From Equipment Changes	10-21
10-21. Sample Weight and Balance Report	10-20
Figure 10-17. Sample Weight and Balance Report to Determine Empty Weight Center of Gravity	10-22
Figure 10-18. Sample Weight and Balance Report Including an Equipment Change for Aircraft Fully Loaded	10-23
10-22. Installation of Ballast	10-20
Figure 10-19. Permanent Ballast Computation Formula	10-24
10-23. Loading Schedule	10-24
10-24.—10-34. [RESERVED.]	10-24

CHAPTER 11. AIRCRAFT ELECTRICAL SYSTEMS

SECTION 1. INSPECTION AND CARE OF ELECTRICAL SYSTEMS

11-1. General	11-1
11-2. Inspection and Operation Checks	11-1
11-3. Functional Check of Check of Stand-By or Emergency Equipment	11-2
11-4. Cleaning and Preservation	11-2
11-5. Battery Electrolyte Corrosion	11-2
11-6. Adjustment and Repair	11-2
11-7. Insulation of Electrical Equipment	11-2
11-8. Bus Bars	11-3
11-9.—11-14. [RESERVED.]	11-3

CONTENTS (CONTINUED)

Paragraph

Page

SECTION 2. STORAGE BATTERIES

11-15. General.....	11-5
11-16. Battery Charging.....	11-5
11-17. Battery Freezing	11-5
Table 11-1. Lead-Acid Battery Electrolyte Freezing Points	11-5
11-18. Temperature Correction	11-5
Table 11-2. Sulfuric Acid Temperature Correction	11-6
11-19. Battery Maintenance.....	11-6
11-20. Electrolyte Spillage.....	11-8
11-21. Noxious Fumes.....	11-9
11-22. Installation Practices.....	11-9
Figure 11-1. Battery Ventilating Systems.....	11-9
11-23.—11-29. [RESERVED.].....	11-10

SECTION 3. INSPECTION OF EQUIPMENT INSTALLATION

11-30. General.....	11-11
11-31. Installation Clearance Provisions	11-11
11-32. Wires, Wire Bundles, and Circuit Protective Devices.....	11-11
11-33. Alternator Diodes	11-11
11-34. Static Electrical Power Converters.....	11-11
11-35. Acceptable Means of Controlling or Monitoring the Electrical Load	11-12
11-36. Electrical Load Determination	11-12
11-37. Junction Box Construction	11-13
11-38.—11-46. [RESERVED.].....	11-13

SECTION 4. INSPECTION OF CIRCUIT-PROTECTION DEVICES

11-47. General.....	11-15
11-48. Determination of Circuit Breaker Ratings.....	11-15
11-49. DC Circuit Protector Chart.....	11-15
Table 11-3. DC Wire and Circuit Protector Chart	11-15
11-50. Resettable Circuit Protection Devices.....	11-15
11-51. Circuit Breaker Usage	11-15
11-52. Circuit Breaker Maintenance.....	11-16
11-53. Switches	11-16
Table 11-4. Switch Derating Factors.....	11-17
Table 11-5. Selection of Contact Material	11-17
11-54. Relays.....	11-19
11-55. Load Considerations.....	11-20
11-56. Operating Conditions for Switches and Relays.....	11-20
11-57.—11-65. [RESERVED.].....	11-20

SECTION 5. ELECTRICAL WIRE RATING

11-66. General.....	11-21
Table 11-6. Tabulation Chart (Allowable Voltage Drop).....	11-21
Table 11-7. Examples of Determining Required Wire Size Using Figure 11-2.....	11-22
Table 11-8. Examples of Determining Maximum Run Length Using Figure 11-3.....	11-22

CONTENTS (CONTINUED)

Paragraph	Page
Table 11-9. Current Carrying Capacity and Resistance of Copper Wire.....	11-23
Table 11-10. Current Carrying Capacity and Resistance of Aluminum Wire	11-24
11-67. Methods for Determining Current Carrying Capacity of Wires	11-25
11-68. Instructions for Use of Electrical Wire Chart.....	11-25
11-69. Computing Current Carrying Capacity	11-28
Figure 11-2. Conductor Chart, Continuous Flow	11-30
Figure 11-3. Conductor Chart, Intermittent Flow.....	11-31
Figure 11-4a. Single Copper Wire in Free Air.....	11-32
Figure 11-4b. Single Copper Wire in Free Air.....	11-33
Figure 11-5. Bundle Derating Curves.....	11-34
Figure 11-6. Altitude Derating Curve.....	11-34a
11-70.—11-75. [RESERVED.].....	11-34a

SECTION 6. AIRCRAFT ELECTRICAL WIRE SELECTION

11-76. General.....	11-35
11-77. Aircraft Wire Materials	11-35
11-78. Substitutions	11-36
11-79.—11-84. [RESERVED.].....	11-37

SECTION 7. TABLE OF ACCEPTABLE WIRES

11-85. Aircraft Wire Table	11-39
Table 11-11. Open Wiring.....	11-40
Table 11-12. Protected Wiring	11-41
11-86. Open Airframe Interconnecting Wire.....	11-39
11-87. Protected Wire.....	11-39
11-88. Severe Wind and Moisture Problems (SWAMP)	11-39
11-89. Shielded Wire	11-39
11-90.—11-95. [RESERVED.].....	11-41

SECTION 8. WIRING INSTALLATION INSPECTION REQUIREMENTS

11-96. General.....	11-43
11-97. Wiring Replacement.....	11-45
11-98. Terminals and Terminal Blocks	11-46
11-99. Fuses and Fuse Holders	11-47
11-100. Connectors	11-47
11-101. Junction Boxes, Panels, Shields, and Microswitch Housings.....	11-48
11-102. Conduit-Rigid Metallic, Flexible Metallic and Rigid Nonmetallic	11-48
11-103. Junctions	11-48
11-104. Circuit Breakers.....	11-49
11-105. System Separation	11-49
11-106. Electromagnetic Interference (EMI)	11-49
11-107. Interference Tests	11-49
11-108. Identification Stencils and Placards On Electrical Equipment	11-50
11-109.—11-114. [RESERVED.].....	11-50

CONTENTS (CONTINUED)

Paragraph

Page

SECTION 9. ENVIRONMENTAL PROTECTION AND INSPECTION

11-115. Maintenance and Operations	11-51
11-116. Group and Bundle Ties	11-51
Figure 11-7. Group and Bundle Ties	11-51
Figure 11-8. Comb for Straightening Wires in Bundles.....	11-51
11-117. Minimum Wire Bend Radii	11-51
11-118. Slack	11-52
Figure 11-9a. Slack Between Supports	11-52a
11-118a. Drip Loop in Wire Bundle	11-52
Figure 11-9b. Drainage Hole in Low Point of Tubing.....	11-52a
11-119. Power Feeders	11-52
11-120. RF Cable	11-52
11-121. Precautions.....	11-52
11-122. Moisture Protection, Wheel Wells, and Landing Gear Areas.....	11-53
11-123. Protection Against Personnel and Cargo.....	11-53
11-124. Heat Precautions.....	11-53
11-125. Movable Controls Wiring Precautions.....	11-53
11-126. Flammable Fluids and Gases.....	11-53
Figure 11-10. Separation of Wires From Plumbing Lines	11-54
11-127.—11-134. [RESERVED.].....	11-55

SECTION 10. SERVICE LOOP HARNESSSES (PLASTIC TIE STRIPS)

11-135. General.....	11-55
11-136. Support.....	11-55
11-137. Anti-Chafing Material	11-55
11-138. Strain Relief.....	11-55
11-139. Service Loop.....	11-55
11-140.—11-145. [RESERVED.].....	11-56

SECTION 11. CLAMPING

11-146. General.....	11-57
Figure 11-11. Safe Angle for Cable Clamps.....	11-58
Figure 11-12. Typical Mounting Hardware for MS-21919 Cable Clamps.....	11-58
Figure 11-13. Installing Cable Clamp to Structure.....	11-59
Figure 11-14. Clamping at a Bulkhead Hole.....	11-60
11-147. Wire and Cable Clamps Inspection.....	11-57
11-148.—11-154. [RESERVED.].....	11-60

SECTION 12. WIRE INSULATION AND LACING STRING TIE

11-155. General.....	11-61
11-156. Insulation Materials	11-61
11-157. Stripping Insulation	11-61
Table 11-13. Allowable Nicked or Broken Strands	11-62
11-158. Lacing and Ties	11-61
Figure 11-15. Single Cord Lacing	11-63
Figure 11-16. Double Cord Lacing.....	11-63
Figure 11-17. Making Ties	11-64
11-159. Insulation Tape	11-62

CONTENTS (CONTINUED)

Paragraph	Page
11-160.—11-166. [RESERVED.].....	11-64

CONTENTS (CONTINUED)

Paragraph Page

SECTION 13. SPLICING.

11-167	General.....	11-65
	Figure 11-18. Staggered Splices in Wire Bundle	11-65
11-168.—11-173.	[RESERVED.].....	11-65

SECTION 14. TERMINAL REPAIRS

11-174.	General.....	11-67
11-175.	Attachment of Terminals to Studs	11-68
11-176.	Studs and Insulators.....	11-68
11-177.	Wire Terminals and Binding Posts.....	11-69
11-178.	Crimp On Terminal Lugs and Splices.....	11-69
11-179.	Lock Washers for Terminals On Equipment	11-70
11-180.—11-184.	[RESERVED.].....	11-70

SECTION 15. GROUNDING AND BONDING

11-185.	General.....	11-71
11-186.	Grounding.....	11-71
11-187.	Bonding	11-73
11-188.	Bonding Inspection.....	11-73
	Figure 11-19. Millivolt Drop Test	11-74
11-189.	Bonding Jumper Installations	11-75
	Table 11-14. Stud Bonding or Grounding to Flat Surface	11-76
	Table 11-15. Plate Nut Bonding or Grounding to Flat Surface	11-77
	Table 11-16. Bolt and Nut Bonding or Grounding to Flat Surface	11-78
	Figure 11-20. Copper Jumper Connector to Tubular Structure	11-79
	Figure 11-21. Bonding Conduit to Structure.....	11-79
	Figure 11-22. Aluminum Jumper Connection to Tubular Structure	11-79
11-190.	Creepage Distance	11-75
11-191.	Fuel Systems	11-80
11-192.	Electric Shock Prevention Bonding	11-80
11-193.	Lightning Protection Bonding.....	11-80
11-194.	Lightning Protection for Antennas and Air Data Probes	11-81
11-195.	Static-Discharge Devices.....	11-81
11-196.	Cleaning.....	11-81
11-197.	Hardware Assembly	11-81
11-198.—11-204.	[RESERVED.].....	11-81

SECTION 16. WIRE MARKING

11-205.	General.....	11-83
11-206.	Wire Identification.....	11-83
11-207.	Identification and Information Related to the Wire and Wiring Diagrams	11-83
11-208.	Placement of Identification Markings	11-83
	Figure 11-23. Spacing of Printed Identification Marks (Direct Marking)	11-84
	Figure 11-24. Spacing of Printed Identification Marks (Indirect Marking).....	11-84
11-209.	Types of Wire Markings.....	11-83
11-210.	Hot Stamp Marking	11-84

CONTENTS (CONTINUED)

Paragraph	Page
11-211. Dot Matrix Marking	11-85
11-212. Ink Jet Marking.....	11-85
11-213. Laser Marking.....	11-85
11-214. Identification Sleeves	11-85
11-215. Identification Tape.....	11-86
11-216. Operating Conditions.....	11-86
11-217. Installation of Printed Sleeves	11-86
Table 11-17. Recommended Size of Identification Sleeving.....	11-86
11-218. Identification of Wire Bundles and Harnesses.....	11-86
Figure 11-25. Identification of Wire Bundles and Harnesses	11-87
11-219. Terminal Marking Sleeve and Tags	11-87
Figure 11-26. Standard Sleeves (135 °C)	11-87
Figure 11-27. Installation of Heat-Shrinkable Insulation Sleeves	11-87
11-220. Sleeves and Cable Markers Selection	11-87
Figure 11-28. Cable Markers.....	11-88
Figure 11-29. Tie-Down Strap Installation.....	11-88
Table 11-18. Selection Table for Standard Sleeves	11-88
Table 11-19. Selection Table for Thin-Wall Sleeves.....	11-89
Table 11-20. Selection Table for High-Temperature Sleeves.....	11-89
Table 11-21. Selection Table for Cable Markers.....	11-89
Table 11-22. Plastic Tie-Down Straps (MS3367, Type I, Class 1)	11-89
Figure 11-30. Tie-Down Strap Installation Tool.....	11-90
Figure 11-31. Completed Installation	11-90
11-221. Temporary Wire and Cable Marking Procedure.....	11-90
Figure 11-32. Temporary Wire Identification Marker	11-90
11-222. Marker Sleeve Installation After Printing	11-90
Figure 11-33. Inserting Wire Into Marker	11-90a
Figure 11-34. Shrinking Marker on Wire	11-90a
11-223.—11-229. [RESERVED.].....	11-90a

SECTION 17. CONNECTORS

11-230. General.....	11-91
11-231. Selection	11-91
11-232. Types of Connectors.....	11-91
Figure 11-35. Connector Information Example.....	11-91b
Figure 11-36. Different Types of Connectors.....	11-93
Figure 11-37. Coax Cable Connectors.....	11-94
11-233. Voltage and Current Rating.....	11-91a
11-234. Spare Contacts (Future Wiring)	11-97
11-235. Installation	11-97
11-236. Feed-Through Bulkhead Wire Protection.....	11-98
11-237. Special Purpose Connector.....	11-98
11-238. Potting Compounds	11-98
11-239. Potting Connectors	11-98
Figure 11-38. Spare Wires for Potting Connector	11-99
11-240. Through Bolts	11-99
11-241.—11-247. [RESERVED.].....	11-99

CONTENTS (CONTINUED)

Paragraph Page

SECTION 18. CONDUITS

11-248. General.....	11-101
11-249. Size of Conduit	11-101
11-250. Conduit Fittings	11-101
11-251. Conduit Installation	11-101
11-252. Rigid Conduit	11-101
Table 11-23. Bend Radii for Rigid Conduit.....	11-101
11-253. Flexible Conduit	11-102
Table 11-24. Minimum Bending Radii for Flexible Aluminum or Brass Conduit.....	11-102
11-254.—11-259. [RESERVED.].....	11-102

SECTION 19. PROTECTION OF UNUSED CONNECTORS

11-260. General.....	11-103
11-261. Quick Reference Chart	11-103
Table 11-25. Contact Cavity Sealing-Quick Reference	11-104
11-262. Unpressurized Area Connectors.....	11-103
Figure 11-39. Stub Wire Installation	11-104
Table 11-26. Sealing Rod Dimensions	11-104
11-263. Pressurized Areas	11-103
Figure 11-40. Sealing Unused Contact Cavities-Unpressurized Areas-(Cut-Away View)	11-104
11-264.—11-270. [RESERVED.].....	11-104

SECTION 20. ELECTRICAL AND ELECTRONIC SYMBOLS

11-271. General.....	11-105
11-272. Symbols	11-105
Table 11-27. Electronic/Electrical Symbols	11-105
11-273.—11-283. [RESERVED.].....	11-118

CHAPTER 12. AIRCRAFT AVIONICS SYSTEMS

SECTION 1. AVIONICS EQUIPMENT MAINTENANCE

12-1. General.....	12-1
12-2. Handling of Components	12-1
12-3.—12-7. [RESERVED.].....	12-1

SECTION 2. GROUND OPERATIONAL CHECKS FOR AVIONICS EQUIPMENT (ELECTRICAL)

12-8. General.....	12-3
12-9. Inspection of Avionics System	12-3
12-10. Communication Systems	12-4

CONTENTS (CONTINUED)

Paragraph	Page
12-11. VHF Omni-Directional Range (VOR)	12-5
12-12. Distance Measuring Equipment (DME).....	12-5
12-13. Automatic Direction Finder (ADF).....	12-5
12-14. Instrument Landing Systems (ILS).....	12-6
12-15. Marker Beacon	12-6
12-16. Long Range Navigation (LORAN).....	12-6
12-17. Global Positioning System (GPS).....	12-7
12-18. Autopilot Systems.....	12-7
12-19. Altimeters.....	12-7
12-20. Transponders	12-8
12-21. Emergency Locator Transmitters (ELT).....	12-8
12-22. Inspection of ELT.....	12-8
12-23. Flight Data Recorder	12-9
12-24. Cockpit Voice Recorders (CVR)	12-10
12-25. Weather Radar.....	12-10
12-26. Radome Inspection.	12-11
12-27. Data Bus	12-11
12-28.—12-36. [RESERVED.].....	12-11

SECTION 3. GROUND OPERATIONAL CHECKS FOR AVIONICS EQUIPMENT (NON ELECTRICAL)

12-37. Compass Check Swing	12-13
12-38. Pneumatic Gyros.....	12-15
Figure 12-1. Venturi System For Providing Airflow Through Gyro Instruments	12-15
Figure 12-2. Instrument Vacuum System Using a Wet-Type Vacuum Pump	12-17
Figure 12-3. Instrument Vacuum system Using a Dry-Type Air Pump.....	12-18
Figure 12-4. Instrument Pressure System Using a Dry-Type Air Pump	12-18
12-39.—12-50. [RESERVED.].....	12-18

SECTION 4. PITOT/STATIC SYSTEM

12-51. General.....	12-19
12-52. System Components	12-19
Figure 12-5. Pitot/Static system for a small aircraft.....	12-19
12-53. Pitot/Static Tubes and Lines.....	12-19
12-54. Static Ports.....	12-19
12-55. Heater Elements.....	12-19
Figure 12-6. Pitot/Static Tube Head.....	12-20
12-56. System Inspection.....	12-19
12-57. System Leak Test.....	12-20
12-58. Static System Test.....	12-21
12-59. Test Pitot System	12-21
12-60. Maintenance Precautions.....	12-21
12-61. Replacing Lines	12-22
12-62. Relocation of Pitot Tube.....	12-22
12-63. Troubleshooting the Pitot/Static Pressure System	12-22
Table 12-1. color codes for pitot-static systems.....	12-23
12-64.—12-69. [RESERVED.].....	12-23

CONTENTS (CONTINUED)

Paragraph Page

SECTION 5. AVIONICS TEST EQUIPMENT

12-70. General.....	12-25
12-71. Test Equipment Calibration Standards.....	12-25
12-72. Test Equipment Calibration.....	12-25
12-73.—12-83. [RESERVED.].....	12-25

CHAPTER 13. HUMAN FACTORS

13-1. Human Factors Influence on Mechanic's Performance.....	13-1
13-2 The FAA Aviation Safety Program.....	13-1
Figure 13-1. Personal Minimum's Checklist	13-1

APPENDIX 1. GLOSSARY. (10 PAGES)

APPENDIX 2. ACRONYMS AND ABBREVIATIONS (4 PAGES)

APPENDIX 3.METRIC-BASED PREFIXES AND POWERS OF 10 (1 PAGE)

CHAPTER 1. WOOD STRUCTURE

SECTION 1. MATERIALS AND PRACTICES

1-1. GENERAL. Wood aircraft construction dates back to the early days of certificated aircraft. Today only a limited number of wood aircraft structures are produced. However, many of the older airframes remain in service. With proper care, airframes from the 1930's through the 1950's have held up remarkably well considering the state of technology and long term experience available at that time. It is the responsibility of the mechanic to carefully inspect such structures for deterioration and continuing airworthiness.

1-2. WOODS.

a. Quality of Wood. All wood and plywood used in the repair of aircraft structures should be of aircraft quality (reference Army Navy Commerce Department Bulletin ANC-19, Wood Aircraft Inspection and Fabrication). Table 1-1 lists some permissible variations in characteristics and properties of aircraft wood. However, selection and approval of woodstock for aircraft structural use are specialized skills and should be done by personnel who are thoroughly familiar with inspection criteria and methods.

b. Substitution of Original Wood. The wood species used to repair a part should be the same as that of the original whenever possible; however, some permissible substitutes are given in table 1-1. Obtain approval from the airframe manufacturer or the Federal Aviation Administration (FAA) for the replacement of modified woods or other non-wood products with a substitute material.

c. Effects of Shrinkage. When the moisture content of a wooden part is lowered, the part shrinks. Since the shrinkage is not equal in all directions, the mechanic should consider the effect that the repair may have on the completed structure. The shrinkage is greatest in a tangential direction (across the fibers and parallel to the growth rings), somewhat less in a radial direction (across the fibers and perpendicular to the growth rings), and is negligible in a longitudinal direction (parallel to the fibers). Figure 1-1 illustrates the different grain directions and the effects of shrinkage on the shape of a part. These dimensional changes can have several detrimental effects upon a wood structure, such as loosening of

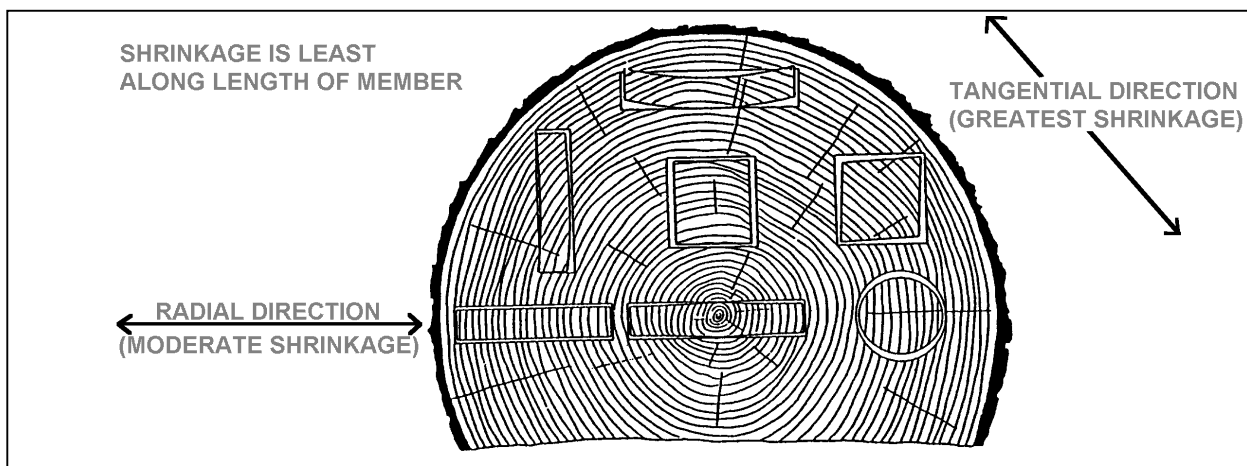


FIGURE 1-1. Relative shrinkage of wood members due to drying.

TABLE 1-1. Selection and Properties of Aircraft Wood. (See notes following table.)

Species of Wood	Strength properties as compared to spruce	Maximum permissible grain deviation (slope of grain)	Remarks
1.	2.	3.	4.
Spruce(Picea) Sitka (P. Sitchensis) Red (P. Rubra) White (P. Glauca).	100%	1:15	Excellent for all uses. Considered as standard for this table.
Douglas Fir (Pseudotsuga Taxifolia).	Exceeds spruce.	1:15	May be used as substitute for spruce in same sizes or in slightly reduced sizes providing reductions are substantiated. Difficult to work with handtools. Some tendency to split and splinter during fabrication and considerable more care in manufacture is necessary. Large solid pieces should be avoided due to inspection difficulties. Gluing satisfactory.
Noble Fir (Abies Nobiles).	Slightly exceeds spruce except 8% deficient in shear.	1:15	Satisfactory characteristics with respect to workability, warping, and splitting. May be used as direct substitute for spruce in same sizes providing shear does not become critical. Hardness somewhat less than spruce. Gluing satisfactory.
Western Hemlock (Tsuga Heterpphylla).	Slightly exceeds spruce.	1:15	Less uniform in texture than spruce. May be used as direct substitute for spruce. Upland growth superior to lowland growth. Gluing satisfactory.
Pine, Northern White (Pinus Strobus).	Properties between 85 % and 96 % those of spruce.	1:15	Excellent working qualities and uniform in properties, but somewhat low in hardness and shock-resisting capacity. Cannot be used as substitute for spruce without increase in sizes to compensate for lesser strength. Gluing satisfactory.
White Cedar, Port Orford (Charaecyparis Lawsoniana).	Exceeds spruce.	1:15	May be used as substitute for spruce in same sizes or in slightly reduced sizes providing reductions are substantiated. Easy to work with handtools. Gluing difficult, but satisfactory joints can be obtained if suitable precautions are taken.
Poplar, Yellow (Liriodendrow Tulipifera).	Slightly less than spruce except in compression (crushing) and shear.	1:15	Excellent working qualities. Should not be used as a direct substitute for spruce without carefully accounting for slightly reduced strength properties. Somewhat low in shock-resisting capacity. Gluing satisfactory.

Notes for Table 1-1**1. Defects Permitted.**

a. Cross grain. Spiral grain, diagonal grain, or a combination of the two is acceptable providing the grain does not diverge from the longitudinal axis of the material more than specified in column 3. A check of all four faces of the board is necessary to determine the amount of divergence. The direction of free-flowing ink will frequently assist in determining grain direction.

b. Wavy, curly, and interlocked grain. Acceptable, if local irregularities do not exceed limitations specified for spiral and diagonal grain.

c. Hard knots. Sound, hard knots up to 3/8 inch in maximum diameter are acceptable providing: (1) they are not projecting portions of I-beams, along the edges of rectangular or beveled unrouted beams, or along the edges of flanges of box beams (except in lowly stressed portions); (2) they do not cause grain divergence **at the edges of** the board or in the flanges of a beam more than specified in column 3; and (3) they are in the center third of the beam and are not closer than 20 inches to another knot or other defect (pertains to 3/8 inch knots—smaller knots may be proportionately closer). Knots greater than 1/4 inch must be used with caution.

d. Pin knot clusters. Small clusters are acceptable providing they produce only a small effect on grain direction.

e. Pitch pockets. Acceptable in center portion of a beam providing they are at least 14 inches apart when they lie in the same growth ring and do not exceed 1-1/2 inches length by 1/8 inch width by 1/8 inch depth, and providing they are not along the projecting portions of I-beams, along the edges of rectangular or beveled unrouted beams, or along the edges of the flanges of box beams.

f. Mineral streaks. Acceptable, providing careful inspection fails to reveal any decay.

TABLE 1-1. Selection and Properties of Aircraft Wood. (See notes following table.) (continued)**2. Defects Not Permitted.**

- a. Cross grain.** Not acceptable, unless within limitations noted in 1a.
- b. Wavy, curly, and interlocked grain.** Not acceptable, unless within limitations noted in 1b.
- c. Hard knots.** Not acceptable, unless within limitations noted in 1c.
- d. Pin knot clusters.** Not acceptable, if they produce large effect on grain direction.
- e. Spike knots.** These are knots running completely through the depth of a beam perpendicular to the annual rings and appear most frequently in quarter-sawn lumber. Reject wood containing this defect.
- f. Pitch pockets.** Not acceptable, unless within limitations noted in 1e.
- g. Mineral streaks.** Not acceptable, if accompanied by decay (see 1f).
- h. Checks, shakes, and splits.** Checks are longitudinal cracks extending, in general, across the annual rings. Shakes are longitudinal cracks usually between two annual rings. Splits are longitudinal cracks induced by artificially induced stress. Reject wood containing these defects.
- i. Compression wood.** This defect is very detrimental to strength and is difficult to recognize readily. It is characterized by high specific gravity, has the appearance of an excessive growth of summer wood, and in most species shows little contrast in color between spring wood and summer wood. In doubtful cases reject the material, or subject samples to toughness machine test to establish the quality of the wood. Reject all material containing compression wood.
- j. Compression failures.** This defect is caused from the wood being overstressed in compression due to natural forces during the growth of the tree, felling trees on rough or irregular ground, or rough handling of logs or lumber. Compression failures are characterized by a buckling of the fibers that appear as streaks on the surface of the piece substantially at right angles to the grain, and vary from pronounced failures to very fine hairlines that require close inspection to detect. Reject wood containing obvious failures. In doubtful cases reject the wood, or make a further inspection in the form of microscopic examination or toughness test, the latter means being the more reliable.
- k. Decay.** Examine all stains and discoloration carefully to determine whether or not they are harmless, or in a stage of preliminary or advanced decay. All pieces must be free from rot, dote, red heart, purple heart, and all other forms of decay.

fittings and wire bracing and checking or splitting of wood members. A few suggestions for minimizing these shrinkage effects are:

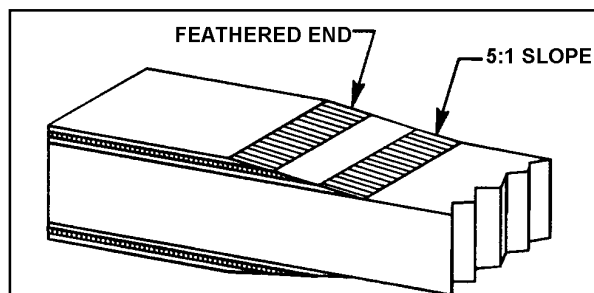
(1) Use bushings that are slightly short so that when the wood member shrinks the bushings do not protrude and the fittings may be tightened firmly against the member.

(2) Gradually drop off plywood faceplates by feathering as shown in figure 1-2.

(3) Thoroughly seal all wood surfaces, particularly end grain and bolt holes, with varnish, epoxy, or other acceptable sealer to slow or prevent moisture changes in the member. (See Section 5. Finishing Wood Structures.)

1-3. MODIFIED WOOD PRODUCTS.

The most common forms of modified woods found in aircraft construction are plywood. Although not a wood product, Phenolic parts are sometimes incorporated into structures. These products are used whenever the manu-

**FIGURE 1-2.** Tapering of faceplate.

facturer requires specialized strength or durability characteristics.

1-4. ADHESIVES. Because of the critical role played by adhesives in aircraft structure, the mechanic must employ only those types of adhesives that meet all of the performance requirements necessary for use in certificated civil aircraft. Use each product strictly in accordance with the aircraft and adhesive manufacturer's instructions.

a. Adhesives acceptable to the FAA can be identified in the following ways:

(1) Refer to the aircraft maintenance or repair manual for specific instructions on acceptable adhesive selection for use on that type aircraft.

(2) Adhesives meeting the requirements of a Military Specification (Mil Spec), Aerospace Material Specification (AMS), or Technical Standard Order (TSO) for wooden aircraft structures are satisfactory providing they are found to be compatible with existing structural materials in the aircraft and the fabrication methods to be used in the repair.

b. Common types of adhesives that are or have been used in aircraft structure fall into two general groups: casein and synthetic-resins. Adhesive technology continues to evolve, and new types (meeting the requirements of paragraph 1-4a) may become available in the future.

(1) Casein adhesive performance is generally considered inferior to other products available today, modern adhesives should be considered first.

CAUTION: Casein adhesive deteriorates over the years after exposure to moisture in the air and temperature variations. Some modern adhesives are incompatible with casein adhesive. If a joint that has previously been bonded with casein is to be rebonded with another type adhesive, all traces of the casein must be scraped off before the new adhesive is applied. If any casein adhesive is left, residual alkalinity may cause the new adhesive to fail to cure properly.

(2) Synthetic-resin adhesives comprise a broad family which includes plastic resin glue, resorcinol, hot-pressed Phenol, and epoxy.

(3) Plastic resin glue (urea-formaldehyde resin glue) has been used in wood aircraft for many years. Caution should be used due to possible rapid deterioration (more rapidly than wood) of plastic resin glue in hot, moist environments and under cyclic swell-shrink stress. For these reasons, urea-formaldehyde should be considered obsolete for all repairs. Any proposed use of this type adhesive should be discussed with the appropriate FAA office prior to using on certificated aircraft.

(4) Federal Specification MMM-A-181D and Military Specification MIL-A-22397 both describe a required series of tests that verify the chemical and mechanical properties of resorcinol. Resorcinol is the only known adhesive recommended and approved for use in wooden aircraft structure and fully meets necessary strength and durability requirements. Resorcinol adhesive (resorcinol-formaldehyde resin) is a two-part synthetic resin adhesive consisting of resin and a hardener. The appropriate amount of hardener (per manufacturer's instruction) is added to the resin, and it is stirred until it is uniformly mixed; the adhesive is now ready for immediate use. Quality of fit and proper clamping pressure are both critical to the achievement of full joint strength. The adhesive bond lines must be very thin and uniform in order to achieve full joint strength.

CAUTION: Read and observe material safety data. Be sure to follow the manufacturer's instructions regarding mixing, open assembly and close assembly times, and usable temperature ranges.

(5) Phenol-formaldehyde adhesive is commonly used in the manufacturing of aircraft grade plywood. This product is cured at elevated temperature and pressure; therefore, it is not practical for use in structural repair.

(6) Epoxy adhesives are a two-part synthetic resin product, and are acceptable providing they meet the requirements of paragraph 1-4a. Many new epoxy resin systems appear to have excellent working properties. They have been found to be much less critical of joint quality and clamping pressure. They penetrate well into wood and plywood. However, joint durability in the presence of elevated temperature or moisture is inadequate in many epoxies. The epoxy adhesives generally consist of a resin and a hardener that are mixed together in the proportions specified by the manufacturer. Depending on the type of epoxy, pot life may vary from a few minutes to an hour. Cure times vary between products.

CAUTION: Some epoxies may have unacceptable thermal or other hidden characteristics not obvious in a shop test. It is essential that only those products meeting the requirements of paragraph 1-4a be used in aircraft repair. Do not vary the resin-to-hardener ratio in an attempt to alter the cure time. Strength, thermal, and chemical resistance will be adversely affected. Read and observe material safety data. Be sure to follow the adhesive manufacturer's instructions regarding mixing, open and closed curing time, and usable temperature ranges.

1-5. BONDING PRECAUTIONS. Satisfactory bond joints in aircraft will develop the full strength of wood under all conditions of stress. To produce this result, the bonding operation must be carefully controlled to obtain a continuous thin and uniform film of solid adhesive in the joint with adequate adhesion and penetration to both surfaces of the wood. Some of the more important conditions involve:

a. Properly prepared wood surfaces.

b. Adhesive of good quality, properly prepared, and properly selected for the task at hand.

c. Good bonding technique, consistent with the adhesive manufacturer's instructions for the specific application.

1-6. PREPARATION OF WOOD SURFACES FOR BONDING. It is recommended that no more time than necessary be permitted to elapse between final surfacing and bonding. Keep prepared surfaces covered with a clean plastic sheet or other material to maintain cleanliness prior to the bonding operation. The mating surfaces should be machined smooth and true with planers, joiners, or special miter saws. Planer marks, chipped or loosened grain, and other surface irregularities are not permitted. Sandpaper must never be used to smooth softwood surfaces that are to be bonded. Sawn surfaces must approach well-planed surfaces in uniformity, smoothness, and freedom from crushed fibers. It is advisable to clean both joint surfaces with a vacuum cleaner just prior to adhesive application. Wood surfaces ready for bonding must be free from oil, wax, varnish, shellac, lacquer, enamel, dope, sealers, paint, dust, dirt, adhesive, crayon marks, and other extraneous materials.

a. Roughening smooth, well-planed surfaces of normal wood before bonding is not recommended. Such treatment of well-planed wood surfaces may result in local irregularities and objectionable rounding of edges. When surfaces cannot be freshly machined before bonding, such as plywood or inaccessible members, very slight sanding of the surface with a fine grit such as 220, greatly improves penetration by the adhesive of aged or polished

surfaces. Sanding should never be continued to the extent that it alters the flatness of the surface. Very light sanding may also improve the wetting of the adhesive to very hard or resinous materials.

b. Wetting tests are useful as a means of detecting the presence of wax, old adhesive, and finish. A drop of water placed on a surface that is difficult to wet and thus difficult to bond will not spread or wet the wood rapidly (in seconds or minutes). The surface may be difficult to wet due to the presence of wax, exposure of the surface to heat and pressure as in the manufacture of hot press bonded plywood, the presence of synthetic resins or wood extractives, or simply chemical or physical changes in the wood surface with time. Good wettability is only an indication that a surface can be bonded satisfactorily. After performing wetting tests, allow adequate time for wood to dry before bonding. Preliminary bonding tests and tests for bond strength are the only positive means of actually determining the bonding characteristics of the adhesive and material combinations. (See paragraph 1-29h.)

1-7. APPLYING THE ADHESIVE. To make a satisfactory bonded joint, spread the adhesive in a thin, even layer on both surfaces to be joined. It is recommended that a clean brush be used and care taken to see that all surfaces are covered. Spreading of adhesive on only one of the two surfaces is not recommended. Be sure to read and follow the adhesive manufacturer's application instructions.

1-8. ASSEMBLY TIME IN BONDING. Resorcinol, epoxy, and other adhesives cure as a result of a chemical reaction. Time is an important consideration in the bonding process. Specific time constraints are as follows:

a. Pot life is the usable life of the adhesive from the time that it is mixed until it must be

spread onto the wood surface. Once pot life has expired, the remaining adhesive must be discarded. Do not add thinning agents to the adhesive to extend the life of the batch.

b. Open assembly time is the period from the moment the adhesive is spread until the parts are clamped together. Where surfaces are coated and exposed freely to the air, some adhesives experience a much more rapid change in consistency than when the parts are laid together as soon as the spreading has been completed.

c. Closed assembly time is the period from the moment that the structure parts are placed together until clamping pressure is applied. The consistency of the adhesive does not change as rapidly when the parts are laid together.

d. Pressing (or clamping) time is the period during which the parts are pressed tightly together and the adhesive cures. The pressing time must be sufficient to ensure that joint strength is adequate before handling or machining the bonded structure.

NOTE: Follow the adhesive manufacturer's instructions for all time limits in the bonding process. If the recommended open or closed assembly periods are exceeded, the bond process should not be continued. Discard the parts if feasible. If the parts cannot be discarded, remove the partially cured adhesive and clean the bond line per adhesive manufacturer's instructions before application of new adhesive.

1-9. BONDING TEMPERATURE. Temperature of the bond line affects the cure rate of the adhesive. Some adhesive types, such as resorcinol, require a minimum temperature which must be maintained throughout the

curing process. Each type of adhesive requires a specific temperature during the cure cycle, and the manufacturer's recommendations should be followed.

1-10. CLAMPING PRESSURE.

a. Use the recommended pressure to squeeze adhesive out into a thin, continuous film between the wood layers. This forces air from the joint and brings the wood surfaces into intimate contact. Pressure should be applied to the joint before the adhesive becomes too thick to flow and is accomplished by means of clamps, presses, or other mechanical devices.

b. Nonuniform clamping pressure commonly results in weak and strong areas in the same joint. The amount of pressure required to produce strong joints in aircraft assembly operations varies with the type of adhesive used and the type of wood to be bonded. Typical pressures when using resorcinol may vary from 125 to 150 pounds per square inch for softwoods and 150 to 200 pounds per square inch for hardwoods. Insufficient pressure or poorly machined wood surfaces usually result in thick bond lines, which indicate a weak joint, and should be carefully guarded against. Some epoxy adhesives require much less clamping pressure to produce acceptable joint strength. Be sure to read and follow the manufacturer's instructions in all cases.

1-11. METHOD OF APPLYING PRESSURE. The methods of applying pressure to joints in aircraft bonding operations range from the use of brads, nails, small screws, and clamps; to the use of hydraulic and electrical power presses. The selection of appropriate clamping means is important to achieving sound bond joints.

a. Hand nailing is used rather extensively in the bonding of ribs and in the application of plywood skins to the wing, control surfaces, and fuselage frames. Small brass screws may also be used advantageously when the particular parts to be bonded are relatively small and do not allow application of pressure by means of clamps. Both nails and screws produce adverse after effects. There is considerable risk of splitting small parts when installing nails or screws. Metal fasteners also provide vulnerable points for moisture to enter during service.

b. On small joints using thin plywood for gussets or where plywood is used as an outer skin, the pressure is usually applied by nailing or stapling. Thin plywood nailing strips are often used to spread the nailing pressure over a larger area and to facilitate removal of the nails after the adhesive has cured.

c. The size of the nails must vary with the size of the members. If multiple rows of nails are required, the nails should be 1 inch apart in rows spaced 1/2 inch apart. The nails in adjacent rows should be staggered. In no case should the nails in adjacent rows be more than 3/4 inch from the nearest nail. The length of the nails should be such that they penetrate the wood below the joint at least 3/8 inch. In the case of small members, the end of the nail should not protrude through the member below the joint. Hit the nails with several light strokes, just seating the head into the surface of the gusset. Be careful not to crush the wood with a heavy hammer blow.

d. In some cases the nails are removed after adhesive cure, while in others the nails are left in place. The nails are employed for clamping pressure during adhesive cure and must not be expected to hold members together in service. In deciding whether to re

move nails after assembly, the mechanic should examine adjacent structure to see whether nails remain from original manufacture.

e. On larger members (spar repairs for example), apply pressure by means of screw clamps, such as a cabinet-maker's bar or "C-clamps." Strips or blocks should be used to distribute clamping pressure and protect members from local crushing due to the

limited pressure area of the clamps, especially when one member is thin (such as plywood). The strip or block should be at least twice as thick as the thinner member being bonded.

f. Immediately after clamping or nailing a member, the mechanic must examine the entire joint to assure uniform part contact and adhesive squeeze-out. Wipe away excess adhesive.

1-12.—1-17. [RESERVED.]

SECTION 2. HEALTH AND SAFETY

1-18. GENERAL. The possibility of an injury is an important consideration when working with wooden aircraft structures. The tools and machines used to shape wooden members can be very dangerous. In addition, there are potential health hazards in working with adhesives and finishes. The mechanic should follow manufacturer's instructions wherever applicable to prevent injury. Federal law mandates that individual chemical manufacturers are to provide Material Safety Data Sheets (MSDS) with health hazard data to all consumers. First aid information and handling precautions must also be identified. Most of the products used in wooden aircraft construction are flammable. Some, such as dope and paint, may be highly flammable.

1-19. SANDING IN AREAS OF EXISTING BOND JOINTS AND FINISHES. Some adhesives used in wooden aircraft construction contain biocides. A commonsense precaution when machining or sanding existing structure is to wear a respirator to avoid inhaling dust products. To lower potential fire hazards avoid using electric sanders around dope, paints, and adhesives.

1-20. HANDLING OF ADHESIVES AND FINISHES. Most adhesives and finish products present at least some toxic potential to users. Injury may occur from skin or eye contact, inhalation, or accidental ingestion. Users should be aware of the manufacturer's instructions and MSDS.

a. Appropriate skin, eye, ear, and respiratory protection should be worn whenever indicated.

b. Shop cleanliness is essential for health and fire safety.

c. Shop personnel should maintain awareness of others in the work area to assure that bystanders are not injured.

d. Proper shop ventilation is essential to disperse fumes emitted from adhesives such as resorcinol and epoxy.

1-21.—1-26. [RESERVED.]

SECTION 3. INSPECTION

1-27. GENERAL. Inspection of wooden structure includes some methods, equipment, and awareness of failure modes which are unique to wooden aircraft.

1-28. TYPES OF DETERIORATION AND DAMAGE.

a. Wood Decay. Wood is an organic product which is subject to attack by fungi. Fungi are plants that grow on and in wood. The moisture content of the wood nominally will have to be 20 percent or greater to sustain fungus growth. The result of this growth is called decay. Decayed wood exhibits softness, swelling if still wet, excessive shrinkage when dry, cracking, and discoloration. Repair or replace wood if any amount or form of decay is found.

b. Splitting. Splits or cracks in wooden members occur along grain lines. When the moisture content of wood is lowered, its dimensions decrease. The dimensional change is greatest in a tangential direction (across the fibers and parallel to the growth rings), somewhat less in a radial direction (across the fibers and perpendicular to the growth rings), and is negligible in a longitudinal direction (parallel to the fibers). These dimensional changes can have detrimental effects upon a wood structure, particularly when two parts are bonded together with grains in different directions. This effect can often be seen where a plywood doubler is bonded to a spruce member. As the spruce member dries, it attempts to shrink, but is restrained by the plywood, which shrinks less. The resulting stress in the spruce member exceeds its cross-grain strength, and a split occurs.

c. Bond Failure. Bond joint failure is generally due to improper fabrication technique or prolonged exposure to moisture in

service. Although none of the older adhesives have been specifically found to fail by simple aging, the mechanic is advised to inspect all accessible joints carefully.

d. Finish Failure. The finish coat on wood structure (usually varnish) is the last line of defense to prevent water entry into wood and the resulting decay. Finish failure can be the result of prolonged water exposure, wood splitting, ultraviolet light exposure, or surface abrasion.

e. Damage. Stress, impact, or mechanical damage to a wood structure is caused by excessive aerodynamic loads or impact loads occurring while the aircraft is on the ground. Overtightening of fittings can also cause crushing of the underlying wood member and possible bending of the metal fitting.

1-29. INSPECTION METHODS. Whenever possible, the aircraft should be kept in a dry, well-ventilated hangar, with all inspection covers, access panels, etc., removed for as long as possible before final inspection. The aircraft should be given a preliminary inspection when first removing the inspection covers and access panels and inspected with a moisture meter at this time. If the moisture content is high, the aircraft should be thoroughly dried. If the aircraft is dry, this will facilitate later inspection, especially when determining the condition of bonded joints.

a. Likely locations for wood structure deterioration should be given special attention. Most damage is caused by external influence such as moisture, temperature extremes, or sunlight. Care should be taken to note all possible entry points for moisture, (i.e., cracks or breaks in the finish, fastener holes, inspection/access openings, control system openings, drain holes, and the interfaces of metal fittings

and the wood structure). The mechanic should also look for evidence of swelling or warpage of the aircraft's wood structure, which would indicate underlying damage or decay. Particular attention should be paid to the wood structure immediately beneath the upper surfaces, especially under areas that are finished in dark colors, for signs of deteriorating adhesives. Cracks in wood spars are often hidden under metal fittings or metal rib flanges and leading edge skins. Any time a reinforcement plate exists that is not feathered out on its ends, a stress riser exists at the ends of the plate. A failure of the primary structure can be expected to occur at this point.

b. Tapping the wood structure with a light plastic hammer or screwdriver handle should produce a sharp solid report. If the suspect area sounds hollow and soft, further inspection is warranted by the following methods.

c. Probe the area in question, if accessible, with a sharp metal tool. The wood structure should be solid and firm. If the suspect area feels soft and mushy the mechanic should assume that the area is rotted. Disassembly of the structure is warranted at this point.

d. Prying the area of a bond joint will reveal any mechanical separation of the joint. If the mechanic detects any relative movement between two adjacent wood members, a failure of the bond is evident. Any loose fittings should arouse the mechanic's suspicion, and the fittings should be removed to check for elongated bolt holes. Disassembly is warranted for further inspection.

e. Odor is an important indicator of possible deterioration. During the initial inspection, as the access panels are being removed from the structure, the mechanic should be aware of any areas that smell musty or moldy.

These odors are indicative of the presence of moisture and associated fungal growth and decay.

f. Visual inspection requires looking at the wood structure both externally and internally for visual signs of decay or physical damage. Any accumulations of dirt, bird nests, or rodent nests are likely places to hold moisture and promote decay.

(1) The mechanic should remove any such accumulations that are found and inspect the area for signs of decay. Decay will appear as a dark discoloration or gray stains running along the grain and often a swelling of the wood member if still wet. Fittings will be imbedded in the wood instead of flush.

(2) Highly suspected structurally damaged areas are shown in figure 1-3. A list of most likely areas to incur structural damage include the following:

(a) Check front and rear spars for compression cracks adjacent to the plywood reinforcing plates, where the lift struts attach, and at the rib attach points on either side of the strut attach points. Triple-check these areas and the spar to fuselage attach points for cracks if the wingtip has contacted the ground, a hangar wall, etc.

(b) Check all metal fittings which attach to wooden structure for looseness, corrosion, cracks, or warps. Areas of particular interest are strut attach fittings, spar butt fittings, aileron and flap hinges, jury strut fittings, compression struts, pulley brackets, and any landing gear fittings.

(c) Check front and rear spars for longitudinal cracks at the ends of the plywood reinforcement plates where the lift struts attach. Triple-check this area if the wing has encountered any kind of ground strike.

(d) Check ribs on either side of strut attach points for missing or loose rib-to-spar attach nails.

(e) Check ribs on either side of strut attach points for cracks where the cap strips pass over and under the spars.

(f) Check for cracked leading edge skin and/or failed nose ribs in the area directly in front of the jury strut.

(g) Check the brackets which attach the struts to the spars for cracks.

(h) Check the aileron, flap hinge, and hinge brackets for cracks and loose or missing rivets.

(i) Check all exposed end grain wood, particularly the spar butts, for cracking or checking. Checking, or splitting, of wood spar butts is common on aircraft based in arid areas.

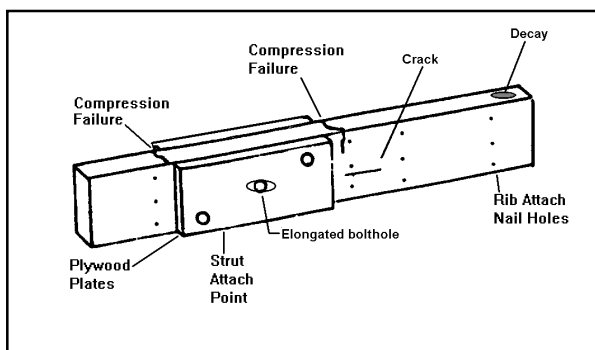


FIGURE 1-3. Likely areas to incur structural damage.

(j) Also check for any cracks that indicate a bond line failure or structural failure of the wood member. Any evidence of movement of fittings, bushings, or fasteners should be cause for concern, and further inspection is warranted. Splits in fabric covering the plywood, especially on upper surfaces exposed to ultraviolet light and water, dictate that the mechanic remove the fabric around the split so the underlying plywood may be inspected for

physical damage or decay. When removing metal fasteners from wood, check for evidence of corrosion. Any corrosion present indicates the presence of moisture and the strong probability of decay in the adjoining wood structure.

(k) Any wooden member that has been overstressed is subject to compression failure (e.g., ground loop). Compression cracking and failure of the wood spars in certain aircraft are a continuing problem. Compression failures are defined as failure of wood fibers on a plane perpendicular to the wood fiber's longitudinal axis. If undetected, compression failures may result in structural failure of the wing during flight. Compression cracks have been found emanating from the upper surfaces of the wing spars and progressing downward.

(l) The usual locations for cracks have been the front spar at both ends of the reinforcement plate for the lift strut and the front spar rib attach points, both inboard and outboard of the spar reinforcement plate; and the rear spar lift strut and rib attach points. An inspection of both the front and rear spars for compression cracks is recommended.

(m) The two areas where it is possible to identify a compression crack are on the face and top surface of the spar. Using a borescope through existing inspection holes is one method of inspection. An alternate method is to cut inspection holes in the skin. If inspection holes are cut, they should be made on the aft side of the front spar and the forward side of the rear spar. This will allow the fabric to be peeled away from the spar. Longitudinal cracks may also be detected during this inspection. Loose or missing rib nails may indicate further damage and should be thoroughly investigated. The mechanic may shine a light, at a low angle and parallel with the grain, in the area of the member

subjected to the compression load. An area of grain waviness would indicate a potential compression failure. In all cases the manufacturer's inspection data should be followed.

g. Moisture Meters are effective tools for detection of excessive moisture content in wood members. An instrument such as this allows the mechanic to insert a probe into the wood member and read its moisture content directly off the meter. A correction chart usually accompanies the instrument to correct for temperature and species of wood. Any reading over 20 percent indicates the probability of fungus growth in the member. Moisture content of the wood should be 8-16 percent, preferably in the 10-12 percent range (this range is during inspection). Where plywood skin covers the spar and the spar would be inaccessible without removing the skin, the moisture meter probe can be inserted through the plywood skin and into the spar to check the moisture content of the spar. The small holes made by the probe are easily sealed.

h. Destructive testing of sample bonded joints whenever a new bond joint is made, a sample joint should be made with the adhesive from the same batch used on the repair and scraps of wood left over from the repair. After curing, the sample joint should be destructively tested to ensure proper bonding of the two wood pieces. Any failure in the bond line indicates a cohesive failure of the adhesive. Any failure along the bond line indicates an adhesive failure, which is indicative of poor bonding. The ideal situation is when wood fibers are observed on both sides of the fracture surface. This indicates a failure in the wood, and indicates the bond joint is actually stronger than the wood.

1-30.—1-35. [RESERVED.]

SECTION 4. REPAIRS

1-36. GENERAL. The basic standard for any aircraft repair is that the repaired structure must be as strong as the original structure and be equivalent to the original in rigidity and aerodynamic shape. Repairs should be made in accordance with manufacturer specifications whenever such data is available.

1-37. REPLACEMENT OF DRAIN HOLES AND SKIN STIFFENERS. Whenever repairs are made that require replacing a portion that includes drain holes, skin stiffeners, or any other items, the repaired portion must be provided with similar drain holes, skin stiffeners, or items of the same dimensions in the same location. Additional drain holes may be required if reinforcement under a skin repair interferes with waterflow to existing drain holes. Make any additional drain holes the same diameter as originals, usually 1/4 inch.

1-38. CONTROL SURFACE FLUTTER PRECAUTIONS. When repairing or refinishing control surfaces, especially on high-performance airplanes, care must be exercised that the repairs do not involve the addition of weight aft of the hinge line. Such a procedure may adversely affect the balance of the surface to a degree that could induce flutter. As a general rule, it will be necessary to repair control surfaces in such a manner that the structure is identical to the original, and that the stiffness, weight distribution, and mass balance are not affected in any way. Consult the aircraft maintenance manual or seek manufacturer's direction for specific requirements on checking control surface balance after repair and refinishing of any control surface.

1-39. SCARF JOINTS. The scarf joint is the most satisfactory method of making an end joint between two solid wood members. Cut

both parts accurately. The strength of the joints depends upon good joint design and a thin, uniform bond line. Make the scarf cut in the general direction of the grain slope as shown in figure 1-4.

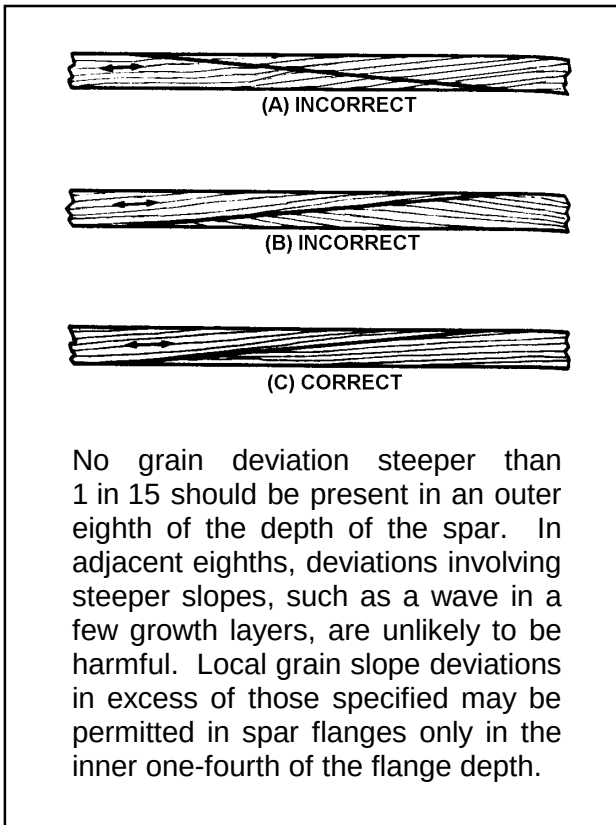


FIGURE 1-4. Consideration of grain direction when making scarf joints.

1-40. SPLICING OF SPARS. Unless otherwise specified by the manufacturer, a spar may be spliced at any point except under the wing attachment fittings, landing gear fittings, engine mount fittings, or lift and interplane strut fittings. These fittings may not overlap any part of the splice. A spar splice repair should not be made adjacent to a previous splice or adjacent to a reinforcing plate. Spacing between two splices or between a splice and a reinforcing plate should be no less than three times the length of the longer splice.

Splicing under minor fittings such as drag wire, antidrag wire, or compression strut fittings is acceptable under the following conditions:

a. The reinforcement plates of the splice should not interfere with the proper attachment or alignment of the fittings. Do not alter the locations of pulley support brackets, bellcrank support brackets, or control surface support brackets. Plates are to be tapered off, as depicted in figure 1-2.

b. The reinforcement plate may overlap drag wire, antidrag wire, or compression strut fittings, if the reinforcement plates are on the rear face of the rear spar or the front face of the front spar. In such cases, it will be necessary to install slightly longer bolts. The front face reinforcement plate should not overlap drag strut fittings, except when it does not require sufficient shortening of compression struts or changes in drag-truss geometry, to prevent adjustment for proper rigging. Even though take up is sufficient, it may be necessary to change the angles on the fittings. (Acceptable methods for splicing the various types of spars are shown in figure 1-4 through figure 1-9.) Reinforcement plates must be used as indicated on all scarf repairs to spars and the slopes of scarves shown are minimum slopes.

1-41. SPAR REPLACEMENT. Replacement of spars is a major repair. Spars may be replaced by new parts made by the manufacturer or the holder of a Parts Manufacturer Approval (PMA) for that part. Owner-produced spars may be installed providing they are made from a manufacturer-approved drawing. Also, a spar may be made by reference to an existing spar providing sufficient evidence is presented to verify that the existing spar is an original part, and that all materials and dimensions can be determined. The dimensions and type of wood used are critical to the structural strength

of the aircraft. Care should be taken that any replacement spars accurately match the manufacturer's original design.

1-42. SPLICING OF BOX SPAR WEBS.

Always splice and reinforce plywood webs with the same type of plywood as found on the original part. Do not use solid wood to replace plywood webs. Plywood is stronger in shear than solid wood of the same thickness due to the grain direction of the individual plies. The face-grain of plywood replacement webs and reinforcement plates must be in the same direction as the original member to ensure that the new web will have the required strength. (The method of splicing plywood webs is shown in figure 1-9.)

1-43. REPLACING SOLID-TYPE SPARS WITH LAMINATED-TYPE SPARS.

Solid spars may be replaced with laminated spars or vice versa, provided the material is of the same high quality. External reinforcements (plywood or solid) must always be replaced as on the original member.

1-44. SPAR LONGITUDINAL CRACKS AND LOCAL DAMAGE.

Cracked spars (except box spars) may be repaired by bonding plates of spruce or plywood of sufficient thickness to develop the longitudinal shear on both sides of the spar. Extend the plates well beyond the termination of the cracks, as shown in figure 1-10. A method of repairing small local damage to either the top or bottom side of a spar is also shown in figure 1-10.

a. Longitudinal Cracking of Wood Wing Spars of Aircraft Operating in Arid Regions.

Aircraft having wood spars and operating in arid regions may develop longitudinal spar cracks in the vicinity of the plywood reinforcement plates. These cracks result from the tendency of the spar to shrink when drying takes place. Plywood resists this tendency to

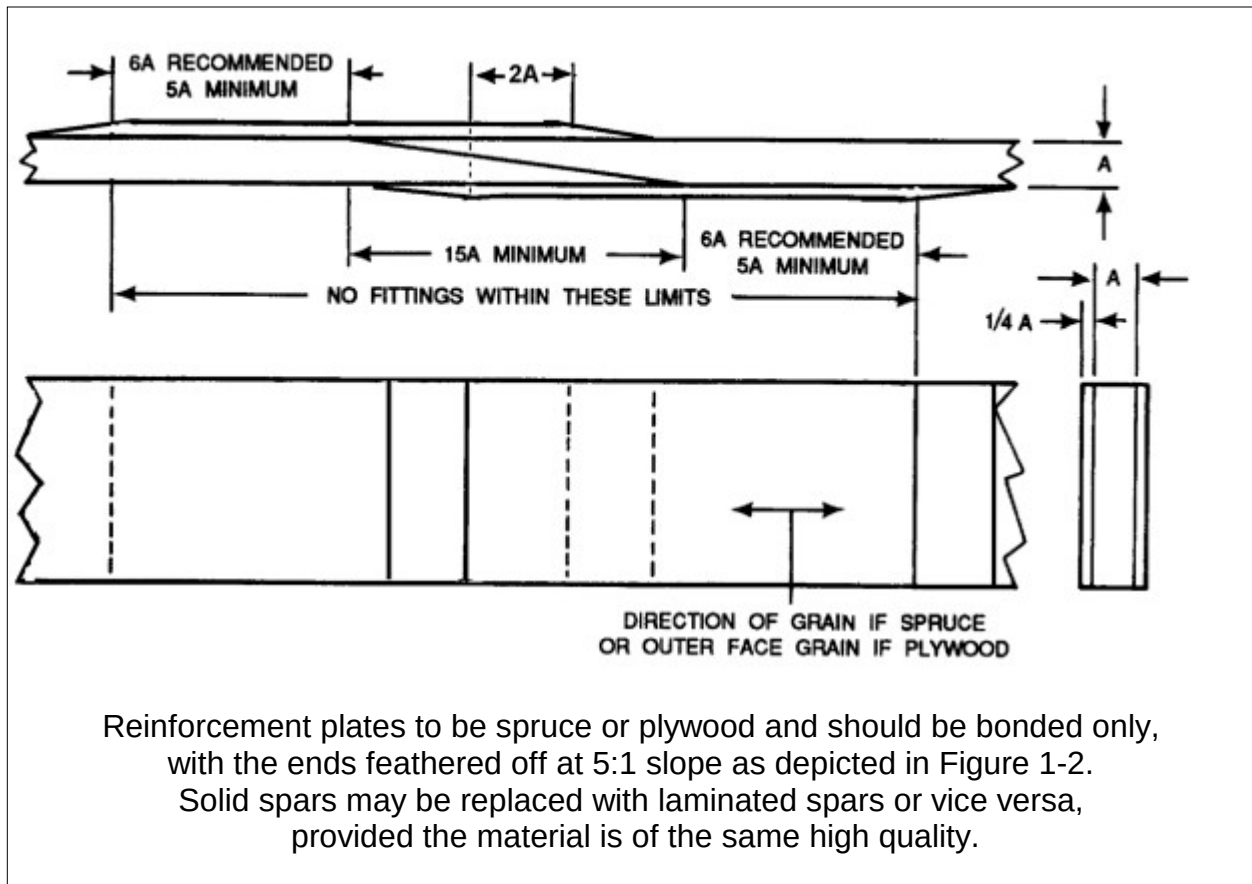


FIGURE 1-5. Method of splicing solid or laminated rectangular spars.

shrink and causes the basic spar stock to split (see paragraph 1-2c). Cracks start under the plywood plates, usually (but not necessarily) at a bolt hole or cutout, and usually spread in each direction until they extend a short distance beyond the ends of the plates where the resistance to spar shrinkage disappears. Cracks have also been found in the butt end of spars. Other factors, which have been found conducive to the formation of cracks are poor protective finishes, large cutouts, and metal fittings that utilize two lines of large diameter bolts.

b. Repairing Cracks Versus Installing a New Spar. The presence of cracks does not necessarily mean that the spar must be discarded. If the crack is not too long or too close to either edge and can be reinforced properly, it will probably be more economical and

satisfactory to perform repair rather than install a new spar or section. However, a generally acceptable procedure suitable for all airplane models is not available. Because of the possibility of strength deficiencies contact the manufacturer. In absence of the manufacturer, the FAA should be contacted for approval before making repairs not in accordance with the manufacturer-approved instructions or the recommendations of this advisory circular. Longitudinal cracking or the recurrence of cracking can be minimized by ensuring that the moisture content of the solid wood portion is within the proper range before bonding. In arid desert areas, during bonding the moisture content should be in the range of 6-8 percent before bonding, but in other areas 10-12 percent is satisfactory. If solid or plywood repair stock is procured from another climatic region, it should be allowed to season,

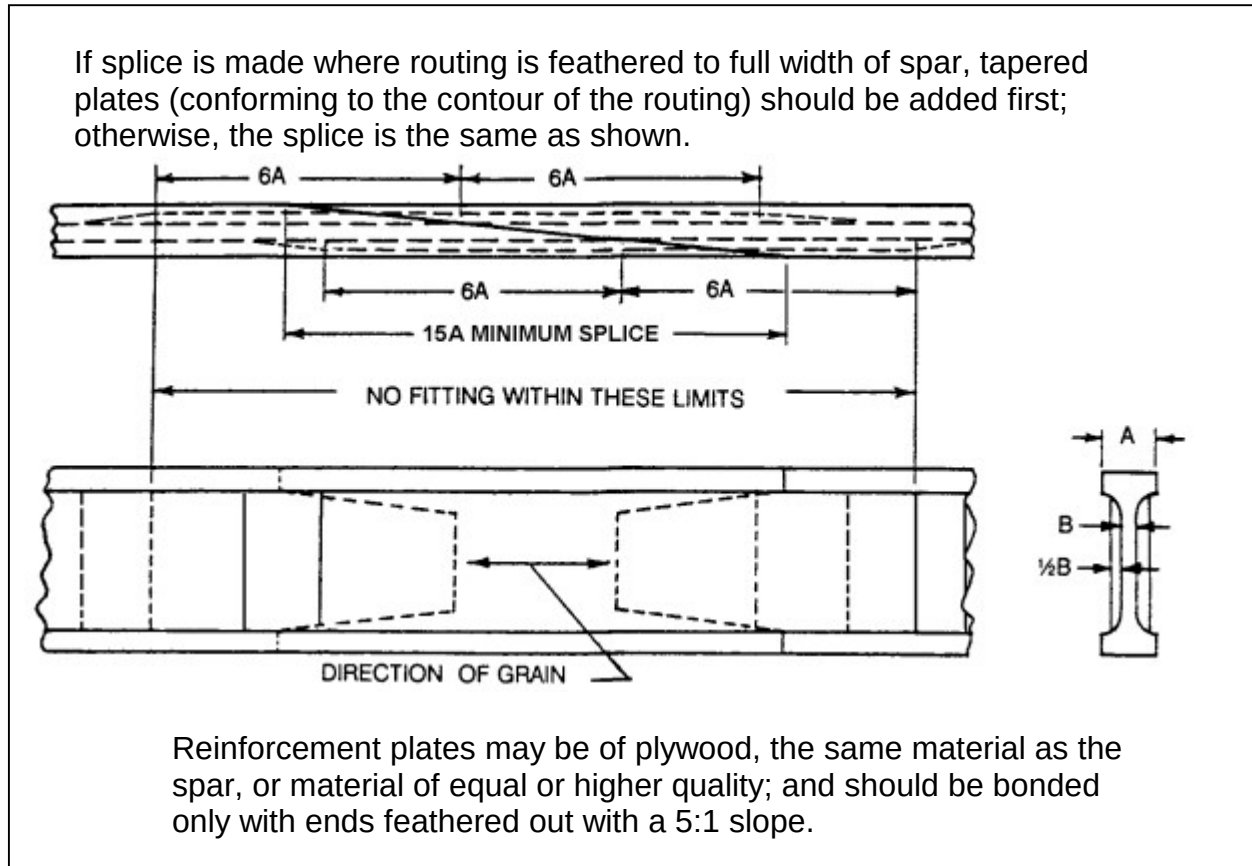


FIGURE 1-6. Method of splicing solid "I" spars.

in the same storage area as the part to be repaired, for no less than 2 weeks.

c. Preventing Cracks. An important step in the prevention of longitudinal cracking, particularly in spar butts, is to ensure that the wood is thoroughly sealed with a penetrating and highly moisture-resistant finish. Application of a thin, slow-curing epoxy adhesive or sealer can be very effective in slowing or preventing moisture changes in spar butts.

1-45. ELONGATED HOLES IN SPARS.

In cases of elongated bolt holes in a spar, or cracks in the vicinity of bolt holes, splice in a new section of spar, or replace the spar entirely. If hole elongation or cracking is minimal and the bolt holes are for noncritical fittings, repair (rather than replacement) may be feasible. Obtain approval for any such repair from the manufacturer or a representative of

the FAA. In many cases, it has been found advantageous to laminate the new section of the spar, particularly if the spar butts are being replaced.

1-46. RIB REPAIRS. Ribs may be replaced by new parts made by the manufacturer or the holder of a PMA for that part. Owner-produced ribs may be installed providing they are made from a manufacturer-approved drawing or by reference to an existing original rib. A rib may be made by reference to an existing rib providing sufficient evidence is presented to verify that the existing rib is an original part and that all materials and dimensions can be determined. The contour of the rib is important to the safe flying qualities of the aircraft, and care should be taken that any replacement ribs accurately match the manufacturer's original design.

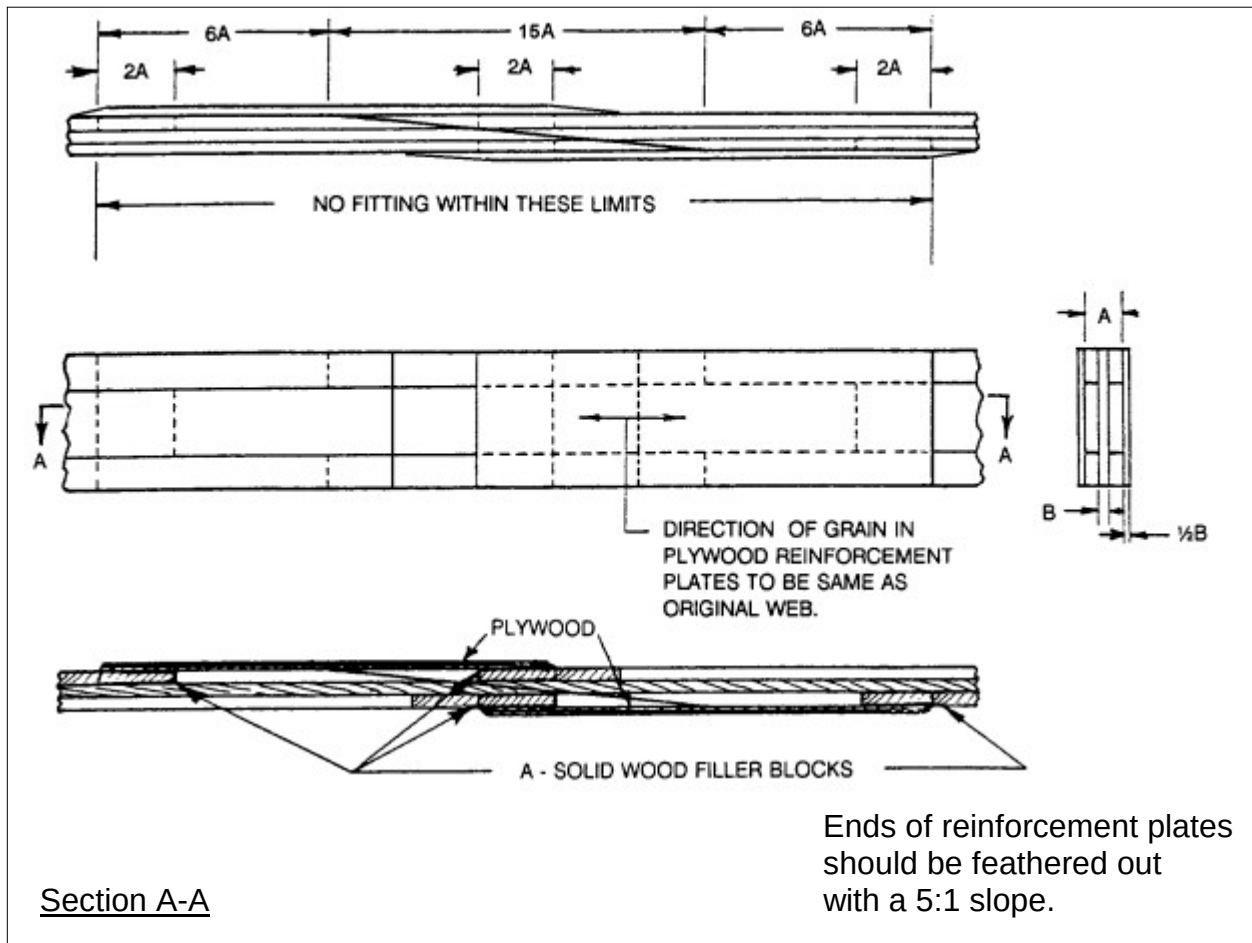


FIGURE 1-7. Repairs to built-up "T" spar.

a. Rib Repair Methods. Acceptable methods of repairing damaged ribs are shown in figure 1-11. Wood ribs should not be nailed to wood spars by driving nails through the rib cap strips, as this weakens the rib materially. The attachment should be by means of adhesive with cement coated, barbed, or spiraled nails driven through the vertical rib members on each face of the spar.

b. Compression Rib Repair. Acceptable methods of repairing damaged compression ribs are shown in figure 1-12.

(1) Figure 1-12(A) illustrates the repair of a compression rib of the "T" section type; i.e., wide, shallow cap strips, and a center plywood web with a rectangular compression member on each side of the web. The rib is

assumed to be cracked through the cap strips, web member, and compression member in the illustration. Cut the compression member as shown in figure 1-12(D). Cut and replace the aft portion of the cap strips, and reinforce as shown in figure 1-11. The plywood side plates are bonded on, as indicated in figure 1-12(A). These plates are added to reinforce the damaged web.

(2) Figure 1-12(B) illustrates a compression rib of the type that is basically a standard rib with rectangular compression members added to one side and plywood web to the other side. The method used in this repair is essentially the same as in figure 1-12(A) except that the plywood reinforcement plate,

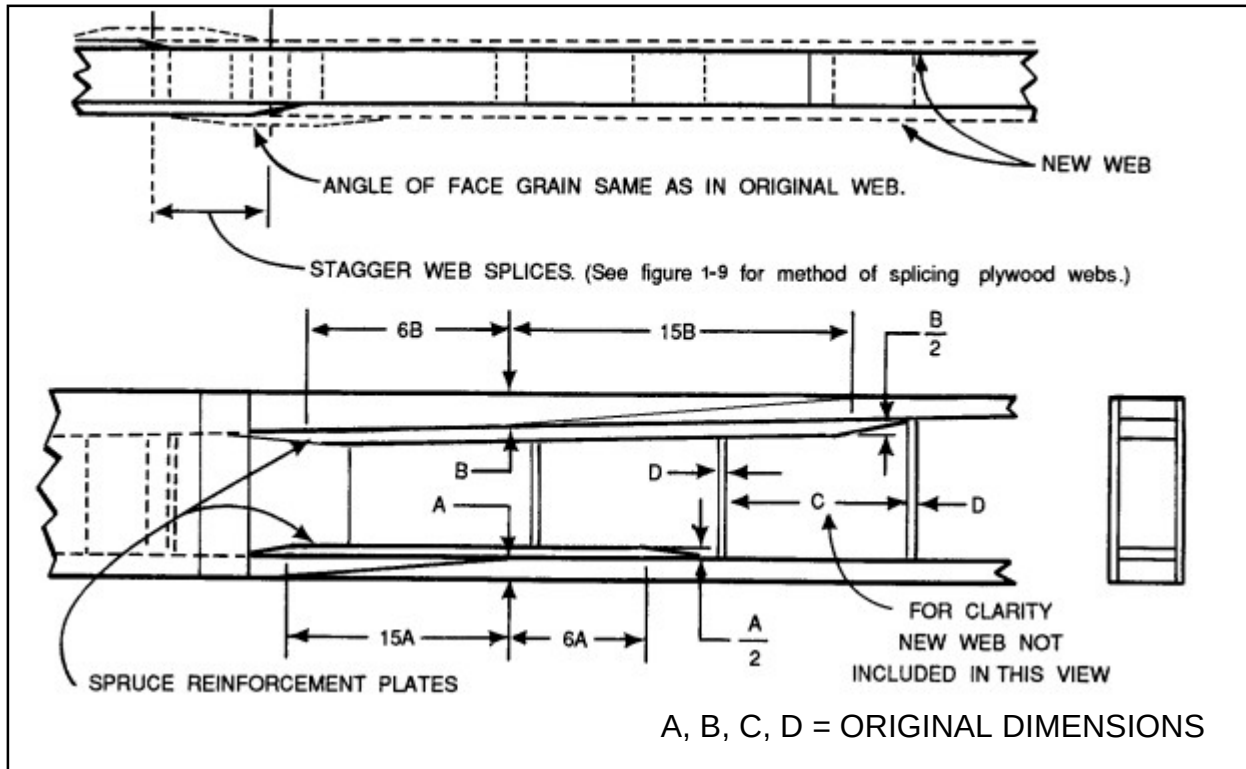


FIGURE 1-8. Method of splicing box spar flanges (plate method).

shown in section B-B, is continued the full distance between spars.

(3) Figure 1-12(C) illustrates a compression rib of the "I" type with a rectangular vertical member on each side of the web. The method of repair is essentially the same as in figure 1-12(A) except the plywood reinforcement plates on each side, shown as striped blocks in section C-C, are continued the full distance between spars.

1-47. PLYWOOD SKIN REPAIR. Make extensive repairs to damaged stressed skin plywood structures in accordance with specific recommendations from the aircraft manufacturer. It is recommended that repairs be made by replacing the entire panel, from one structural member to the next, if damage is very extensive. When damaged plywood skin is repaired, carefully inspect the adjacent internal structure for possible hidden damage. Repair any defective frame members prior to making skin repairs.

1-48. DETERMINATION OF SINGLE OR DOUBLE CURVATURE. Much of the outside surface of plywood aircraft is curved. On such areas, plywood used for repairs to the skin must be similarly curved. Curved skins are either of single curvature or of double (compound) curvature. A simple test to determine which type of curvature exists may be made by laying a sheet of heavy paper on the surface in question. If the sheet can be made to conform to the surface without wrinkling, the surface is either flat or single curvature. If the sheet cannot be made to conform to the surface without wrinkling, the surface is of double curvature.

1-49. REPAIRS TO SINGLE CURVATURE PLYWOOD SKIN. Repairs to single curvature plywood skin may usually be formed from flat plywood, either by bending it dry or after soaking it in hot water. The degree of curvature to which a piece of plywood can be bent will depend upon the direction of the grain and the thickness. Table 1-2 is a guide

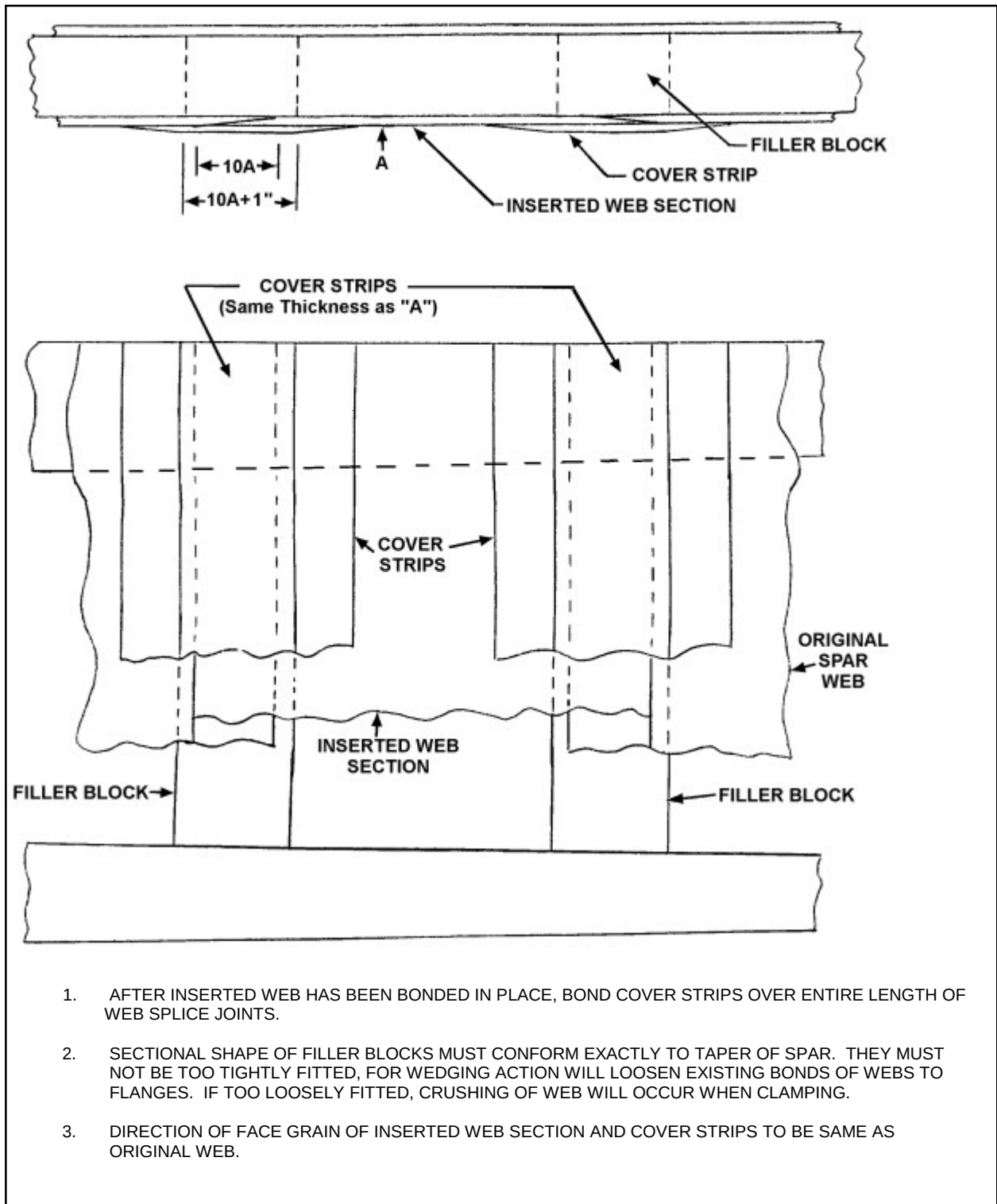


FIGURE 1-9. Method of splicing box spar webs.

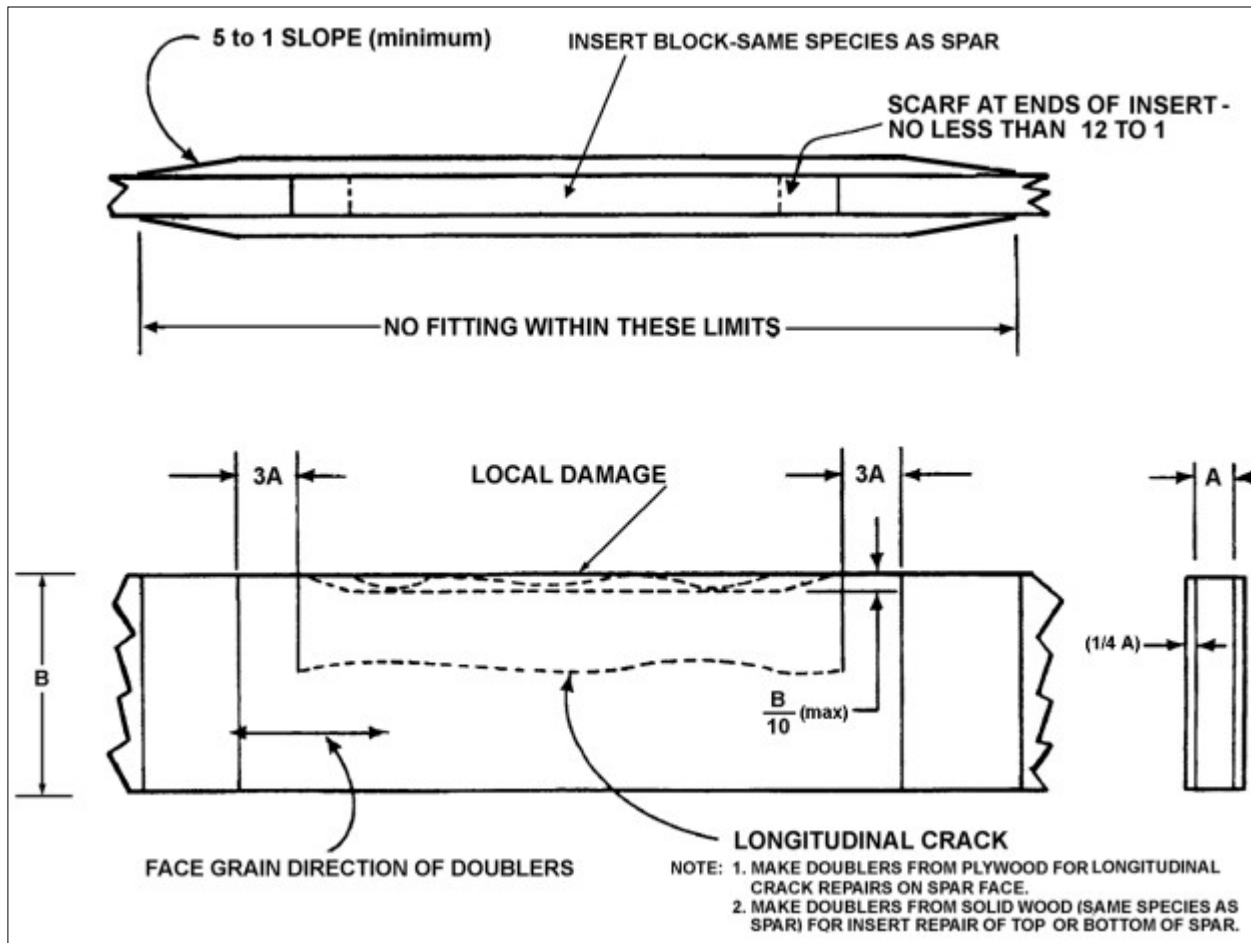


FIGURE 1-10. Method of reinforcing a longitudinal crack and/or local damage in a solid spar.

for determining which process of bending should be used for the curvature being considered.

a. Plywood, after softening, may be bent on a cold ventilated form, or it may be bent over the leading edge near the area being patched if space permits. In either method the repair part should be allowed to dry completely on the form. When bending plywood over a leading edge, drying may be hastened by laying a piece of coarse burlap over the leading edge before using it as a bending form. To speed drying, a fan may be used to circulate air around the repair part.

b. In bending pieces of small radii or to speed up the bending of a large number of parts of the same curvature, it may be necessary to use a heated bending form. The surface

temperature of this form may be as high as 149 °C (300 °F), if necessary, without danger of damage to the plywood. The plywood should be left on the heated form only long enough to dry to room conditions.

1-50. REPAIRS TO DOUBLE CURVATURE PLYWOOD SKIN.

The molded plywood necessary for a repair to a damaged plywood skin of double curvature cannot be made from flat plywood unless the area to be repaired is very small or is of exceedingly slight double curvature; therefore, molded plywood of the proper curvature must be on hand before the repair can be made. If molded plywood of the proper curvature is available, the repair may be made using the same procedure as on single curvature skins.

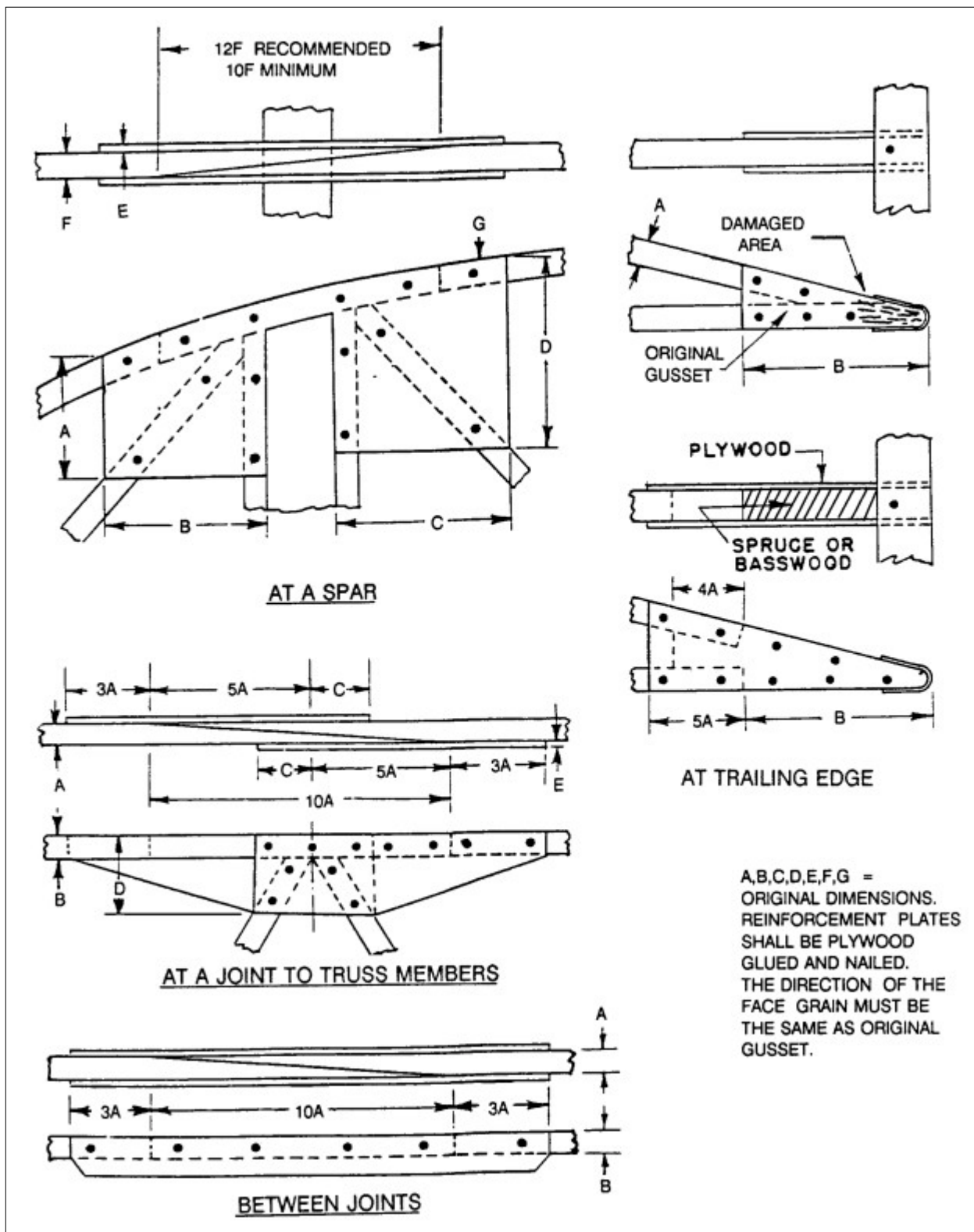


FIGURE 1-11. Repair of wood ribs.

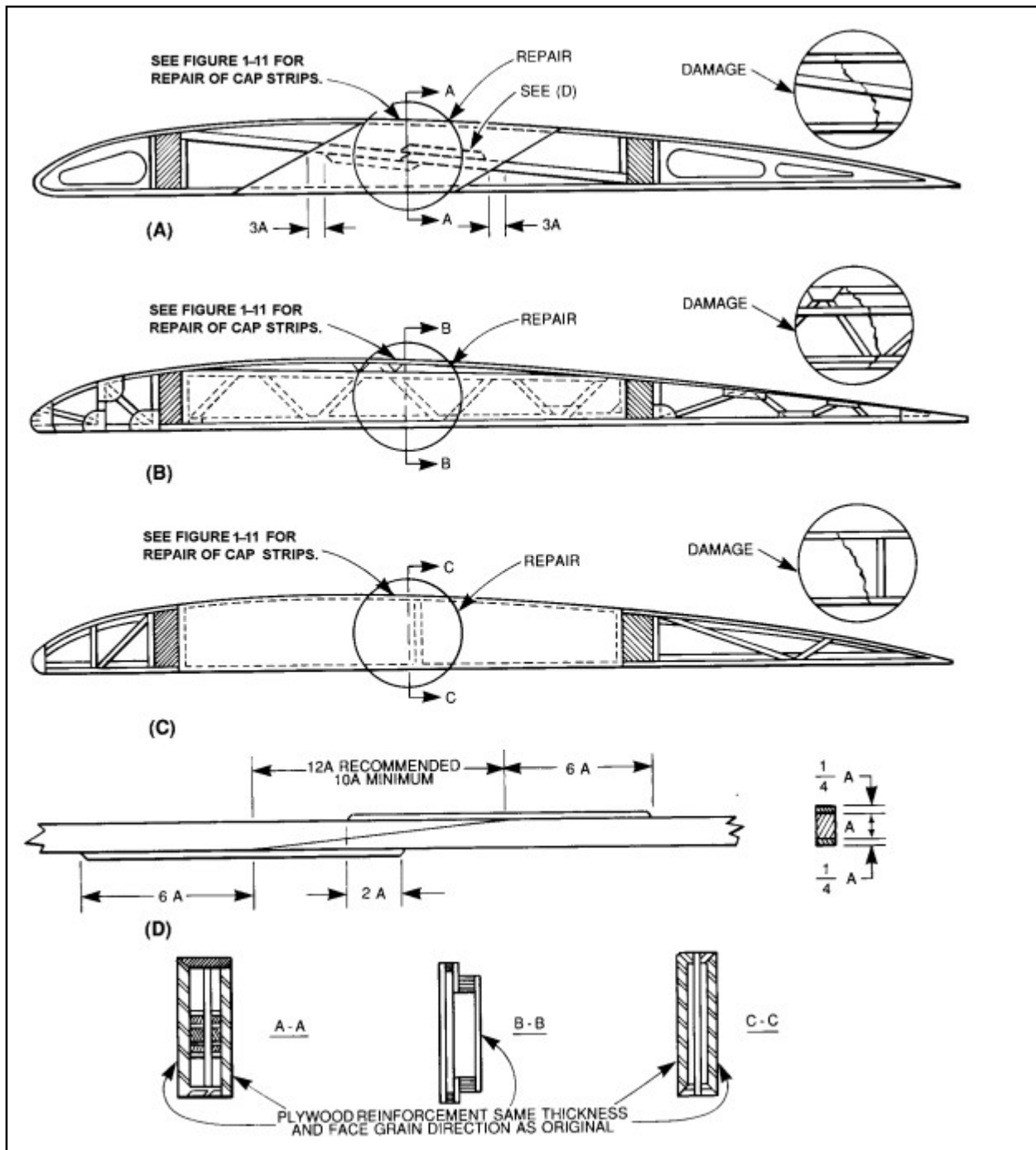
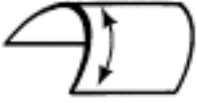
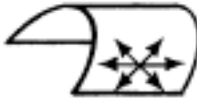
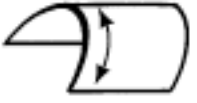
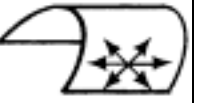


FIGURE 1-12. Typical wing compression rib repairs.

TABLE 1-2. Minimum recommended bend radii for aircraft plywood.

		10 PERCENT MOISTURE CONTENT, BENT ON COLD MANDRELS		THOROUGHLY SOAKED IN HOT WATER AND BENT ON COLD MANDRELS	
PLYWOOD CHARACTERISTICS		AT 90° TO FACE GRAIN	AT 0° OR 45° TO FACE GRAIN	AT 90° TO FACE GRAIN	AT 0° OR 45° TO FACE GRAIN
					
THICKNESS (INCHES)	NUMBER OF PLIES	MINIMUM BEND RADIUS (INCHES)			
.035	3	2.0	1.1	0.5	0.1
.070	3	5.2	3.2	1.5	0.4
.100	3	8.6	5.2	2.6	0.8
.125	3	12	7.1	3.8	1.2
.155	3	16	10	5.3	1.8
.185	3	20	13	7.1	2.6
.160	5	17	11	6	2
.190	5	21	14	7	3
.225	5	27	17	10	4
.250	5	31	20	12	5
.315	5	43	28	16	7
.375	5	54	36	21	10

1-51. TYPES OF PATCHES. There are four types of patches: splayed patch, surface (or overlay) patch, scarf patch, and plug patch. They are all acceptable for repairing plywood skins.

a. Splayed Patch. Small holes with their largest dimensions not over 15 times the skin thickness, in skins not more than 1/10 inch in thickness, may be repaired by using a circular splayed patch as illustrated in figure 1-13. The term “splayed” is used to denote that the edges of the patch are tapered, but the slope is steeper than is allowed in scarfing operations.

(1) Lay out the patch according to figure 1-13. Tack a small piece of plywood over the hole for a center point and draw two circles with a divider, the inner circle to be the size of the hole and the outer circle marking the limits

of the taper. The difference between the radii is 5T (5 times the thickness of the skin). If one leg of the dividers has been sharpened to a chisel edge, the dividers may be used to cut the inner circle.

(2) Taper the hole evenly to the outer mark with a chisel, knife, or rasp.

(3) Prepare a circular tapered patch to fit the prepared hole, and bond the patch into place with face-grain direction matching that of the original surface.

(4) Use waxed paper or plastic wrap, (cut larger than the size of the patch) between the patch and the plywood pressure plate. This prevents excess adhesive from bonding the pressure plate to the skin. Center the pressure plate carefully over the patch.

(5) As there is no reinforcement behind this patch, care must be used so that pressure is not great enough to crack the skin. On horizontal surfaces, weights or sandbags will be sufficient. On patches too far from any edge for the use of standard hand clamps, jaws of greater length may be improvised. Table 1-2, columns (1) and (3), may also be used for determining the maximum thickness of single laminations for curved members.

(6) Fill, sand, and refinish the patch.

b. Surface Patch. Plywood skins that are damaged between or along framing members may be repaired by surface or overlay patches as shown in figure 1-14. Surface patches located entirely aft of the 10 percent chord line, or which wrap around the leading edge and terminate aft of the 10 percent chord line, are permissible. Surface patches may have as much as a 50 inch perimeter and may cover as much as 1 frame (or rib) space. Trim the damaged skin to a rectangular or triangular shape and round the corners. The radius of rounded corners must be at least 5 times the skin thickness. Bevel the forward edges of patches located entirely aft of the 10 percent chord line to 4 times the skin thickness. The face-grain direction must be the same as the original skin. Cover completed surface patches with fabric to match surrounding area. The fabric must overlap the original fabric at least 2 inches.

c. Scarf Patch. A properly prepared and inserted scarf patch is the best repair for damaged plywood skins and is preferred for most skin repairs. Figure 1-15 shows the details and dimensions to be used when installing typical scarf skin patches, when the back of the skin is accessible. Follow figure 1-16 when the back of the skin is not accessible. The scarf slope of 1 in 12, shown in both figures, is the steepest slope permitted for all kinds of plywood. If the radius of curvature of the skin at all points

on the trimmed opening is greater than 100 times the skin thickness, a scarf patch may be installed.

(1) Scarf cuts in plywood may be made by hand plane, spoke shave, scraper, or accurate sandpaper block. Rased surfaces, except at the corners of scarf patches and sawn surfaces, are not recommended as they are likely to be rough or inaccurate.

(2) Nail strip or small screw clamping is often the only method available for bonding scarf joints in plywood skin repairs. It is essential that all scarf joints in plywood be backed with plywood or solid wood to provide adequate nail holding capacity. The face-grain direction of the plywood patch must be the same as that of the original skin.

(3) If the back of a damaged plywood skin is accessible (such as a fuselage skin), it should be repaired with a scarf patch, following the details shown in figure 1-15. Whenever possible, the edges of the patch should be supported as shown in section C-C of figure 1-15. When the damage follows or extends to a framing member, the scarf may be supported as shown in section B-B of figure 1-15. Damages that do not exceed 25 times the skin thickness in diameter after being trimmed to a circular shape and are not less than 15 times the skin thickness to a framing member, may be repaired as shown in figure 1-15, section D-D.

(a) The backing block is carefully shaped from solid wood and fitted to the inside surface of the skin, and is temporarily held in place with nails.

(b) Use waxed paper or plastic wrap to prevent bonding of the backing block to the skin.

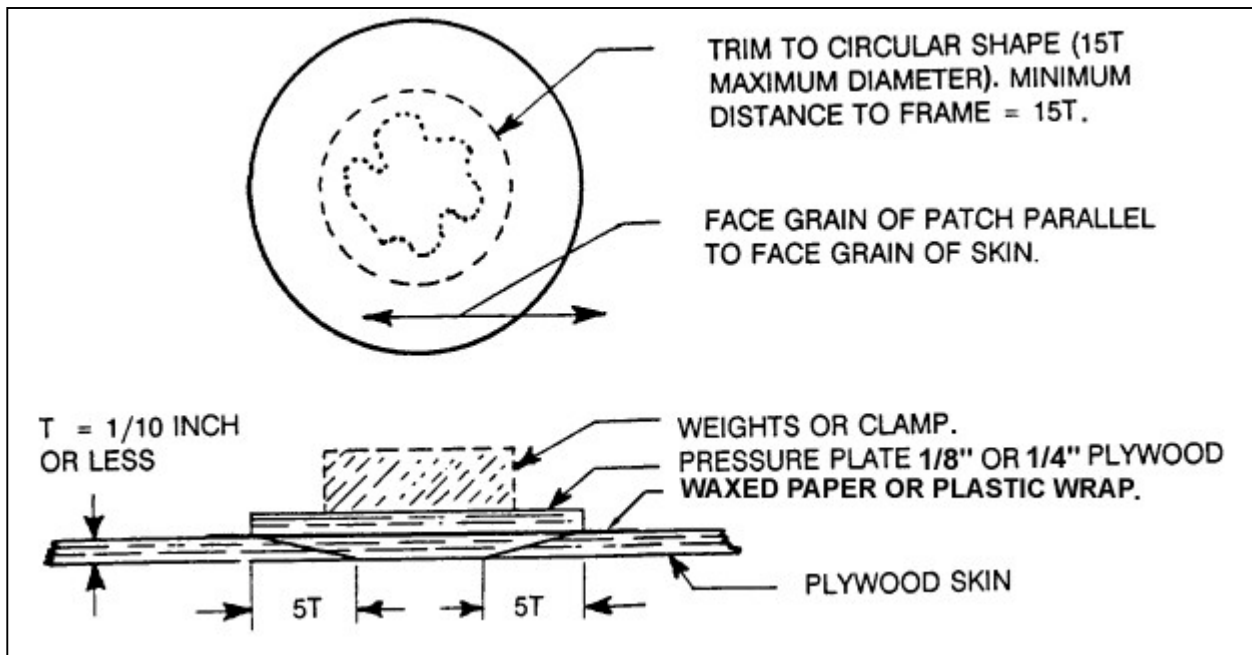


FIGURE 1-13. Splayed patch.

(c) A hole, the exact size of the inside circle of the scarf patch, is made in the block, and is centered over the trimmed area of damage.

(d) The block is removed, after the adhesive on the patch has set, leaving a flush surface to the repaired skin.

(4) Steps in making a scarf patch when the back of the skin is not accessible are as follows:

(a) After removing damaged sections, install backing strips, as shown in figure 1-16, along all edges that are not fully backed by a rib or a spar. To prevent warping of the skin, backing strips should be made of a soft-textured plywood, such as yellow poplar or spruce rather than solid wood. All junctions between backing strips and ribs or spars should have the end of the backing strip supported by a saddle gusset of plywood.

(b) If needed, nail and bond the new gusset plate to rib. It may be necessary to

remove and replace the old gusset plate with a new saddle gusset, or it may be necessary to nail a saddle gusset over the original gusset.

(c) Attach nailing strips to hold backing strips in place while the adhesive sets. Use a bucking bar, where necessary, to provide support for nailing. After the backing strips are fully bonded, install the patch.

d. Plug Patch. Either oval or round plug patches may be used on plywood skins provided the damage can be covered by the patches whose dimensions are given in figure 1-17 and figure 1-18. The plug patch is strictly a skin repair, and should be used only for damage that does not involve the supporting structure under the skin. The face-grain direction of the finished patch must match the surrounding skin.

(1) Steps in making an oval plug patch are as follows:

(a) Explore the area about the hole to be sure it lies at least the width of the oval

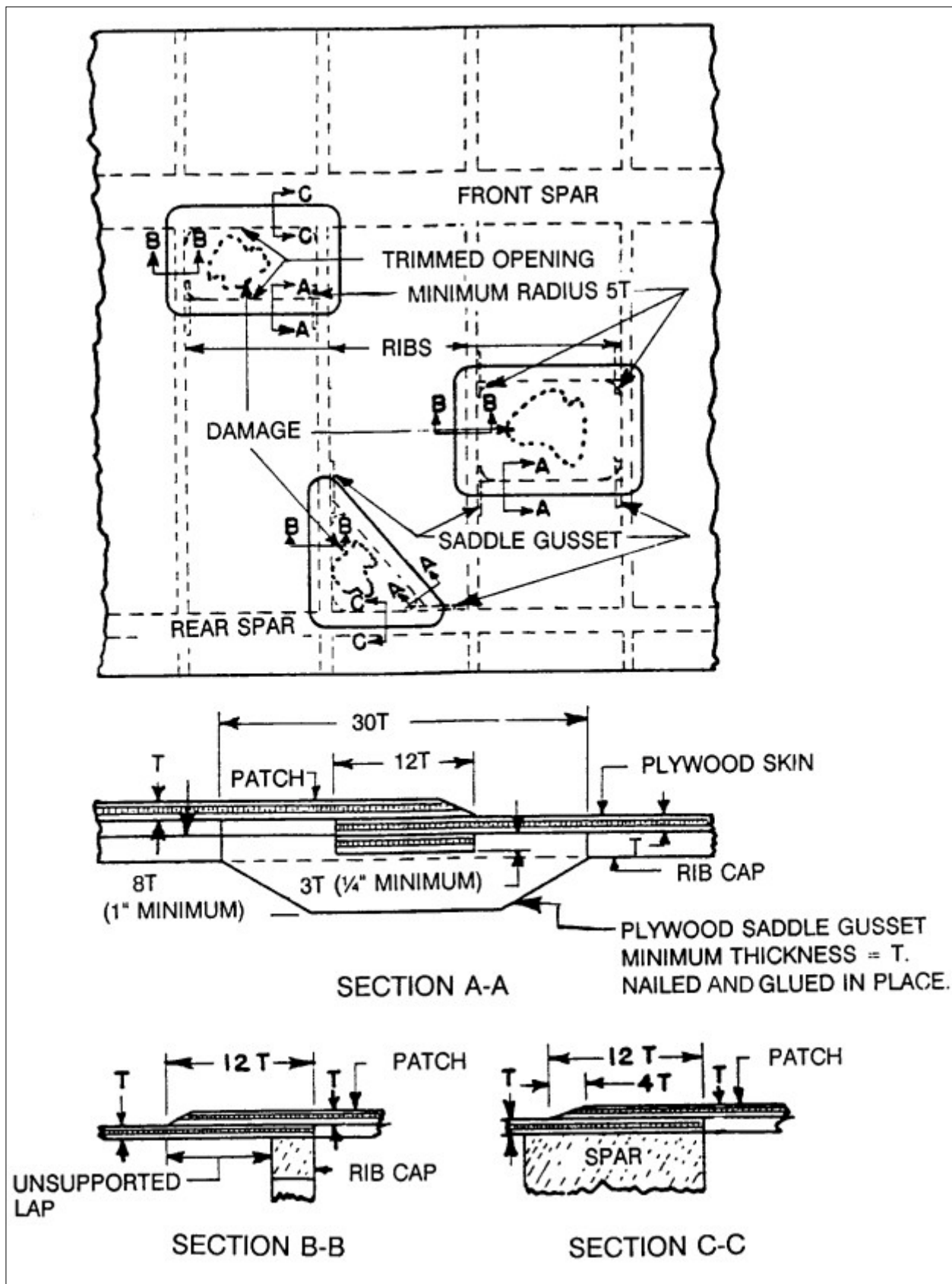


FIGURE 1-14. Surface patches.

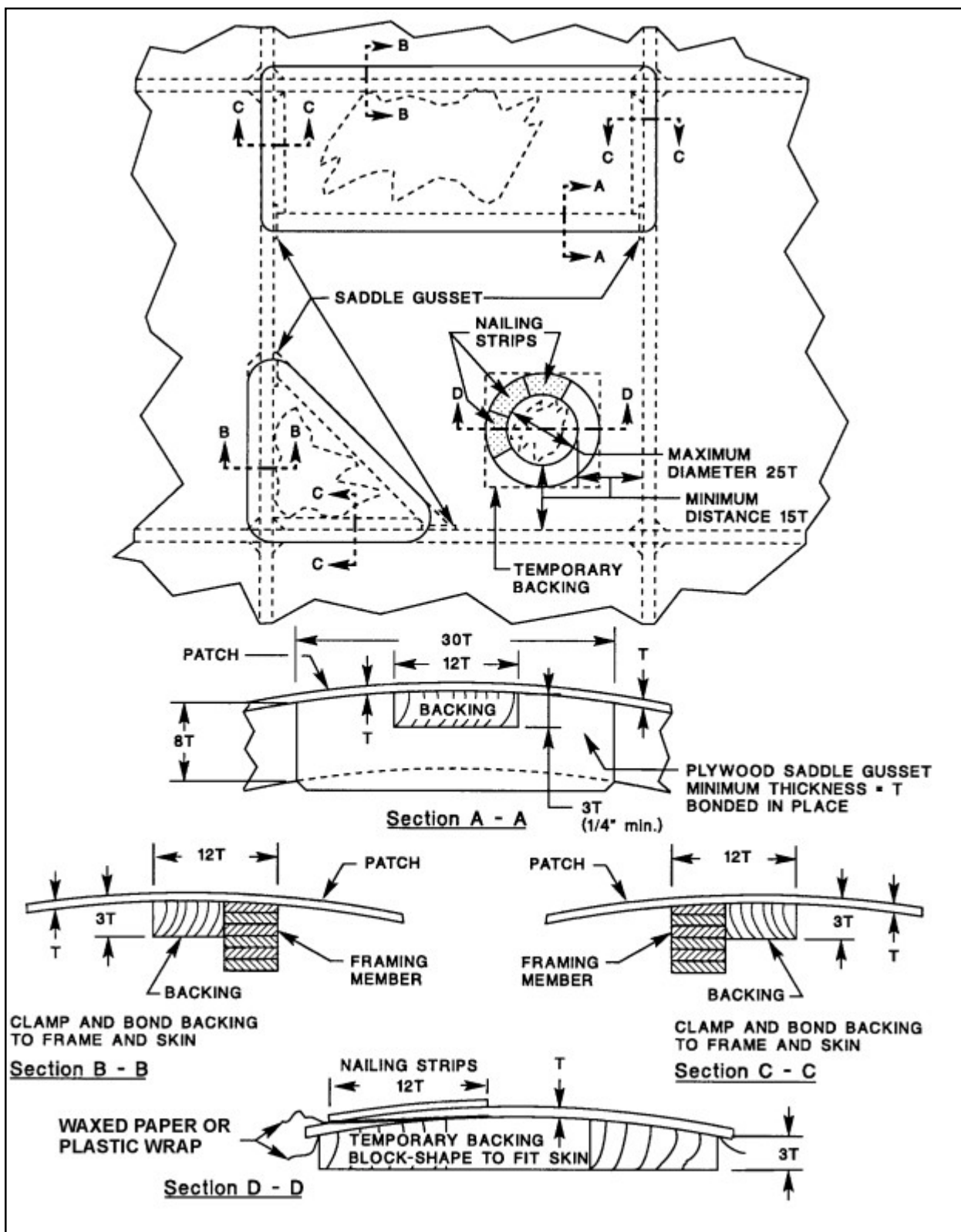


FIGURE 1-15. Scarf patches (back of skin accessible).

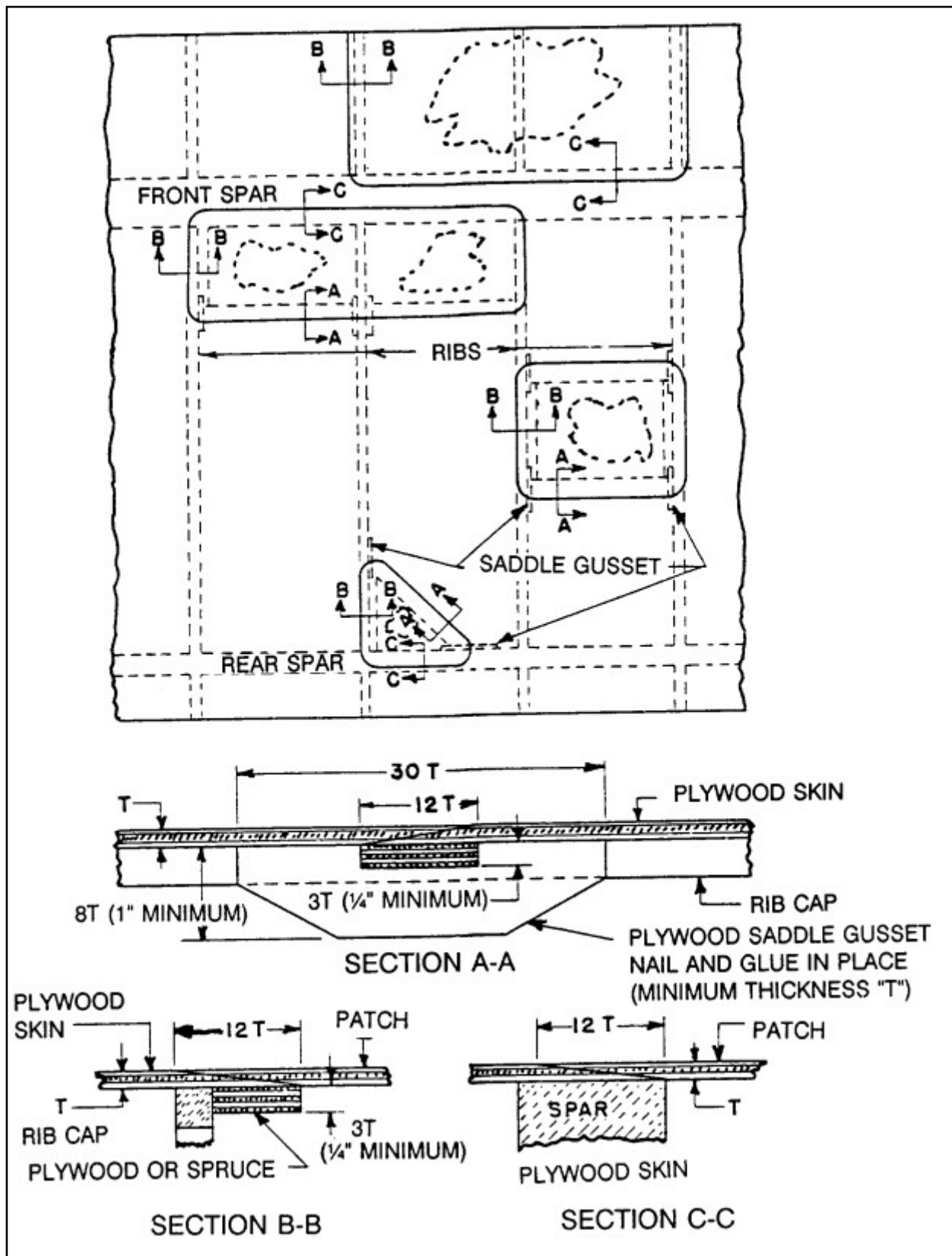


FIGURE 1-16. Scarf patches (back of skin not accessible).

doubler from a rib or a spar. Refer to figure 1-17 for repair details.

(b) Prepare a patch and a doubler of the same species plywood as the surrounding skin using the dimensions shown in figure 1-17.

(c) Lay the oval plug patch over the damage and trace the patch onto the skin. Saw to the line, and trim the hole edges with a knife and sandpaper.

(d) Mark the exact size of the patch on one surface of the oval doubler and apply adhesive to the area outside the line. Insert doubler through the hole and bring it, adhesive side up, to the underside of the skin with the pencil outline of the patch matching the edges of the hole. If the curvature of the surface to be repaired is greater than a rise of 1/8 inch in 6 inches, the doubler should be preformed by hot water or steam bending to the approximate curvature. As an alternative to preforming of the 1/4 inch stock, the doubler may be laminated from two thicknesses of 1/8 inch ply.

(e) Apply nailing strips outlining the hole to apply bonding pressure between doubler and skin. Use a bucking bar to provide support for nailing. When two rows of nails are used, stagger nail spacing. Allow adhesive to cure.

(f) Apply adhesive to remaining surface of the doubler and to the mating surface on the patch. Lay the patch in position over the doubler, and screw the pressure plate to the patch assembly using a small nail to line up the holes that have been previously made with patch and plate matching. No. 4 round head screws are used. Lead holes in the plywood doubler are not necessary. Waxed paper or plastic wrap between the plate and patch prevents adhesive from bonding the plate to the patch. No clamps or further pressure need be applied, as the nailing strips and screws exert ample pressure.

(2) Round plug patches may be made by following the steps in figure 1-18. The steps are identical to those for making the oval patch except for the insertion of the doubler. In using the round patch, where access is from only one side, the round doubler cannot be inserted unless it has been split.

1-52. FABRIC PATCH. Small holes not exceeding 1 inch in diameter, after being trimmed to a smooth outline, may be repaired by doping a fabric patch on the outside of the plywood skin. The edges of the trimmed hole should first be sealed, and the fabric patch should overlap the plywood skin by at least 1 inch. Holes nearer than 1 inch to any frame member, or in the leading edge or frontal area of the fuselage, should not be repaired with fabric patches.

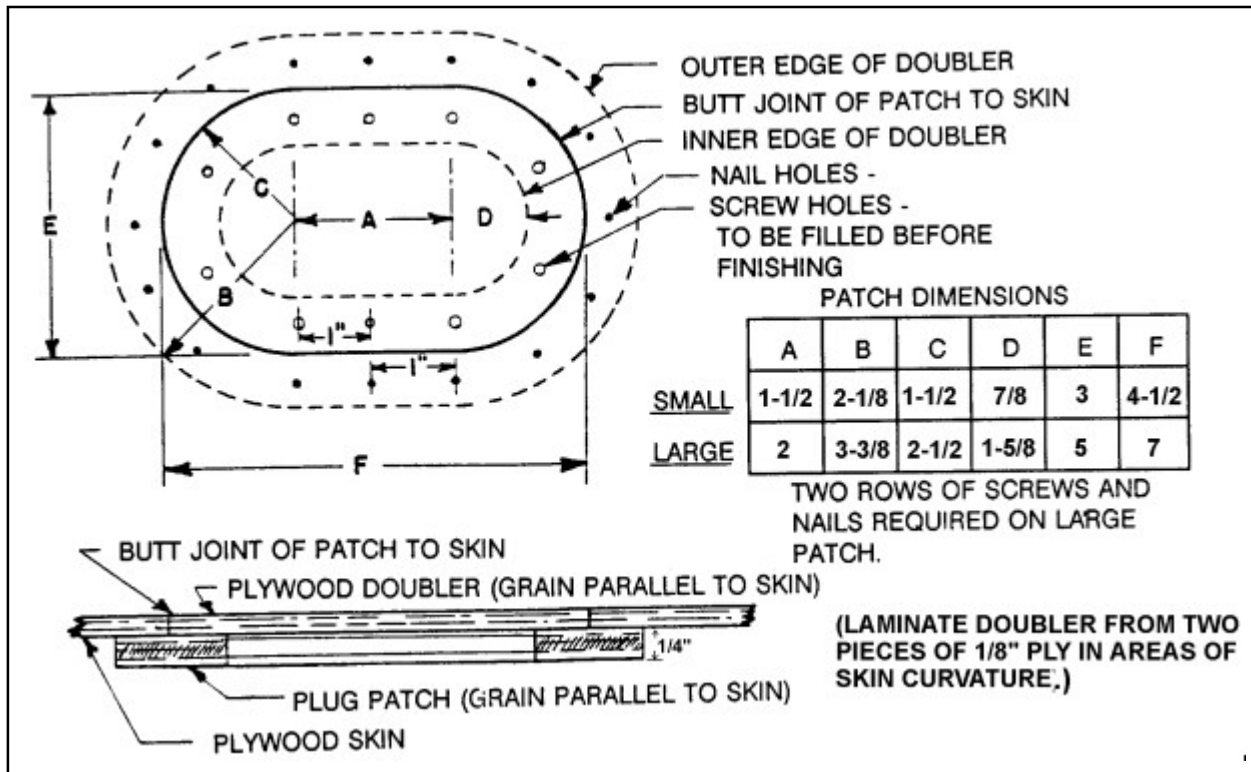


FIGURE 1-17. Oval plug patch assembly.

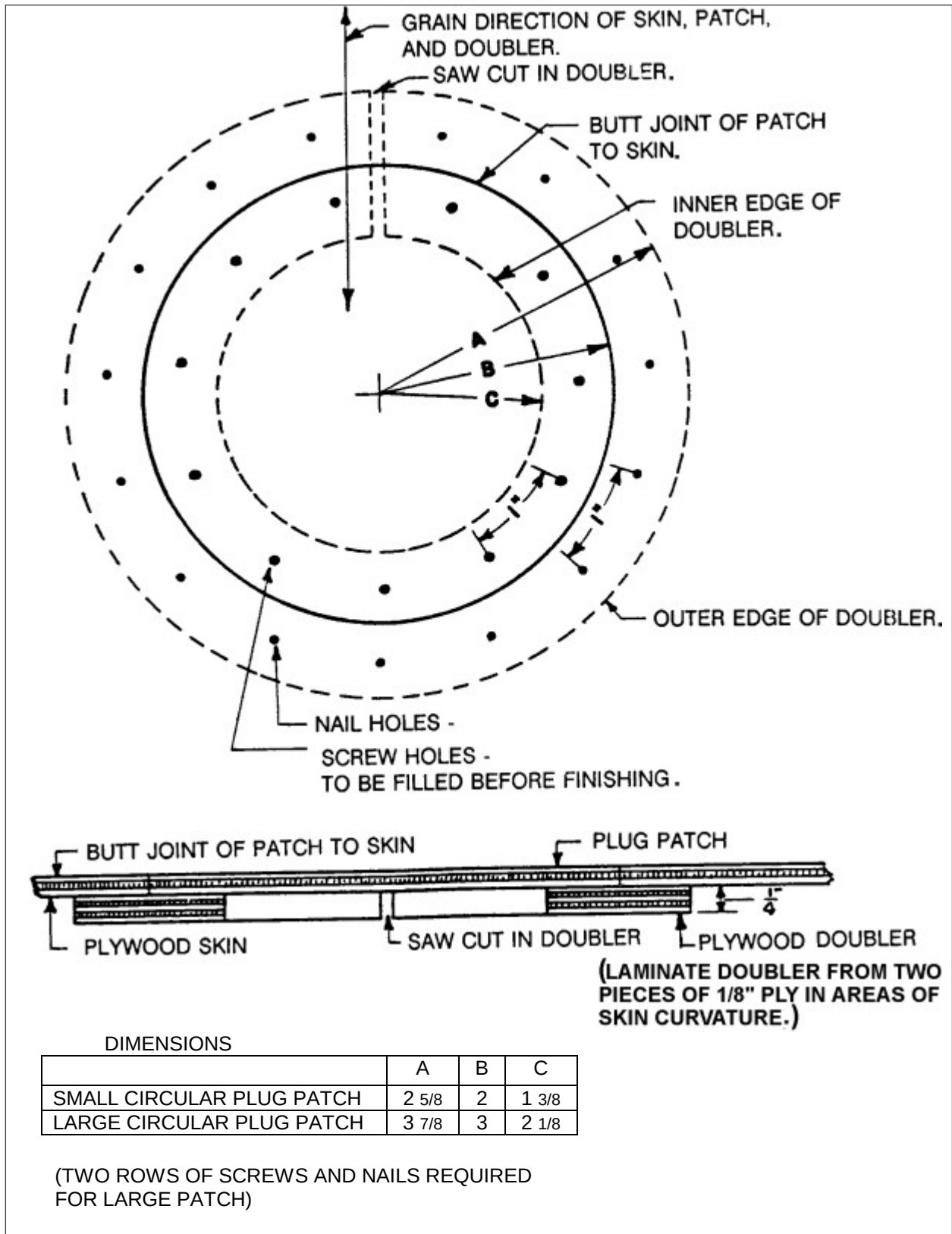
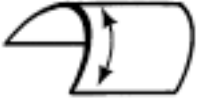
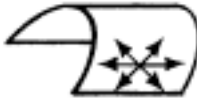
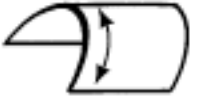
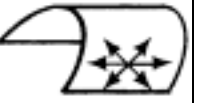


FIGURE 1-18. Round plug patch assembly.

1-53.—1-63. [RESERVED.]

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THICKNESS (INCHES)	NUMBER OF PLIES	MINIMUM BEND RADIUS (INCHES)			
.035	3	2.0	1.1	0.5	0.1
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.125	3	12	7.1	3.8	1.2
.155	3	16	10	5.3	1.8
.185	3	20	13	7.1	2.6
.160	5	17	11	6	2
.190	5	21	14	7	3
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.250	5	31	20	12	5
.315	5	43	28	16	7
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(1) Lay out the patch according to figure 1-13. Tack a small piece of plywood over the hole for a center point and draw two circles with a divider, the inner circle to be the size of the hole and the outer circle marking the limits

of the taper. The difference between the radii is 5T (5 times the thickness of the skin). If one leg of the dividers has been sharpened to a chisel edge, the dividers may be used to cut the inner circle.

(2) Taper the hole evenly to the outer mark with a chisel, knife, or rasp.

(3) Prepare a circular tapered patch to fit the prepared hole, and bond the patch into place with face-grain direction matching that of the original surface.

(4) Use waxed paper or plastic wrap, (cut larger than the size of the patch) between the patch and the plywood pressure plate. This prevents excess adhesive from bonding the pressure plate to the skin. Center the pressure plate carefully over the patch.

(5) As there is no reinforcement behind this patch, care must be used so that pressure is not great enough to crack the skin. On horizontal surfaces, weights or sandbags will be sufficient. On patches too far from any edge for the use of standard hand clamps, jaws of greater length may be improvised. Table 1-2, columns (1) and (3), may also be used for determining the maximum thickness of single laminations for curved members.

(6) Fill, sand, and refinish the patch.

b. Surface Patch. Plywood skins that are damaged between or along framing members may be repaired by surface or overlay patches as shown in figure 1-14. Surface patches located entirely aft of the 10 percent chord line, or which wrap around the leading edge and terminate aft of the 10 percent chord line, are permissible. Surface patches may have as much as a 50 inch perimeter and may cover as much as 1 frame (or rib) space. Trim the damaged skin to a rectangular or triangular shape and round the corners. The radius of rounded corners must be at least 5 times the skin thickness. Bevel the forward edges of patches located entirely aft of the 10 percent chord line to 4 times the skin thickness. The face-grain direction must be the same as the original skin. Cover completed surface patches with fabric to match surrounding area. The fabric must overlap the original fabric at least 2 inches.

c. Scarf Patch. A properly prepared and inserted scarf patch is the best repair for damaged plywood skins and is preferred for most skin repairs. Figure 1-15 shows the details and dimensions to be used when installing typical scarf skin patches, when the back of the skin is accessible. Follow figure 1-16 when the back of the skin is not accessible. The scarf slope of 1 in 12, shown in both figures, is the steepest slope permitted for all kinds of plywood. If the radius of curvature of the skin at all points

on the trimmed opening is greater than 100 times the skin thickness, a scarf patch may be installed.

(1) Scarf cuts in plywood may be made by hand plane, spoke shave, scraper, or accurate sandpaper block. Rased surfaces, except at the corners of scarf patches and sawn surfaces, are not recommended as they are likely to be rough or inaccurate.

(2) Nail strip or small screw clamping is often the only method available for bonding scarf joints in plywood skin repairs. It is essential that all scarf joints in plywood be backed with plywood or solid wood to provide adequate nail holding capacity. The face-grain direction of the plywood patch must be the same as that of the original skin.

(3) If the back of a damaged plywood skin is accessible (such as a fuselage skin), it should be repaired with a scarf patch, following the details shown in figure 1-15. Whenever possible, the edges of the patch should be supported as shown in section C-C of figure 1-15. When the damage follows or extends to a framing member, the scarf may be supported as shown in section B-B of figure 1-15. Damages that do not exceed 25 times the skin thickness in diameter after being trimmed to a circular shape and are not less than 15 times the skin thickness to a framing member, may be repaired as shown in figure 1-15, section D-D.

(a) The backing block is carefully shaped from solid wood and fitted to the inside surface of the skin, and is temporarily held in place with nails.

(b) Use waxed paper or plastic wrap to prevent bonding of the backing block to the skin.

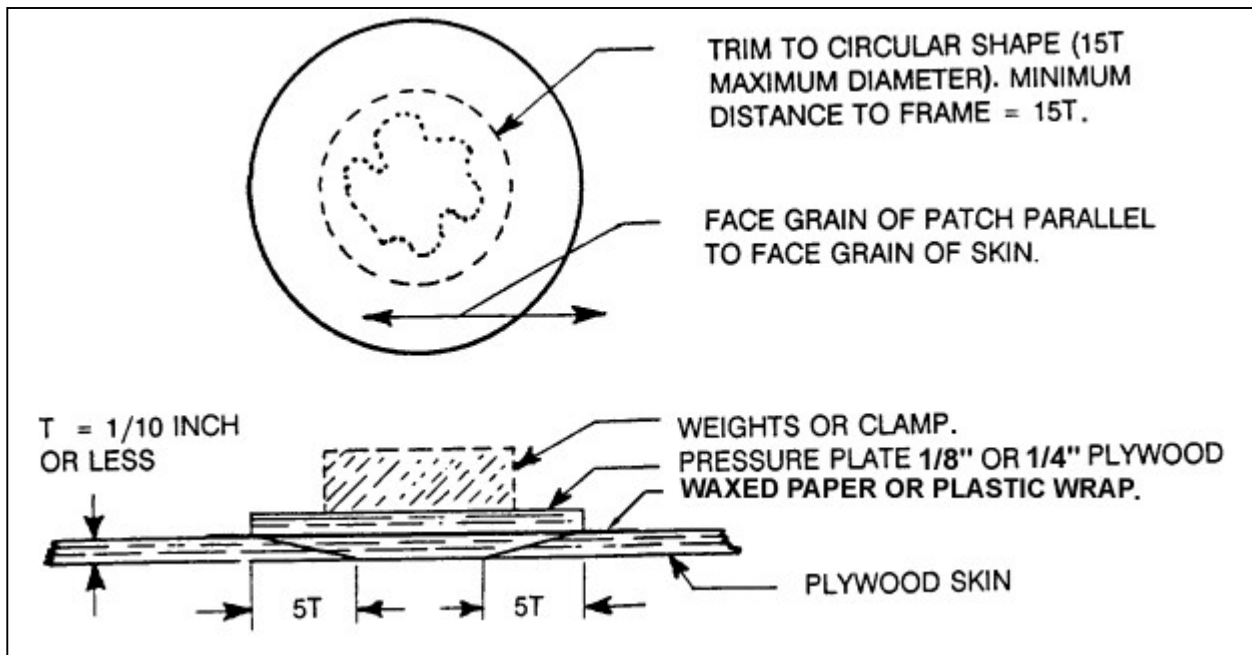


FIGURE 1-13. Splayed patch.

(c) A hole, the exact size of the inside circle of the scarf patch, is made in the block, and is centered over the trimmed area of damage.

(d) The block is removed, after the adhesive on the patch has set, leaving a flush surface to the repaired skin.

(4) Steps in making a scarf patch when the back of the skin is not accessible are as follows:

(a) After removing damaged sections, install backing strips, as shown in figure 1-16, along all edges that are not fully backed by a rib or a spar. To prevent warping of the skin, backing strips should be made of a soft-textured plywood, such as yellow poplar or spruce rather than solid wood. All junctions between backing strips and ribs or spars should have the end of the backing strip supported by a saddle gusset of plywood.

(b) If needed, nail and bond the new gusset plate to rib. It may be necessary to

remove and replace the old gusset plate with a new saddle gusset, or it may be necessary to nail a saddle gusset over the original gusset.

(c) Attach nailing strips to hold backing strips in place while the adhesive sets. Use a bucking bar, where necessary, to provide support for nailing. After the backing strips are fully bonded, install the patch.

d. Plug Patch. Either oval or round plug patches may be used on plywood skins provided the damage can be covered by the patches whose dimensions are given in figure 1-17 and figure 1-18. The plug patch is strictly a skin repair, and should be used only for damage that does not involve the supporting structure under the skin. The face-grain direction of the finished patch must match the surrounding skin.

(1) Steps in making an oval plug patch are as follows:

(a) Explore the area about the hole to be sure it lies at least the width of the oval

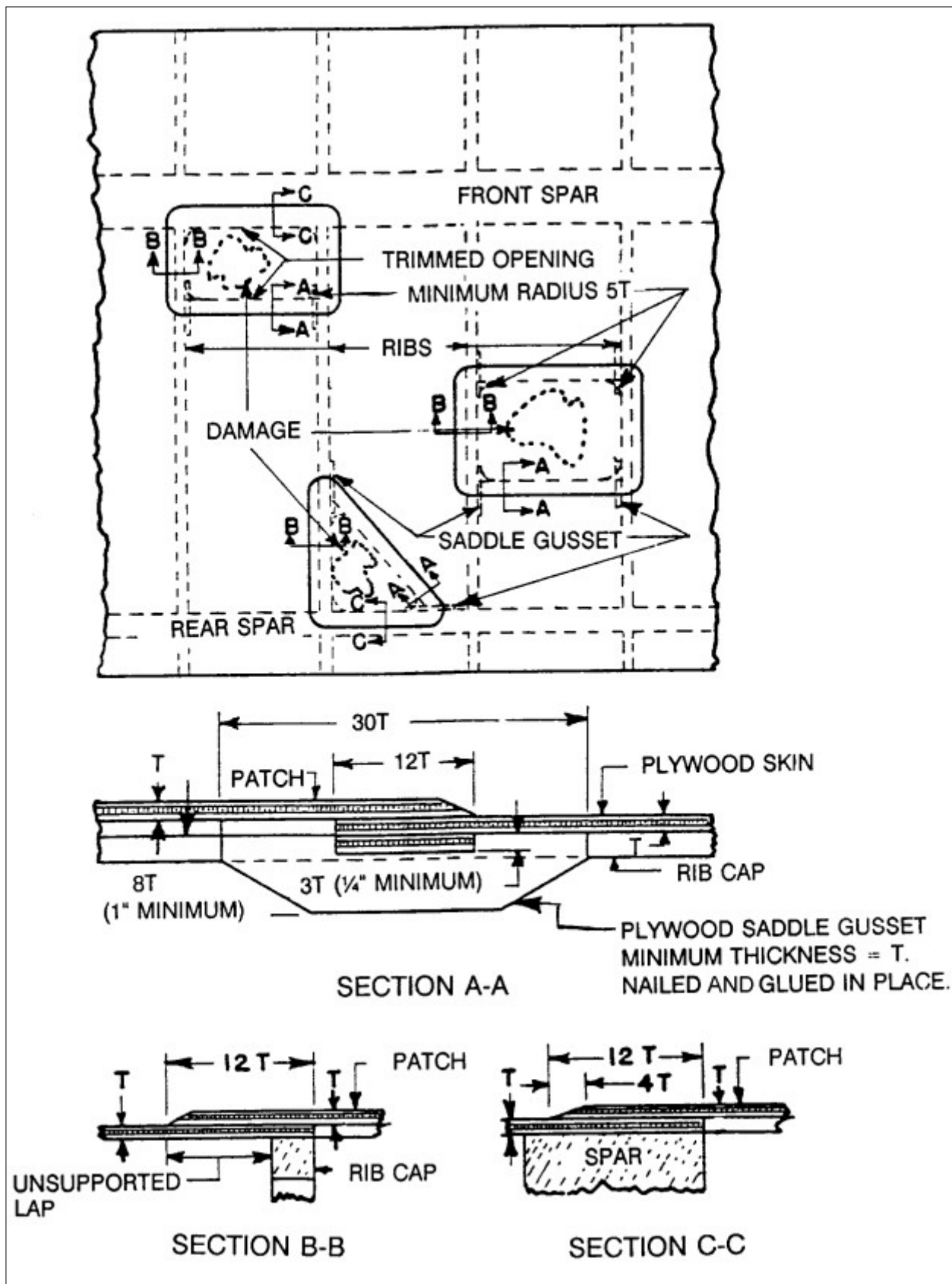


FIGURE 1-14. Surface patches.

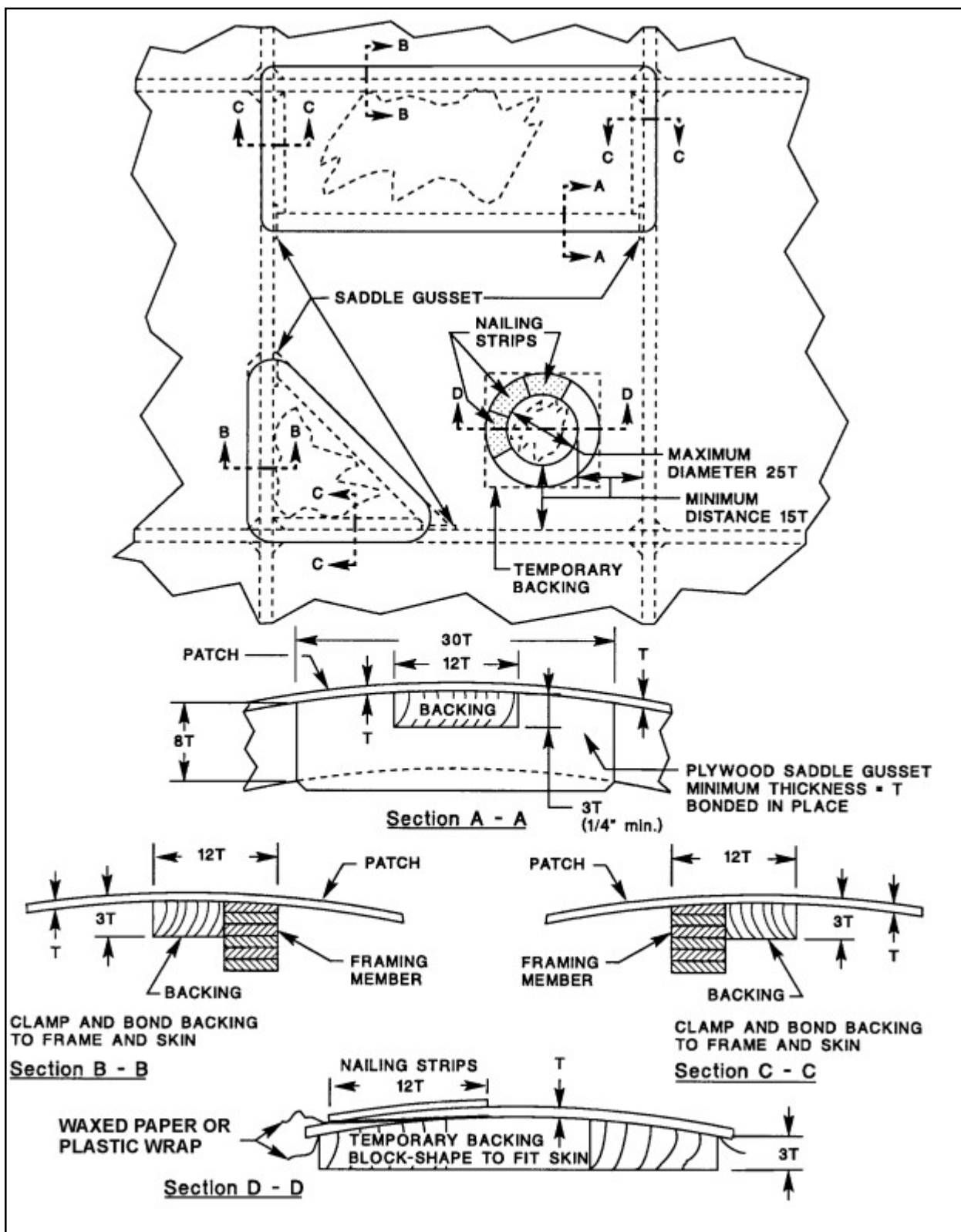


FIGURE 1-15. Scarf patches (back of skin accessible).

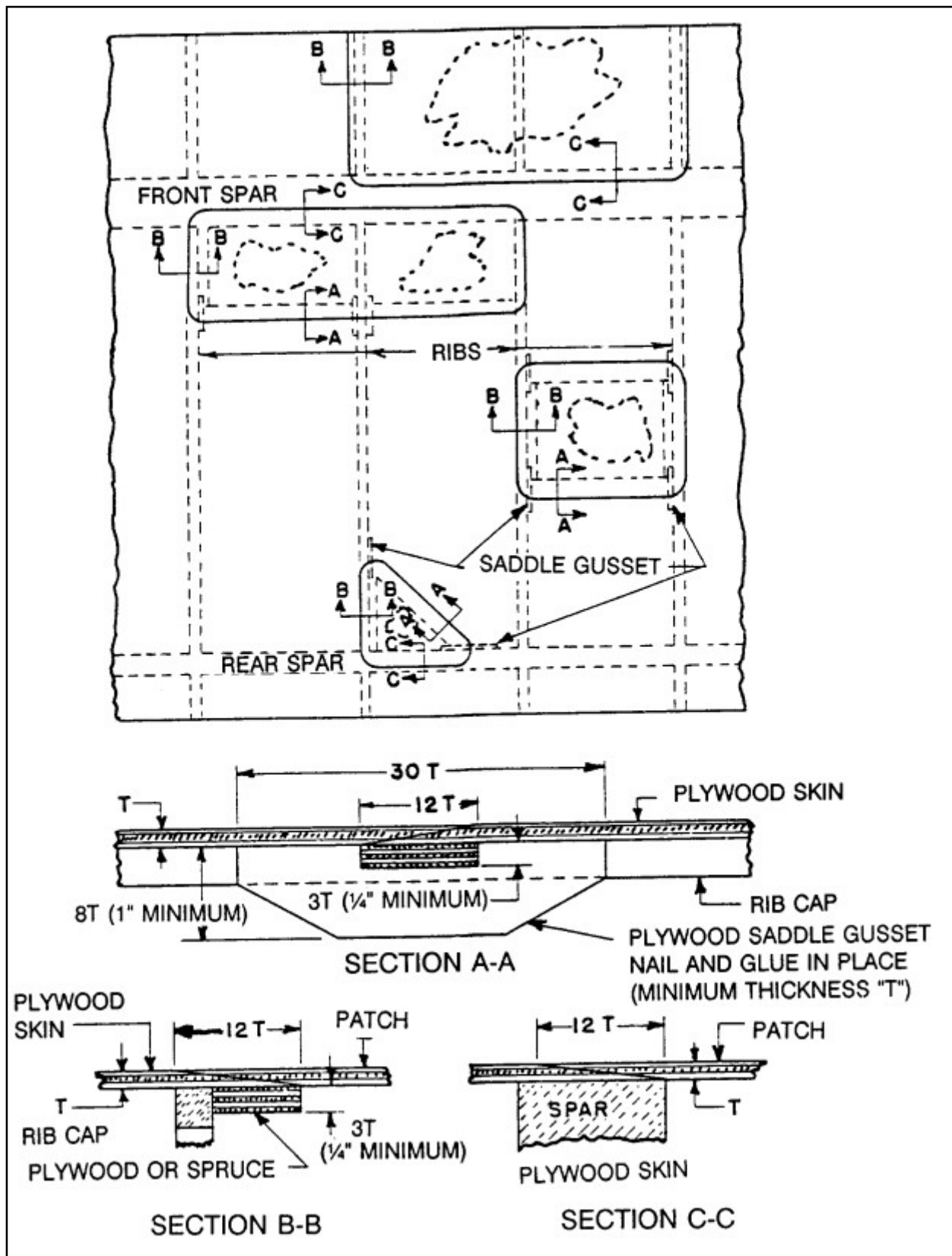


FIGURE 1-16. Scarf patches (back of skin not accessible).

doubler from a rib or a spar. Refer to figure 1-17 for repair details.

(b) Prepare a patch and a doubler of the same species plywood as the surrounding skin using the dimensions shown in figure 1-17.

(c) Lay the oval plug patch over the damage and trace the patch onto the skin. Saw to the line, and trim the hole edges with a knife and sandpaper.

(d) Mark the exact size of the patch on one surface of the oval doubler and apply adhesive to the area outside the line. Insert doubler through the hole and bring it, adhesive side up, to the underside of the skin with the pencil outline of the patch matching the edges of the hole. If the curvature of the surface to be repaired is greater than a rise of 1/8 inch in 6 inches, the doubler should be preformed by hot water or steam bending to the approximate curvature. As an alternative to preforming of the 1/4 inch stock, the doubler may be laminated from two thicknesses of 1/8 inch ply.

(e) Apply nailing strips outlining the hole to apply bonding pressure between doubler and skin. Use a bucking bar to provide support for nailing. When two rows of nails are used, stagger nail spacing. Allow adhesive to cure.

(f) Apply adhesive to remaining surface of the doubler and to the mating surface on the patch. Lay the patch in position over the doubler, and screw the pressure plate to the patch assembly using a small nail to line up the holes that have been previously made with patch and plate matching. No. 4 round head screws are used. Lead holes in the plywood doubler are not necessary. Waxed paper or plastic wrap between the plate and patch prevents adhesive from bonding the plate to the patch. No clamps or further pressure need be applied, as the nailing strips and screws exert ample pressure.

(2) Round plug patches may be made by following the steps in figure 1-18. The steps are identical to those for making the oval patch except for the insertion of the doubler. In using the round patch, where access is from only one side, the round doubler cannot be inserted unless it has been split.

1-52. FABRIC PATCH. Small holes not exceeding 1 inch in diameter, after being trimmed to a smooth outline, may be repaired by doping a fabric patch on the outside of the plywood skin. The edges of the trimmed hole should first be sealed, and the fabric patch should overlap the plywood skin by at least 1 inch. Holes nearer than 1 inch to any frame member, or in the leading edge or frontal area of the fuselage, should not be repaired with fabric patches.

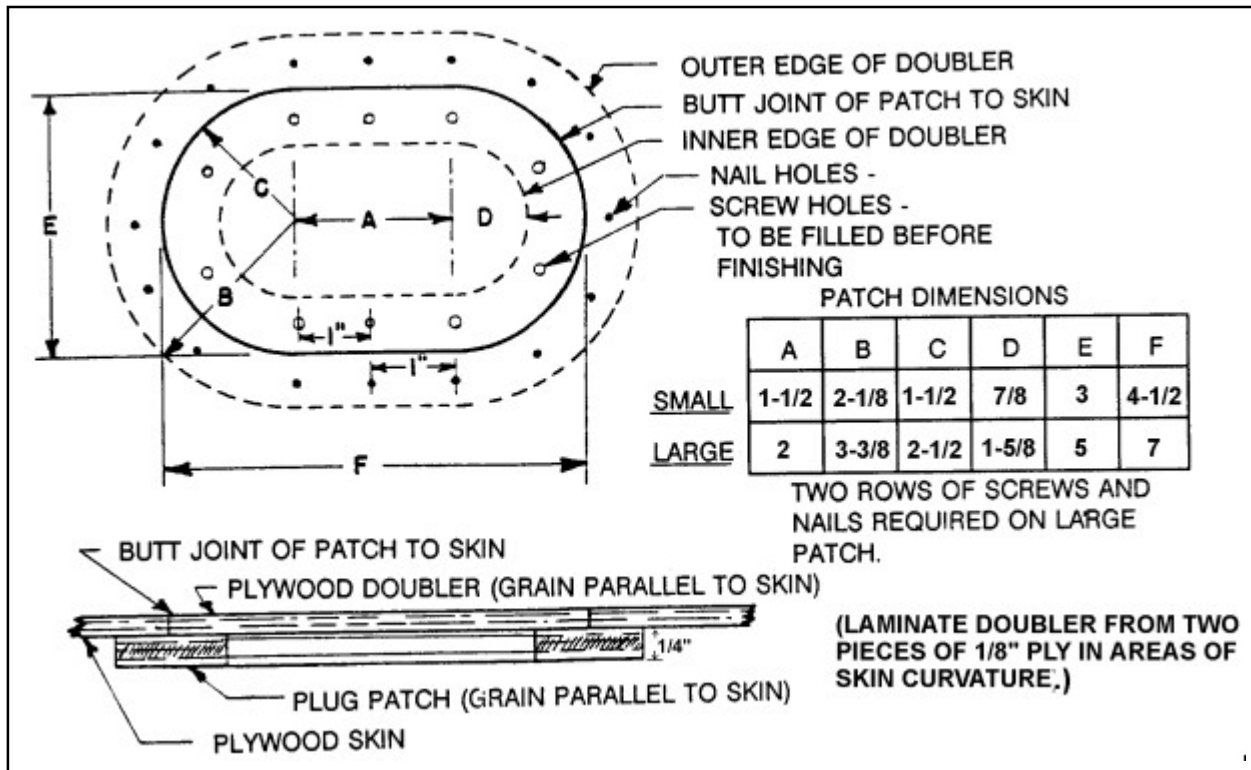


FIGURE 1-17. Oval plug patch assembly.

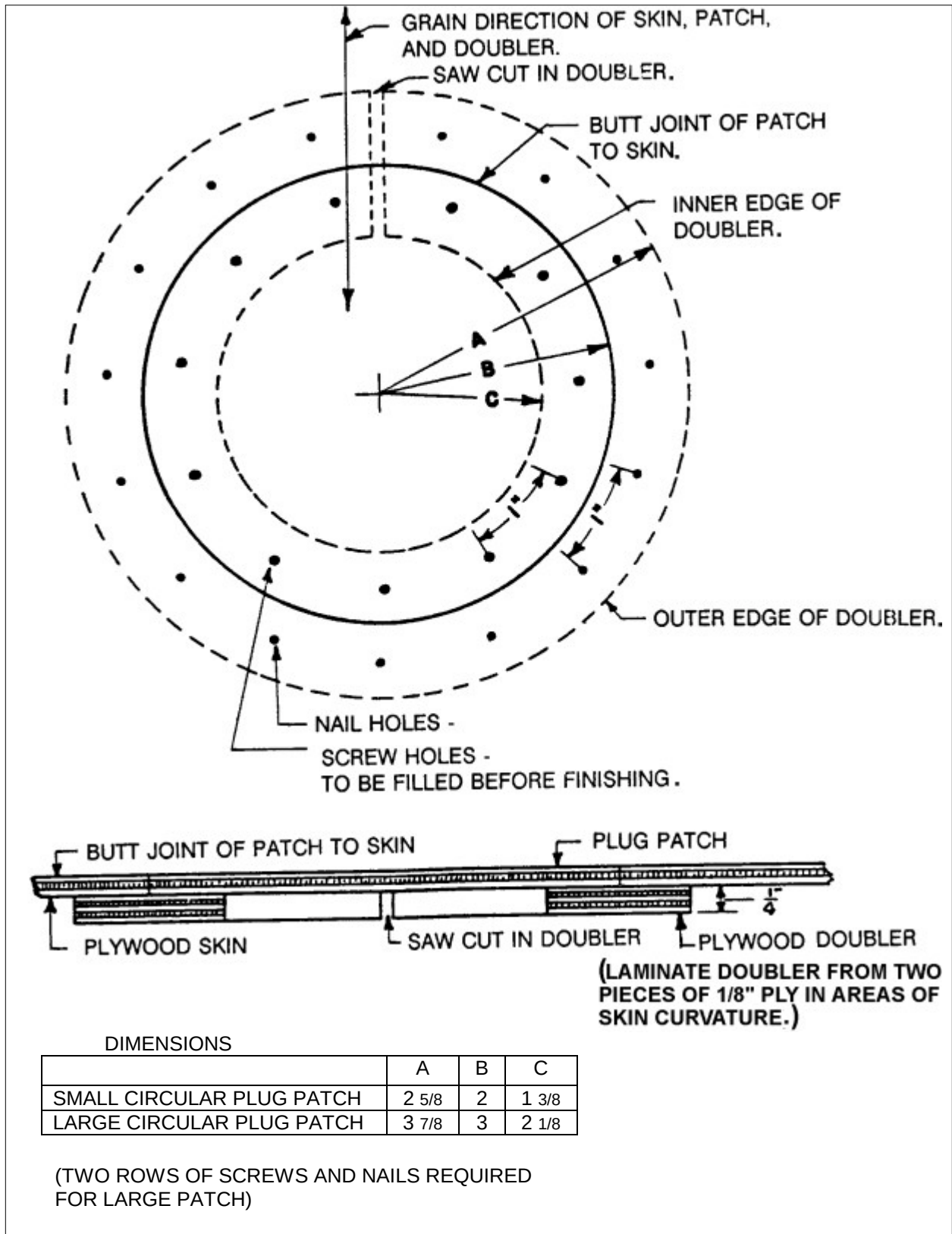


FIGURE 1-18. Round plug patch assembly.

1-53.—1-63. [RESERVED.]

SECTION 5. FINISHING WOOD STRUCTURES

1-64. GENERAL. Any repair to spars, ribs, skin surfaces, or other structural parts of the airframe involves finishing as the final step of the job. The surface finish is the final line of defense to prevent the destructive effects of moisture entry into the structure. The time and effort spent during the preparatory phase of the refinishing process will be reflected in the appearance and longevity of the finished surface. Adherence to the instructions issued by the finish manufacturer is necessary to obtain the appearance desired and protective characteristics for the product used. The primary objective of interior finishes is to afford protection of the wood against serious change in moisture content when exposed to damp air or to water that gains entrance to closed spaces by condensation or by penetration of rain, mist, or fog. Coatings, on contact areas between wood and metal protect the metal against corrosion from moisture in the wood. The primary objectives of the exterior finish are to protect the wood against weathering, provide a suitable appearance, and present a smooth surface in flight.

1-65. ACCEPTABLE FINISHES. Any varnish conforming to Federal Specification TT-V-109, as amended, or other coating approved by the airframe manufacturer or the FAA is acceptable. Exterior surfaces must be further protected from the effects of abrasion, weather, and sunlight. A number of systems for exterior finishing have STC approval and are manufactured under a PMA. (See Chapter 2, Fabric Covering.) Low viscosity epoxy adhesive (meeting the requirements of paragraph 1-4 for that purpose) may be used as an internal surface finish when subsequent bonding is necessary.

1-66. PRECAUTIONS.

a. When making repairs, avoid excessive contamination of surfaces with adhesive squeeze-out at joints and on all surfaces. Excess adhesive should always be removed before applying finish. Because many paints and adhesives are incompatible, even a slight amount of adhesive underneath the finish may cause premature deterioration of the finish.

b. Soiling substances, such as oil and grease, should be removed as completely as possible. Naphtha may be used to remove surface deposits of oil and grease; however, thinned residue may penetrate into any unprotected wood. In areas where minor amounts of oil or grease have penetrated the wood surface, removal may be accomplished by use of an absorbent type of cleaner such as gunsmith's whiting or a clothing spot lifter such as K2r™. Marks that are made by grease pencils or lumber crayons containing wax are harmful and should be removed, but marks made by ordinary soft graphite pencils and nonoily stamp pad inks may be safely finished over. All dust, dirt, and other solid particles should be removed.

c. Sawdust, shavings, and chips must be removed from enclosed spaces before they are sealed off by replacement of skin. A vacuum cleaner is useful for such cleaning.

d. Since most adhesives will not bond satisfactorily to sealers, it is necessary to avoid applying sealer over the areas where adhesive will be applied. Mark off areas to receive adhesive, and allow an additional 1/4 inch on each side of the adhesive area to provide for

misalignment when mating the parts. It is preferable to leave some unsealed areas rather than risk weakening the joint by accidental overlap of the sealer into the bonded areas. Wherever possible, apply sealer to the margins after the adhesive has cured. As an example, the lower skin of a wing bay would be installed first, leaving access from above to apply sealer. All low spots (where moisture would collect) are well sealed. The top skin would be installed last, so that the only unsealed margins would be on upper surfaces where moisture is least likely to collect.

e. An alternative to the previous paragraph is to use an approved epoxy coating and compatible epoxy adhesive. Apply the coating, allowing 1/4 inch margins as in the previous paragraph. After the coating has cured, apply epoxy adhesive to joint surfaces, and overlap the adhesive onto the sealer. Close joint and clamp. The epoxy adhesive will bond satisfactorily to the coating and ensure a complete coverage of the wood surfaces. Use only approved and compatible adhesives and coatings for this method.

1-67. FINISHING OF INTERIOR SURFACES. Finish repaired ribs, spars, interior of plywood skin, and other internal members, including areas of contact between metal and wood, by applying one thinned coat (for penetration into wood grain) of varnish or other acceptable finish, followed by two full coats. Protect built-up box spars and similar closed structures on the interior in the same way. Where better protection is required, as on the surfaces of wheel wells and the bottoms of hulls below the floor boards, an additional coat of aluminized sealer consisting of 12 to 16 ounces of aluminum paste per gallon of sealer, may be applied.

1-68. FINISHING OF EXTERIOR SURFACES. Exterior surfaces require more protection than interior areas due to the effects of

abrasion, weather, and sunlight. (See chapter 2.) Tests have shown that the interior temperature of wooden aircraft structures can reach 185 °F when the aircraft is finished in a dark color and parked outdoors on a hot, still day. Exposure to prolonged high temperature is detrimental to wood, adhesives, and finishes. Wood loses approximately 25 percent of its strength at 125 °F. For this reason, the mechanic should consider temperature effects when selecting finish colors or looking for areas of likely deterioration. The lowest temperatures are found when the aircraft is finished in white or very light colors, while darker colors produce higher temperatures. A general trend toward higher temperatures may be seen when exterior colors are yellow, pink, light blue, aluminum, purple, blue, light green, orange, tan, red, green, brown, and black. A lighter shade of a particular color helps to reduce temperatures.

1-69. FINISHING OF END GRAIN SURFACES. End grain portions of wooden members are much more absorbent than side grain. Because of this extreme vulnerability to moisture entry, it is necessary to take extra precautions to seal end grain.

a. Apply at least one thinned coat of acceptable sealer to ensure maximum penetration, and then follow with as many full strength coats as necessary to achieve a smooth, glossy coating. Depending on the type wood to be finished, two to four full coats will be required. A final coat of aluminized varnish may be applied to end grain surfaces. If the surfaces are to be finished with dope or lacquer, a dope-proof sealer, similar to Federal Specification TT-V-109, or epoxy sealer should be used.

b. Exposed end grain includes such surfaces as spar butts, skin edges, areas around vent holes, inspection holes, fittings, and exposed scarfed or tapered surfaces.

1-70. FINISHING WITH FABRIC OR TAPE. A number of systems for exterior finishing have STC approval and are manufactured under a PMA. Follow the product manufacturer's instructions for the system used.

a. If the finish surrounding the repair is a traditional dope system, seal the wood grain with a suitable solvent resistant one-part varnish, commonly described as "dope proof," or a two-part epoxy varnish. Follow with two coats of clear dope, and allow sufficient drying time between coats.

b. Apply a third coat of clear dope and lay a piece of pinked-edge airplane cloth into the wet film. All air bubbles should be worked out by brushing to ensure maximum adhesion. When dry, apply one brush coat, to ensure

proper penetration, and at least one spray coat of clear dope. The dried spray coat may be sanded with fine sandpaper to obtain a smoother finish. Complete the refinishing of the surface by application of a topcoat as required to match the adjacent area.

1-71. SEALING OF BOLT HOLES. Bolt holes in wooden structure provide a vulnerable entry point for moisture. Variations in moisture content around bolt holes can lead to decay or splitting. In addition, excessive moisture at bolt holes promotes corrosion of the bolts. Sealing of the wood surfaces in bolt holes can be accomplished by application of varnish or other acceptable sealer into the open hole. The sealer must be allowed to dry or cure thoroughly prior to bolt installation.

1-72.—1-79. [RESERVED.]